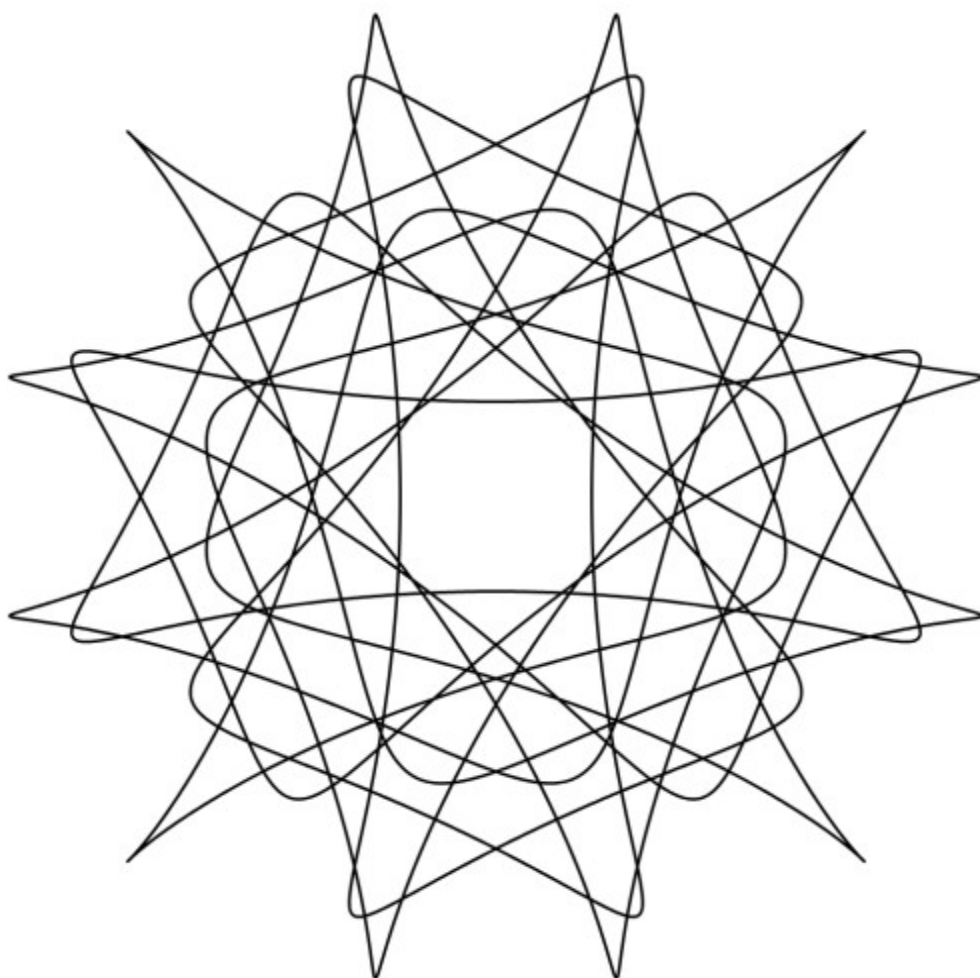

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E P I C Y C L E S

o f

INEVITABILITY

The Refusal to Pretend to Know



Monitum: De Diversis Opinionibus

A quarter-century has passed, yet accounts of how the Twin Towers collapsed still diverge into a chorus of incompatible certainties. Even among credentialed voices, one finds a shifting set of explanations, each presented as self-evident:

- *“It was a hammer-blow of mass.”*
- *“It was a zipper-like sequence of connections.”*
- *“The core was hollowed out.”*
- *“The core was redundant and robust.”*
- *“It was basic Newtonian $F = ma$.”*
- *“It was emergent behavior beyond simple reduction.”*
- *“The steel sagged.”*
- *“The steel fractured.”*

Complex failures admit multiple descriptive lenses. They do not admit mutually exclusive sufficiencies. When incompatible mechanisms are each treated as decisive, the explanatory frame has become elastic enough to accommodate any outcome. “Inevitability,” used this way, does more argumentative work than any single mechanism has earned — a placeholder for demonstration rather than its result.

Scientific claims earn their standing through demonstrated mechanism. Consensus follows demonstration; it does not substitute for it. Where inevitability is asserted, the burden is mechanical: to show that the proposed processes, under stated assumptions and across realistic variance, reproduce the observed progression with the claimed generality.

Epicycles of Inevitability undertakes a narrow audit. It identifies the operative mechanical claims embedded in prevailing explanations and examines whether those claims are supported by validation proportional to the strength of the conclusions drawn from them. The issue is not whether collapse can be produced under selected assumptions. The issue is whether inevitability has been earned by the evidence.

Unresolved questions are not an embarrassment to inquiry. They are its working condition. This project does not seek to displace existing models. It seeks to discipline their scope: to distinguish what has been established, what has been provisionally inferred, and what remains unshown.

... sì perché l'autorità dell'opinione di mille nelle scienze non val per una scintilla di ragione di un solo, sì perché le presenti osservazioni spogliano d'autorità i decreti de' passati scrittori, i quali se vedute l'avessero, avrebbero diversamente determinato.

— [Galileo Galilei](#)

Epistemic Limits

(*The Refusal to Pretend to Know: Pt. I*)

Abstract

Post-event analyses of the World Trade Center (WTC) collapses frequently characterize the global collapse of the towers as *inevitable* once initiation occurred, while also acknowledging that the collapses were a profound surprise to experienced structural engineers. Part I of *Epicycles of Inevitability* examines the logical and epistemic tension between surprise and inevitability, and between claims of design specificity and claims of generic collapse behavior. The paper demonstrates that confidence in inevitability claims exceeds demonstrated validation. Retrospective certainty has outpaced demonstrated understanding, and that greater epistemic humility and curiosity are warranted in the interpretation of rare, catastrophic structural failures.

1. Introduction

The collapses of the WTC towers represent among the most consequential structural failures in modern history. Considerable analytical effort has been devoted to reconstructing the initiating mechanisms and explaining the subsequent global collapse. These efforts have yielded plausible and internally consistent narratives grounded in accepted principles of structural mechanics, fire effects, and gravity-driven progression.

However, beyond plausibility, authoritative reports go further, asserting that collapse was *inevitable* once initiation occurred, and that detailed modeling of collapse progression was therefore unnecessary. At the same time, the collapses are widely described—by investigators and researchers alike—as unprecedented and surprising.

These positions cannot be held simultaneously without contradiction. The level of confidence expressed is not justified by the degree of validation achieved.

2. Surprise as Epistemic Evidence

In engineering failure analysis, *surprise* is not merely a psychological reaction; it is an epistemic signal. When experienced practitioners are collectively surprised by an outcome in their own domain, this signals either:

1. Operation in a regime outside normal design and analytical assumptions, or
2. Incompleteness in prevailing models or conceptual frameworks.

In the case of the WTC collapses, surprise was widespread. Statements by structural engineers, emergency managers, and investigators indicate that global progressive collapse was not anticipated in real time, even after aircraft impact and prolonged fire exposure. [BAZANT-466](#)

If collapse progression was truly obvious and inevitable under the observed conditions, then this collective surprise demands explanation. The least informative explanation—that experts simply failed to recognize the obvious—offers little insight into structural behavior and undermines confidence in professional judgment. The surprise indicates that the governing failure mechanisms were not transparent within the analytical frameworks available at the time.

Treating surprise as irrelevant discards information about limits of prior understanding.

3. Retrospective Inevitability and Predictive Power

Claims of inevitability in engineering carry specific implications. An outcome described as inevitable is generally understood to follow deterministically from initial conditions across realistic variations in parameters, geometry, and material properties. Such claims are strongest when supported by models that demonstrate robustness, sensitivity bounds, and predictive capability.

In the WTC case, post-event analyses demonstrate that gravity-driven collapse is *compatible* with energy considerations and plausible failure sequences. However, compatibility is not equivalent to inevitability. Models calibrated to observed outcomes do not establish that alternative outcomes were impossible under realistic parameter variance.

The absence of detailed progression modeling and systematic arrest analysis limits the strength of inevitability claims. Small variations produce qualitatively different outcomes in nonlinear, threshold-sensitive systems. Without ensemble analysis across such variations, inevitability remains an assertion rather than a demonstrated property.

4. Design Specificity and Generality

Explanations of the WTC collapses simultaneously invoke the towers' "unique" design features to account for the rarity of the event *and* assert that progressive collapse, once initiated, would occur in essentially any tall building due to basic physics.

These two positions imply different regions of applicability. If design specifics are critical, then the collapse occupies a narrow region of design and damage space, and generalization to other structures is limited. If, instead, collapse progression is largely design-independent, then similar failures would be expected more frequently, which empirical experience does not support.

Initiation and progression cannot be cleanly separated in this way. Structural configuration, load paths, redundancy, and reserve capacity influence both. Appeals to uniqueness when explaining rarity, and to generality when asserting inevitability, are rhetorical conveniences, not resolved understanding.

5. Modeling Scope and Epistemic Closure

A central methodological decision in the official investigation was to terminate detailed analysis at the point of global collapse initiation. The justification offered was that progression was inevitable and therefore did not require further study [NCSTAR-1, BAZANT-466](#).

From an epistemic standpoint, this reasoning is circular. Inevitability of progression is precisely the claim that would require demonstration through progression modeling, resistance quantification, and arrest analysis. Declaring inevitability in order to justify not modeling progression reverses the normal logic of engineering inquiry.

When modeling stops short of the phenomenon under discussion, claims about that phenomenon lack evidential support.

5.5 The Evidential Burden of Inevitability Claims

NIST terminated detailed analysis at collapse initiation, stating in footnotes¹ that progression modeling was unnecessary because “collapse became inevitable.” [NCSTAR-1, pp. xxxvii, 82](#)

NIST did not model:

- Story-by-story resistance through the lower structure
- Arrest conditions or probabilities
- Sensitivity to structural or thermal variance
- Partial collapse outcomes

Analytical treatment of progression is found in NCSTAR 1-6, p. 323, where NIST reviews Bažant & Zhou (2002) and states: “NIST agrees with the assessment.”

5.5.1 The Duhem-Quine Insulation of Core Hypotheses

The reliance on a singular, idealized progression framework to declare global collapse a certainty illustrates a classic vulnerability identified by the **Duhem-Quine thesis** in the philosophy of science. The thesis posits that an isolated hypothesis cannot be tested in a vacuum because it is always insulated by a network of “auxiliary assumptions.” When an empirical model faces potential refutation or extreme boundary sensitivity, the core hypothesis can be artificially protected by adjusting these auxiliary variables.

In this context, the core hypothesis—that *gravity-driven progressive collapse was mathematically inevitable*—is systematically insulated within Bažant & Zhou (2002).

The paper states its aim explicitly: [BAZANT-405](#)

“We are not attempting to model the details of the real failure mechanism but seek only to prove that the towers must have collapsed and do so in the way seen.”

1 The present analysis focuses on the evidential basis available at the time NIST reached its conclusions. Later analytical refinements by the same authors (e.g., Bažant & Le, 2008-2017) extend the original framework but do not retroactively establish ensemble-level validation or address the absence of progression modeling in the original investigation. If anything, the iterative refinement of the model to address criticisms confirms that the original framework was incomplete.

Bažant & Zhou state this as a methodological choice, not a limitation. However, declaring one's aim as 'proof of inevitability' rather than 'evaluation of whether inevitable conditions were met' reveals the epistemic direction: conclusion-first, analysis-second.

Method:

- Energy-balance model of repeated impacts causing multi-floor buckling
- Assumes rigid upper block ("important hypothesis" to maximize resistance)
- Uses "optimistic assumptions" for stiffness and force distribution
- Calculates $W_g/W_p \approx 8.4$ initially, dropping to $\sim 1\%$ dissipation during descent
- Concludes arrest impossible, collapse "almost free fall"

Under Duhem-Quine analysis, what this demonstrates is not an inherent property of physical reality, but rather the internal consistency of an insulated system:

- Under stated assumptions, energy favors progression
- Energy imbalance worsens as mass accumulates

What this does not demonstrate:

- Robustness across parameter variance
- Sensitivity to assumptions (rigid block, stiffness, dissipation modes)
- Arrest probability under heterogeneous conditions
- Effect of geometric or material scatter

The methodology is sound. The insulation is not. By refusing to test sensitivity across the parameter space that the auxiliary assumptions occupy, the analysis protects its core claim through structural safeguarding, not empirical robustness.

5.5.2 Epistemic Closure by Administrative Decision

By adopting this insulated algebraic framework as an administrative endpoint, the regulatory investigation transitions from an exercise in forensic discovery to a structural production of missing data: a decision, whatever its motivation, that leaves a permanent gap in the record.

When an agency treats a dynamic phase as "inevitable" in order to justify not modeling it, the result — regardless of intent — is a gap in the record: parameter uncertainty that never gets tested against evidence, because the test was declared unnecessary before it was run.

This agnotological boundary is explicitly reinforced in later work. Bažant & Verdure (2007) characterize progression modeling as "superfluous" for proving inevitability, stating it is useful for "learning from demolitions" and "dispelling myths." [BAZANT-466](#)

Rather than executing open-ended validation, the model is calibrated post-hoc to observed collapse time and characteristics (*Dialectic Inventory*, Chapter 6). This treats the known historical outcome as an invariant baseline, retroactively adjusting auxiliary assumptions until the algebraic model mirrors the visual record.

The result is that what should have been an empirical question—does progression arrest under realistic variance?—was converted into an administrative answer: no further modeling needed.

5.5.3 Evidential Status

- **NIST:** Progression not modeled because inevitable

- **Bažant/Zhou:** Inevitability from energy balance under single parameter set
- **Neither:** Ensemble analysis, variance testing, or arrest statistics

Inevitability claims rest on:

- One energy-balance model
- Assumptions stated to maximize resistance
- No demonstration of robustness across uncertainty
- Calibration to observation (not independent validation)

An outcome can be energetically favored under specific assumptions without being inevitable across realistic variability. The latter requires ensemble demonstration, which has not been provided.

6. Retrospective Obviousness and Historical Precedent

Historical precedents in structural engineering illustrate a different response to surprise. The collapse of the Tacoma Narrows Bridge, for example, revealed a poorly understood aeroelastic instability. Surprise prompted extensive experimental, theoretical, and code-level developments, and the event is now recognized as exposing a genuine gap in prior understanding.

In contrast, the WTC collapses are often described as retrospectively obvious, despite comparable expressions of surprise at the time. This asymmetry is significant. In one case, surprise motivated deeper inquiry and framework revision; in the other, surprise is treated as a transient misperception, corrected by post-hoc reasoning.

Such retrospective obviousness conflates explanation with understanding. **Compatibility with known facts is not evidence of predictability.**

7. Intellectual Honesty and Standards of Confidence

Scientific and engineering integrity depend not only on producing plausible explanations, but on calibrating confidence to evidence. Overstated certainty can be as misleading as unwarranted doubt.

A more epistemically cautious position requires distinguishing between:

- **What is demonstrated:** plausibility, internal consistency, and compatibility with known physics
- **What remains uncertain:** robustness across variance, arrest probabilities, and generalizability to other structures

Acknowledging uncertainty does not weaken the explanatory framework; rather, it strengthens it by aligning claims with demonstrated scope. [FEYNMAN-1974](#)

8. Implications for Engineering Practice

The way the WTC collapses are framed has implications beyond historical understanding. Claims of inevitability influence:

- risk assessment and emergency decision-making
- design philosophy regarding robustness and disproportionate collapse

- research priorities in fire-structure interaction and progressive collapse

If collapse progression is truly generic and inevitable, then prevention must focus almost entirely on avoiding initiation. If, instead, progression depends sensitively on design and conditions, then understanding arrest mechanisms becomes equally important.

At present, the strength of inevitability claims exceeds the evidentiary basis required to resolve this distinction.

9. Conclusion

The WTC collapses revealed structural behavior that surprised experienced engineers. That surprise is evidence of epistemic limits at the time of the event. Post-event analyses have produced plausible explanations consistent with accepted mechanics, but plausibility should not be conflated with inevitability.

Intellectual honesty in engineering requires that confidence be proportional to validation, and that surprise be treated as a prompt for deeper inquiry rather than something to be explained away. A posture of epistemic humility—recognizing what is understood, what is inferred, and what remains uncertain—better serves both science and public trust than retrospective claims of obviousness.

The enduring lesson of the WTC collapses is not that the outcome was inevitable, but that rare failures expose the boundaries of understanding. Respecting those boundaries is a prerequisite for extending them.

Summary Table: Epistemic Tensions

Paradox	Position A	Position B	What It Reveals
Modeling Scope	“Most detailed investigation ever” (NIST/EPN)	“Includes little analysis... after the conditions for collapse initiation were reached” (NCSTAR 1, pp. xxxvii, 82)	Delimited Scope applied to support working hypothesis
Core Description	47 box columns (NIST)	“Hollow shaft” (9/11 Commission Report)	Fundamental structural misunderstanding
Symmetry	Asymmetric damage	Symmetric collapse	Centering mechanism not identified
Kinematic Claims	“Almost free fall” (Bažant 2002); “essentially free fall” (NIST 2005); “only 17% longer” (Bažant 2007)	“65.5% longer than free fall” (Bažant 2008)	Kinematic characterization is not stable within official literature; these claims cannot all be simultaneously true
Oscillation Period	“Critically damaged”	“Nearly undamaged frequency”	Sudden transition from robust to failed unexplained
Temperature Evidence	“Fires weakened steel”	No examined sample exceeded 600 °C (NCSTAR 1, p. 90)	Physical evidence doesn’t support thermal narrative
Jet Fuel	“Jet fuel critical”	“Jet fuel fires themselves lasted at most a few minutes” (NCSTAR 1, p. 182)	Early explanations fundamentally wrong
Steel Preservation	“Largest mass murder”	229 of 236 retained samples $\approx$ 0.25–0.5% of 200,000 t (NCSTAR 1, p. 87)	Investigation scope didn’t match event magnitude
Residue Testing	“Found no corroborating evidence for alternative hypotheses” (NCSTAR 1)	“Did not test for the residue of these compounds” (NIST 29)	Absence of testing treated as negative result

Actioni contrariam semper & aequalem esse reactionem: sive corporum duorum actiones in se mutuo semper esse aequales & in partes contrarias dirigi.

— [Isaac Newton](#)

Dialectic Inventory

(The Refusal to Pretend to Know: Pt. II)

Preamble

This document audits the mechanical and methodological foundations of prevailing explanations for the progressive collapse of tall steel-framed structures, evaluating a single claim: **that the observed outcomes followed from known mechanisms in a manner that is physically, methodologically, and epistemically well-established**. The aim is not to dispute plausibility, but to examine whether the evidentiary standards required for claims of inevitability and robustness have been met.

History offers reminders that scientific closure can occur before evidentiary standards are fully satisfied. Avoiding premature closure is a methodological challenge rather than a question of consensus or credential. The standard adopted here is straightforward:

Claims about inevitability must demonstrate inevitability. Claims about robustness must demonstrate robustness. Claims about prediction must demonstrate predictive power.

These are not rhetorical demands but ordinary scientific criteria in near-threshold dynamical systems, where permissibility does not determine regime occupancy.

This assessment evaluates existing explanations against their stated standards and scope. The absence of fully developed alternative frameworks lies outside the present analysis.

Scope of Analysis

This document addresses the mechanics of downward progressive collapse through largely intact structural steel once vertical motion is underway.

It does **not** address fire initiation scenarios, detailed temperature reconstructions, or debates over ignition sources. Those questions are orthogonal to the mechanical sufficiency of the propagation phase.

The central question is strictly mechanical:

Given the observed progression, were the structural resistances and energy transfers quantitatively sufficient to produce the descent characteristics through predominantly cold steel?

This is where the diagnostic difficulty arises, and where the analysis is focused.



Figure 1. North Tower collapses. Credit: Semendinger.

Chapter 1 — Framing the Problem

1.1 The Claim Under Examination

The core claim: Given aircraft impact damage and fires, progressive gravitational collapse was **inevitable**, or nearly so, once downward movement initiated. Accompanying assertions: The mechanism is **well understood** and **fully explained** within conventional structural mechanics.

This document evaluates whether those assertions meet ordinary standards for complex failure analysis, focusing on:

- Mathematical structure of models
- Transparency of inputs and parameters
- Calibration vs. validation distinction
- Variance and uncertainty treatment
- Ensemble analysis presence
- Asserted inevitability vs. demonstrated robustness
- Consistency of energy, momentum, and resistance

The evaluation concerns **method**, not belief.

1.2 What Would Constitute Adequate Support

Adequate support for inevitability or robustness requires:

- full publication of governing equations and boundary conditions
- public availability of model inputs or executable simulation code
- sensitivity analyses showing parameter dependence
- ensemble modeling across realistic material and geometric variance
- explicit reporting of arrest frequencies, partial failures, and runaways
- clear separation between assumptions, calibration, and inference
- quantification of uncertainty, not rhetorical confidence

These standards mirror normal practice in fracture mechanics and reliability engineering, and structural risk assessment.

1.3 What We Actually Observe in the Literature

Prominent analytical treatments exhibit recurring features: [BAZANT-405](#), [BAZANT-466](#), [BAZANT-476](#), [BAZANT-499](#), [BAZANT-585](#), [BAZANT-625](#), [BAZANT-EPN](#)

- **upper blocks treated as rigid bodies**
- **force–displacement functions assumed or idealized rather than measured**
- **known unknowns replaced by constant surrogates**
- **energy dissipation localized to advancing crush fronts**
- **calibration to observed duration and mass ejection to determine parameters**
- **progression treated as one-dimensional and centrally aligned**
- **sensitivity studies largely absent or fragmentary**
- **key resistive terms introduced or omitted between papers without systematic reconciliation**

These define their epistemic status:

plausibility constructions conditioned on assumptions, not yet fully validated dynamical accounts.

When such models are used to claim **inevitability**, that is where the methodological tension arises.

1.4 The Central Question

The central technical question is this:

Does progressive floor-by-floor collapse occur across a broad region of realistic parameter space, or only within narrow, tuned conditions?

This distinguishes:

- **robust outcome** vs. **sensitive trajectory**
- **“could”** vs. **“must”**
- **explanation** vs. **curve-match**

The answer depends on:

- energy balance
- momentum transfer
- post-yield connection behavior
- heterogeneity
- stochastic variance
- dynamic scaling with height and mass
- ensemble statistics

At present, assertions of robustness exceed what has been empirically demonstrated.

1.5 Scope: Claims Evaluated vs. Claims Outside Analysis

This document argues	This document does not argue
That claims of inevitability require demonstration of robustness under variance	That controlled demolition or any specific alternative mechanism occurred
That ensemble modeling across realistic parameter distribution is necessary to support strong inevitability claims	That collapse was impossible under gravity and damage alone
That existing models rely on calibration to observed outcomes rather than independent validation	That investigators acted with intent, malice, or conspiracy
That observed progression characteristics require mechanistic explanation beyond general energy sufficiency	That specific alternative values or mechanisms can be determined from available evidence
That epistemic closure preceded methodological demonstration	That absence of ensemble validation proves official explanations false
That current confidence levels exceed demonstrated robustness	That current models are necessarily incorrect

1.6 The Null Hypothesis and Burden of Proof

In scientific reasoning, the null position is not “collapse happened, therefore the explanation is valid,” but rather:

Progressive collapse is not inevitable under realistic variance—robustness must be demonstrated, not presumed.

This null hypothesis does not deny possibility. It restricts claims of necessity to standards commensurate with their strength.

1.7 Scope

This work evaluates the evidentiary support for claims of inevitability within a gravity-driven collapse framework. It does not evaluate, weight, or endorse alternative explanations.

Chapter 2 — The Replication Difficulty

2.1 Why Replication Matters

Replication is diagnostic because robust outcomes recur under variation. A mechanism claimed to be generic and dynamically inevitable should not require delicate tuning to reproduce or depend on narrow corridors of coincidence.

Replication here means:

- explicit assumptions
- published equations
- publicly testable implementations
- ensemble runs rather than hand-selected cases
- reporting of failure and arrest rates

The following sections assess the literature against that standard.

2.2 What Should Be Demonstrable

If progressive gravitational collapse under damage and fire is a **broad, stable attractor**, models should reveal:

- reproducible downward progression without requiring fine adjustment
- arrest events occurring at quantifiable rates under realistic variance
- partial collapses appearing as common outcomes
- asymmetries developing naturally in heterogeneous structures
- collapse duration varying in structured ways tied to inputs

In other words: a **population of behaviors**, not a **single narrative trajectory**.

This does not require perfect realism. It requires honesty about what the mechanism does under perturbation.

2.3 What We Actually See

Instead, the record shows:

- calibration to the historical event
- selection of parameters to fit observations
- idealized post-yield behavior
- homogenization of heterogeneity

- systematic reduction of lateral DOFs
- sparse sensitivity analyses
- absence of large-scale ensemble studies
- visually persuasive but method-opaque animations

This does not falsify prevailing explanations. It shows that claims of inevitability remain unsubstantiated.

2.4 The Limits of Post Hoc Calibration

Fitting a single trajectory after parameter selection establishes compatibility, not necessity.

A model may match observation while being:

- sensitive to initial conditions
- dependent on tuned thresholds
- critically reliant on assumptions
- fragile under variance

Calibration is retrodiction. Validation requires **independent predictive survival**.

2.5 Heterogeneity and the Space of Competing Outcomes

Tall steel structures are highly redundant, heavily overdesigned against gravity, and structurally heterogeneous. [SALVADORI-1979](#), [SALVADORI-1982](#) Fire does not homogenize a structure; it increases irregularity. Damage does not align failure fronts; it amplifies asymmetry.

Systems with:

- distributed load paths
- staggered weakening
- imperfect symmetry
- variable local capacity

have documented, real-world instances of partial failure, arrest, and irregular progression under comparable damage and fire exposure. [WP-WINDSOR](#), [WP-PLASCO](#), [WP-ONE-MERIDIAN-PLAZA](#), [WP-RONAN-POINT](#) These features enlarge the space of mechanically available outcomes, so a claim of robust, symmetric, non-arresting progression requires demonstrating why those competing modes are dynamically suppressed.

If such a dominant attractor exists, it must be demonstrated — not assumed.

2.6 The Razor-Edge Regime

Many treatments implicitly assume what must be shown:

- that global collapse follows once downward motion begins
- that resistance rapidly falls below required stopping force
- that momentum accumulation overwhelms heterogeneity
- that failure propagates faster than compensatory redistribution

But when these claims are tested analytically, they often resolve to **threshold conditions**: collapse proceeds if resistance falls below a tuned function of mass, height, and velocity; if connections behave in specific

ways; if post-yield behavior conforms to assumed idealizations; if internal redundancy does not meaningfully absorb impulse.

Threshold dynamics are acceptable when presented as conditional. Problems arise when they are presented as robust inevitability without ensemble justification.

2.7 Absence of Ensemble Is Not Neutral

If inevitability is claimed, robustness must be demonstrated. If demonstration is expensive, certainty is expensive.

Absence of ensemble modeling does not support closure. It indicates unresolved status.

2.8 Epistemic Consequences of Replication Difficulty

Replication difficulty does not imply impossibility or intent. It means:

the mechanism has not yet been demonstrated to be generically robust.

Chapter 3 — Constraints, Fine-Tuning, and Sequential Failure

3.1 Sequential Collapse as a Constrained, Near-Threshold Process

Progressive, floor-by-floor gravitational collapse is not merely “downward motion.” It is a **coordinated sequence of mechanical events** that must satisfy multiple constraints simultaneously over many stories:

1. the descending mass must retain coherence long enough to transmit impulse
2. each successive floor must fail dynamically on first engagement
3. resistance must not increase faster than available momentum
4. lateral instability must not divert or arrest progression
5. redistribution mechanisms must not stabilize the structure
6. asymmetries must not convert descent into tipping or partial collapse

Each of these conditions is individually non-trivial. Their **simultaneous satisfaction across dozens of stories** is what elevates the problem.

This requirement reveals a structural tension built into high-rise design. Each story must satisfy two contradictory demands:

- **high static capacity and redundancy** under service conditions, and
- **low effective dynamic resistance** under impact from above.

Safety factors, ductility, and redundancy are introduced precisely to **prevent** dynamic failure cascades. Columns are not designed as fuses. For sequential crushing to proceed without arrest, the structure must therefore reside near a **critical threshold** in parameter space: static reserve capacity must remain large enough to support service loads, while dynamic resistance must collapse rapidly enough to be overwhelmed repeatedly by the descending mass.

This combination places the system near a **razor-edge regime**. Small changes in:

- post-yield deformation capacity,
- connection behavior,
- load redistribution efficiency,
- mass accretion efficiency, or
- lateral degrees of freedom
- dissipative capacity relative to comminution demand, or the rate of resistance decay relative to gravitational acceleration

can qualitatively change the outcome from sustained progression to partial failure or arrest ².

The razor-edge cuts in both directions: if stabilizing tendencies dominate, the collapse arrests or becomes partial; if destabilizing tendencies dominate too strongly, the descending mass loses coherence and breaks up, which equally prevents the clean, sequential crushing the mechanism requires.

This is the diagnostic signature of **fine-tuning**: outcomes that depend sensitively on coordinated parameter values rather than emerging robustly across broad variation. Fine-tuning does not imply intent or artifice. It implies **fragile dynamics** residing near critical boundaries in outcome space.

Claims of inevitability are claims that such fine-tuning is absent—that the collapse trajectory occupies a **broad basin of attraction**. Demonstrating that claim requires showing persistence of the same qualitative outcome across realistic parameter variance. Absent such demonstration, the appropriate epistemic posture is that sequential collapse is **plausible under specific conditions**, not generically inevitable.

3.2 Default Structural Responses to Local Failure

Real structures subjected to severe damage and uneven heating do not normally self-organize into clean, symmetric, floor-by-floor failure modes. They:

- redistribute loads
- develop plastic hinges
- shed nonstructural components
- stabilize in new geometries
- fail partially or asymmetrically
- arrest after partial progression

These behaviors are consequences of redundancy and the prerequisites for the structure having stood at all. A structure that could not redistribute, hinge, shed, and settle into new geometries would never have survived its own erection, decades of service loading, thermal cycling, and progressive loss of section to renovation or corrosion. Redundancy is not a margin appended to the structure's ability to stand; it is the mechanism of that ability. This has an evidential consequence: the observable operation of these same response modes over a building's service life is direct evidence of the processes any account of total failure must nullify.

Redundancy thus enlarges outcome space in both directions: it is what the structure draws on to stand, and what a claimed mechanism must overcome to fall.

A mechanism that claims to produce:

- rapid progression

2 The technical structure of this near-threshold regime—and the parameter couplings that control it—is formalized in Chapter 4, where the severity parameter Γ and associated dissipation, localization, and accretion assumptions make the fine-tuning explicit rather than implicit.

- symmetry
- non-arrest
- high completeness

must therefore identify the **specific dynamical process** that cancels, totally and on a timescale of seconds, the same response modes whose operation kept the structure standing through its entire service life.

3.3 Vertical Reserve Capacity and the “Overbuilt” Reality

High-rise design is governed primarily by:

- lateral drift limits
- global stability under $P-\Delta$ effects
- wind and seismic excitation
- buckling of slender columns

Meeting these constraints forces designers to provide substantial **axial reserve capacity** as a side effect. Columns typically operate far below Euler buckling load; axial failure is seldom the governing limit state.

This reserve is not merely theoretical. Analyses of fire exposure, load redistribution, and thermal weakening often show that even under severe impact and multi-floor fires, significant structural integrity persists for extended durations—consistent with the towers’ actual survival time. [BAILEY](#) The transition from demonstrated robustness to asserted inevitability occurs without demonstration of how redundancy, heterogeneity, and dissipation are dynamically neutralized across many floors. [BAZANT-499](#) The result is an explanatory discontinuity: robustness is documented up to initiation, but inevitability is declared afterward, without demonstrating a *generic pathway* from one to the other. A complete account must bridge this gap by showing that the collapse attractor is broad enough to dominate despite the very mechanisms that enabled prolonged stability.

This means:

- gravity alone is normally insufficient to push tall buildings into runaway axial failure
- substantial additional mechanisms must act to suppress redistribution pathways and redundancy

Total destruction, if gravity-driven, required a **critical regime** in which ordinary reserves were neutralized across many floors. If this regime is claimed to be “inevitable,” its basin of attraction must be shown to be large. That has not yet been demonstrated.

3.5 The Problem of Reference Cases

When mechanisms are generic, reference cases abound:

- progressive collapse in concrete-frame buildings
- partial collapses under fire damage
- asymmetric failures due to eccentric loading
- local buckling without full-height destruction

What is rare — and therefore requires explanation — is:

- full-height global collapse
- rapid near-axial descent
- sequential floor-by-floor failure without arrest

- high comminution across most of the structure
- recurrence of the joint outcome twice in the same complex

Rarity does not invalidate an explanation. It **increases the burden of mechanistic demonstration**.

3.6 Sequential Failure Is Not the Default Attractor

The available analytical record does not map the basin established in §3.1. Instead, it posits the sequential-collapse attractor and then treats its existence as explanatory. The correct order is the reverse: first demonstrate the basin, then claim inevitability if the basin is large, and refrain from such claims if the basin is small or uncharacterized.

3.7 Where the Burden Properly Falls

The central question is not:

- whether collapse could happen
- whether collapse did happen
- whether collapse can be modeled

The question is:

Has it been shown that the specific observed form of collapse is a robust consequence of the known initial conditions, across realistic uncertainties, without fine-tuning?

If yes, inevitability is warranted. If no, it is not.

The remainder of the *Inventory* examines the models that purport to answer “yes.”

Chapter 4 — Naive Bounds, Critical Regimes, and the Limits of Simple Models

Naive bounds do not deny complexity; they enforce it. A steel frame, however intricate, conserves energy and momentum. If observed behavior strains these bounds, the explanation must specify—without rhetorical shortcuts—how heterogeneity channels dynamics into a narrow outcome, not assume it does.

4.1 Relevance of Naive Models

Naive models do not explain complex structural failure. They do something simpler and more important: **they set outer limits on what is even remotely plausible.**

They answer questions of the form:

- How far can a falling block penetrate a comparable structure?
- How quickly can gravity-driven collapse proceed in the absence of added assumptions?
- How much resistance is required to produce arrest?
- And: what kinematic ceiling does the official collision mechanism impose on observable acceleration?

If observed behavior lies **outside** these naïve bounds, the explanation must invoke additional mechanisms. If observed behavior lies **inside**, the bounds do not constrain.

Either way, naive models are diagnostic. They define the **energy–momentum envelope** within which more elaborate accounts must live.

4.2 Impact Depth as an Upper-Bound Estimate

Treat the falling top portion as a single projectile of length L , density ρ_p , impacting a target of density ρ_t .

By simple momentum/energy arguments, the maximum depth of penetration into a comparably dense medium is on the order of the projectile length:

$$s \sim L \cdot (\rho_p / \rho_t)$$

When $\rho_p \approx \rho_t$, the estimate is $s \sim L$.

Interpretation:

- a block of x floors cannot, in naive continuum mechanics, crush many multiples of x floors below it before its momentum is exhausted
- increasing resistance with depth **reduces** penetration below this already generous bound

This is not a full structural model. It is a **ceiling** on what gravity-driven crushing can accomplish without invoking critical-state dynamics, coordinated fracturing, mass shedding, or externally supplied energy.

4.3 Inelastic Collision and Arrest Tendencies

Now treat the interaction not as a continuum penetration problem but as a discrete inelastic collision between:

- a falling mass m_1 with velocity v
- a stationary mass m_2 attached to the Earth

The post-impact velocity is:

$$v' = [m_1 / (m_1 + m_2)] \cdot v$$

As m_2 grows with each engaged floor, v' rapidly decreases. Momentum is transferred downward; acceleration decays; arrest becomes increasingly likely unless resistance is vanishingly small.

Sequential progression without arrest therefore requires simultaneously:

- persistent maintenance of substantial downward momentum
- continuing suppression of rising resistance
- preservation of fall-block coherence despite violent interaction

These are **constraints**. They are met only if the system occupies a narrow dynamical regime.

4.3.1 The Coefficient of Restitution Paradox

The analysis above assumes momentum is conserved during inelastic collision. But observed behavior presents a paradox when examined more carefully.

For velocity to **increase** between impacts (acceleration during descent):

$$v' > v$$

This requires:

$$m_1/(m_1 + m_2) > 1$$

Which is **impossible** for any $m_2 > 0$.

For velocity to **remain constant** (free-fall descent, as approximately observed):

$$v' = v$$

This requires:

$$m_1/(m_1 + m_2) = 1$$

Which implies $m_2 = 0$ (zero effective resistance).

Yet observed behavior shows **near-constant descent acceleration** over multiple stories. This is mechanically possible only if:

1. The **coefficient of restitution** — defined for this collision as $\epsilon \equiv (v_2' - v_1')/(v_1 - v_2)$, the ratio of post-impact separation speed to pre-impact approach speed — is $\epsilon \approx 1$ (nearly elastic collision), **or**
2. **Dynamic resistance is vanishingly small** ($m_2 \approx 0$ effectively), **or**
3. **Energy is added** to the system between impacts

All three contradict known structural behavior:

- Steel exhibits **highly inelastic** collision behavior ($\epsilon \ll 1$)
- Documented column strengths satisfy $P_{static} \gg mg$
- Energy dissipation occurs through:
 - Plastic buckling
 - Connection fracture
 - Material comminution
 - Air displacement

Resolution requires demonstration (not assertion) that:

- Sequential failure produces **effective ϵ near unity** through some unidentified mechanism, **or**
- Dynamic loading reduces resistance by **orders of magnitude** below static capacity, **or**
- The collision model is fundamentally inappropriate for this regime

None of these have been demonstrated.

Bažant's own video analysis reports the cumulative departure from free fall directly: the South Tower descended, after 2 s, "10.5% slower than the free-fall velocity," and the North Tower, after 3 s, "17.4% slower" (Bažant et al. 2008, Fig. 7). How that cumulative deceleration is distributed across individual floor-impact events is disputed. One paper argues the per-impact velocity drop implied by the stated resistance would be too small to detect on amateur video, "far too small to be perceptible... and much smaller than the inevitable error of such video records." [BAZANT-499](#) An independent frame-by-frame measurement of the North Tower roof, tracked through its first nine stories at 33 ms resolution, reports the opposite: using the same stated resistance assumptions, the calculated minimum jolt should have been resolvable, and no discrete deceleration event of any size appears in the measured curve. [MACQUEEN](#) "Jolt" is not the measurement's own term but Bažant's: the addendum to the same 2002 paper, addressing a hypothetical higher-floor impact with

a shorter, more slender upper section, contrasts the scenario actually modeled — “one powerful jolt” — against the alternative, “a series of many small vertical impacts, none of them fatal.” [BAZANT-405](#) That contrast is tied explicitly, in the same passage, to the falling mass’s rigidity — the same assumption examined at §6.3. The dispute over whether the required signature is present is a dispute over whether the actual falling mass produced the fatal case or the non-fatal one.

The paradox is not proof of impossibility. It is **diagnostic evidence** that the mechanism occupies a **narrow, non-obvious region** of parameter space requiring explicit demonstration rather than assumed inevitability.

This mechanical tension is explored quantitatively in the ensemble study below (§4.4).

A separate passage in the same addendum addresses a different question and should not be read as bearing on this one. Responding to whether plastic deformation “cushioned” the initial descent, it models the first story’s own columns buckling beneath the upper section — not an impact against a second, still-intact story — and concludes the resulting descent is “only about 6% longer than the duration of a free fall from that height, which is 0.87 s.” This is a statement about the softness of one story’s own buckling process, not about momentum conservation at the discrete impact the coefficient-of-restitution paradox concerns; it neither addresses, nor was offered to address, whether a velocity discontinuity should appear at the moment the falling mass strikes the floor beneath it.

4.3.2 The Acceleration Ceiling Under Pure Inelastic Accretion

The question

What is the maximum average acceleration achievable under the mechanism NIST specifies: a falling mass strikes a stationary floor, the struck mass joins the falling block, the combined mass falls to the next floor, and the sequence repeats to the ground? No structural resistance is assumed. Columns offer no static capacity; connections offer no dynamic resistance; only inertia retards the motion. This is the most favorable possible assumption — a ceiling, not an estimate.

The continuous formulation

Model the structure as a continuous medium of uniform linear density μ (mass per unit height) — the same accretion-rate quantity used in §4.9.4’s $m(z) = m_0 + \mu z$. A falling block has descended distance y , accumulated mass $m = \mu y$, and travels at velocity v . In the absence of any resistance other than inertia, momentum conservation gives:

$$d(mv)/dt = mg$$

Expanding:

$$m(dv/dt) + v(dm/dt) = mg$$

Since $dm/dt = \mu v$ and $m = \mu y$:

$$\mu y \cdot a + \mu v^2 = \mu y g$$

$$a = g - v^2/\mu$$

This is the equation of motion for continuous inelastic accretion under gravity with no structural resistance. [CHEREpanov](#)

The asymptotic limit

At large y , assume constant acceleration: $v^2 = 2ay$, so $v^2/y = 2a$. Substituting:

$$a = g - 2a$$

$$3a = g \rightarrow a_\infty = g/3 \approx 3.27 \text{ m/s}^2$$

This is confirmed by assuming $v \approx at$ for large t , giving $y \approx \frac{1}{2}at^2$, so $v^2/y = 2a$ — the same result by a different route. That the asymptote is genuinely approached and stable follows from defining $f \equiv v^2/y$: differentiating along the trajectory gives $df/dt = f(2g - 3f)/v$, whose non-zero fixed point is $f = 2g/3$, stable from both sides ($df/dt > 0$ below it, < 0 above). Substituting, $a = g - f = g/3$. The asymptote is approached from above: early acceleration can exceed $g/3$ before the momentum sink grows large enough to assert itself. The limit characterizes sustained descent through many floors, which is the relevant regime for a full-height collapse.

The discrete formulation

The same physics can be expressed as a sequence of discrete perfectly plastic collisions. An upper block of mass M strikes a stationary floor of mass m ; by conservation of momentum:

$$v' = Mv / (M + m)$$

Between collisions the combined mass falls one story height h under gravity. Tracking the energy balance across this repeated cycle for a uniform structure — each floor adding equal mass m to the falling block — and setting $\beta = 1$ (all mass retained within the footprint — the full-retention upper-limit case in Němec's own parameterization, with β the portion of mass hitting the structure below per collision and $(1 - \beta)$ falling aside; distinct from both Bažant's column-force-reduction β and this document's R , §4.10.3.³):

$$a_{\text{max}} = \beta g / (2 + \beta) = g/3$$

This is the Němec result. [NEMEC](#)

Two mathematically distinct formalisms — a continuous differential equation and a discrete collision recurrence — converge on the same ceiling. The convergence is not coincidence: both are formulations of the same physical principle (momentum conservation under pure inelastic accretion) in different limiting regimes. The result is structural, not parametric.

A convergence beyond the WTC literature.*

The $g/3$ ceiling is not a result derived only for this application; it is the resistance-free asymptote of a family of accretion problems in classical mechanics. A structurally identical setup — a chain heaped at the edge of a table, drawn through a hole and pulling more chain into motion from the stationary pile as it falls (Cayley's problem, 1857) — gives the identical result by the identical logic: "A uniform, flexible and frictionless chain falling link by link from a heap by the edge of a table falls with an acceleration $g/3$ if the motion is nonconservative, but $g/2$ if the motion is conservative." [WONG-CAYLEY](#) This has been confirmed experimentally, in an apparatus with no relation to structural collapse: a three-dimensional mechanical analog ("xyz-slider")

3 $\beta = 1$ is the full-retention upper-limit case in Němec's notation ("the portion of the mass βm hitting the motionless mass of the building beneath"). The formula is confirmed at both reported parameter values: Němec reports $\bar{a} = g/3$ at $\beta = 1$ and $\bar{a} = g/5$ at $\beta = 0.5$, both exactly $a_{\text{max}} = \beta g / (2 + \beta)$, with $v(x) = \sqrt{(2g\beta x / (2 + \beta))}$ from eq. (17), and obtains the same closed form from both energy and momentum conservation under the explicit assumption that accreted mass starts from rest — the same one-law-two-readouts identity demonstrated in Note 3.4

measured, after correcting for friction, $a/g = 0.328 \pm 0.004$ against the theoretical $1/3$. [WONG-CAYLEY](#) The same accretion physics appears in the variable-mass-systems literature under the name “inverse rocket problem” — the mirror case of the standard rocket equation, in which a body accretes rather than ejects mass. [MUNGAN](#) None of these sources engage the WTC question in any way; the convergence is with the underlying conservation law, not with any assumption made in this document.

↳ The “raindrop” problem most often cited in physics pedagogy uses a different accretion law — mass growing with surface area rather than with distance fallen — and gives a different asymptote, $g/7$. [EDWARDS](#) That case is not equivalent to this one; the relevant convergence here is specifically with the constant-linear-density, distance-proportional accretion family (chain-from-a-heap, inverse rocket), not the mist-accretion raindrop, and the two should not be cited interchangeably.

The $g/3$ limit is not absent from the official literature. Bažant’s own equation of motion reduces, in the zero-resistance continuum limit, to the Cherepanov equation $a = g - v^2/z$, whose asymptote is $g/3$. He notes explicitly that the momentum accretion force F_m “causes a greater difference from free fall than do forces F_b , F_s , F_a , and F_e combined.” [BAZANT-476](#) The physics of the ceiling was known and included; it was then circumvented by calibrated parameter choices rather than confronted as a constraint.

The diagnostic implication

NIST characterizes the descent of the upper building section as occurring “essentially in free fall.” [NCSTAR-1](#) If this characterization is accurate, observed acceleration approached $g \approx 9.81 \text{ m/s}^2$. The ceiling under the official mechanism, in its most favorable formulation — pure inelastic accretion of stationary mass, zero structural resistance — is $g/3 \approx 3.27 \text{ m/s}^2$.

The discrepancy is a factor of three. It is not sensitive to parameter choices — the $g/3$ ceiling holds at any mass, any density, any story height, provided only that the accretion is perfectly plastic and resistance is absent. Mass ejection does not: at zero vertical exit velocity the vertical momentum sink per collision is reduced, the asymptote remains $g/3$, and partial retention cancels identically from the continuum equation ⁴, so shedding provides no kinematic escape from the ceiling.

Closing this gap within the inelastic accretion framework requires that floors not be stationary at the moment of impact — that pre-failure of the lower structure propagated systematically ahead of the descending mass. This is a specific mechanical claim, a departure from the stated model rather than a consequence of it. It requires independent demonstration. It has not received any.

NIST’s own quantitative figures give a related but distinct comparison. NIST reports approximately 11 s (WTC 1) and 9 s (WTC 2) for “the first exterior panels to strike the ground after the collapse initiated,” measured from the initiating floor (98th, WTC 1; 82nd, WTC 2) to grade. [NIST-FAQ](#) Applied to those distances, this gives average accelerations of roughly 5.9 m/s^2 and 7.4 m/s^2 — a factor of roughly 1.8–2.3 above the $g/3$ ceiling, not three. The factor-of-three figure follows specifically from NIST’s prose characterization (“essentially in free fall,” implying $a \approx g$); NIST’s own measured duration figures give a smaller, though still unreconciled, discrepancy. Both versions point the same direction; the narrower one is the one built on

4 With retention fraction β and ejection at zero vertical exit velocity, the impact momentum balance is $Mv = (M + \beta m)v'$, the accretion rate is $\beta\mu v$, and the block mass is $m(z) = m_0 + \beta\mu z$. Substituting into $d(mv)/dt = mg$ gives $\beta\mu y \cdot a + \beta\mu v^2 = \beta\mu yg$, i.e. $a = g - v^2/y$: β cancels identically, so the sustained-descent asymptote is $g/3$ at every retention fraction. (For ejection with nonzero vertical exit velocity u_z , the balance becomes $Mv = (M + \beta m)v' + (1 - \beta)m \cdot u_z$, and the result is direction-dependent: downward-directed ejection tightens the sink, upward- or horizontally-directed ejection loosens it.

NIST’s own numbers rather than its verbal description of them, and is offered here as the more defensible figure.

NIST states elsewhere that these 11 s / 9 s figures answer a narrower question than total collapse duration: “Neither the duration of the seismic records nor video evidence... are reliable indicators of the total time it took for each building to collapse completely.” [NIST-FAQ](#) The official kinematic record for the full descent is accordingly thinner than the widely-cited 11 s / 9 s figures suggest on their own.

A note on Seffen’s result

Seffen’s treatment yields an asymptotic acceleration of $g/2$. [SEFFEN](#) This arises from applying Lagrange’s equation to a variable-mass system without the Pesce correction — a formulation that implicitly treats accreted mass as already in motion before pickup, and therefore omits the kinetic energy that must be dissipated when a stationary floor is accelerated to the velocity of the falling block. Seffen identifies this omission explicitly, calling the additional term “artificial dissipation,” but the dissipation is physically real: a stationary mass set in motion by an inelastic collision always reduces the kinetic energy of the center-of-mass motion, regardless of whether pickup is modeled as impulsive or continuous. Seffen’s own Newtonian formulation, which does conserve momentum at pickup, recovers the $g/3$ asymptote. The $g/2$ result is therefore not an independent physical model but the consequence of a specific and physically unjustified modeling choice. The ceiling relevant to the present analysis is $g/3$.

This is not an idiosyncratic error specific to this application. The identical $g/2$ -versus- $g/3$ divide, traced to the identical cause — whether accreted mass is credited with pre-existing motion at pickup, i.e. energy conservation substituted for momentum conservation — is independently documented in the falling-chain literature, entirely outside any structural-collapse context. [WONG-CAYLEY](#)

What this does not show

This result does not:

- identify an alternative mechanism
- establish that the observed acceleration is impossible under any physical model
- demonstrate deliberate causation
- conflict with Seffen’s treatment, which addresses a different model

It establishes that the official mechanism as stated — inelastic collision between falling mass and stationary floors — cannot produce the acceleration the official account attributes to the collapse. Resolving this requires either specifying a departure from the stated mechanism, challenging the kinematic characterization, or demonstrating that pre-failure propagated generically through the lower structure ahead of the crush front. None of these resolutions has been provided to the standard of ensemble validation or independent experimental confirmation.

4.3.3 What d’Alembert’s Principle Requires of the Claimed Kinematics

The crush-down equations are themselves built on d’Alembert’s principle — the restatement of Newton’s second law that treats the inertial term as a force, allowing the dynamic problem to be written as an instantaneous equilibrium. [BAZANT-625](#) Applied to the falling upper section without any assumption about how resistance is generated — no accretion mechanism, no hinge count, no localization — this reduces to arithmetic. For an upper mass m at instantaneous acceleration a , the resisting force actually transmitted by the structure below is

$$F_{mob} = m(g - a)$$

Before collapse, $a = 0$ and $F_{mob} = mg$ — a small fraction of the system capacity F_c^{sys} (§6.5.1) available at equilibrium. Once descent is underway, this equation runs the opposite direction from how the model itself is normally used: rather than assuming a resistance and solving for the acceleration it produces, it takes the *claimed* acceleration and reads off the resistance that acceleration requires.

NIST’s own characterization of the descent — “essentially in free fall” (§4.3.2) — is a claim about a , not about F_{mob} . Substituted into the equation above, $a \rightarrow g$ gives $F_{mob} \rightarrow 0$. Applied across eighty-plus intact stories, this is the same demand the “Anti-Newtonian Solid” (Exhibit B) was constructed to name explicitly — a structure offering essentially zero resistance the instant dynamic descent begins, having offered F_c^{sys} -order resistance the instant before.

This is not a derivation competing with §4.3.2’s accretion-based ceiling; it is the same conclusion by a shorter route, requiring no assumption about the collision mechanism at all — only the principle the crush-down equations are themselves built on, and the kinematic characterization the official account itself supplies.

What this does not show. It does not establish what F_{mob} actually was; it establishes what the *claimed* kinematics require it to have been, given nothing but Newton’s second law. Resolving the gap requires either revising the kinematic claim or independently measuring F_{mob} across the descent — precisely the inverse analysis one paper in this corpus itself proposes as a check: “the present idealized model allows simple inverse analysis which can yield the crushing energy per story and other properties of the structure from a precisely recorded history of motion.” [BAZANT-466](#) As far as this document has found, that check has not been carried out and published.

4.3.4 What NIST’s Own Record Says About the Core

Condition 2 of §3.1’s constraint list — that each successive floor fail dynamically on first engagement — is not merely a modeling assumption examined here. NIST’s own account states it directly: “significant portions of the cores of both buildings (roughly 60 stories of WTC 1 and 40 stories of WTC 2) are known to have stood 15 to 25 seconds after collapse initiation before they, too, began to collapse.” [NIST-FAQ](#)

This describes a large fraction of each tower’s core surviving well past the point the perimeter crush front, per the same account, offered “minimal resistance” to the mass moving through it. A mechanism in which the front’s own governing structure sheds resistance on first contact does not, on its face, leave the core standing for fifteen to twenty-five seconds behind that front. NIST reports both claims in the same document without reconciling them. This document does not resolve the reconciliation; it notes that the official record itself contains the tension.

4.4 Ensemble Sensitivity Analysis of the Discrete Floor Model

Qualitative arguments about sensitivity and narrow regimes are useful, but quantitative demonstration carries greater weight. To test the robustness of progressive collapse predictions, the discrete floor energy-balance model introduced by Bazant & Zhou [BAZANT-405](#) and carefully generalized by Schneider [SCHNEIDER](#) was implemented and subjected to large-scale Monte Carlo exploration.

The model idealizes the tower as a sequence of discrete stories. A falling upper block of effective mass M accretes floor mass upon inelastic impact, loses energy to structural dissipation (E_{abs}), and gains gravitational potential as it descends the effective story height $h_{eff} = h(1 - \lambda)$, where λ is the compaction ratio. Arrest

occurs at any story if the available energy (post-impact kinetic plus gravitational) is less than the story's dissipation capacity. Column capacity is tapered downward to reflect the heavier lower stories, and floor-to-floor heterogeneity is introduced via lognormal multiplicative noise.

Implementation details

The simulation exactly reproduces Schneider's analytical arrest threshold at nominal parameters ($M_0 = 33 \times 10^6 \text{ kg}$, $\lambda = 0.18$, $\alpha = 1.0$), confirming correct coding to within 0.1 %. Ten thousand Monte Carlo trials were performed, sampling all input parameters uniformly across ranges drawn **exclusively from the peer-reviewed literature** (no ad-hoc values). Uniform distributions were chosen deliberately to explore sensitivity under documented parameter ignorance rather than to produce a posterior probability.

Parameter ranges

(full justification in supplementary material):

Parameter	Symbol	Uniform Range	Primary Source / Notes
Initial upper block mass	M_0	$30 - 58 \times 10^6 \text{ kg}$	Bazant (2002) vs. Schneider/Urich corrections
Energy dissipation per story	E_{abs}	$250 - 2,000 \text{ MJ}$	Bazant (500), Korol-Sivakumaran (1,500–2,000), Bazant-Le rescaling (1,000–1,300)
Floor mass ratio	m/M	$0.063 - 0.10$	NIST structural data
Initiation free-fall fraction	α	$0.52 - 1.0$	Chandler (2010) video to Bazant-Le
Compaction ratio	λ	$0.15 - 0.22$	Bazant & Verdure (2007), Schneider
Mass shedding fraction	f_{shed}	$0.0 - 0.50$	Poorly constrained
Floor-to-floor c.o.v.	CV_E	$0.0 - 0.40$	Heterogeneity modeling

↳ The E_{abs} range spans figures computed by different methods, not a single directly measured quantity: Bažant's original estimate is a plastic-hinge calculation; the Korol-Sivakumaran-derived figure is Bažant & Le's own extrapolation of a column-test ratio $K\&S$ measured, not a number $K\&S$ report themselves; the Bažant-Le rescaled figure is a further recalibration of that same chain. Sampling uniformly across this range treats three derivation chains as commensurable estimates of the same quantity, which they have not been shown to be.

Key Results

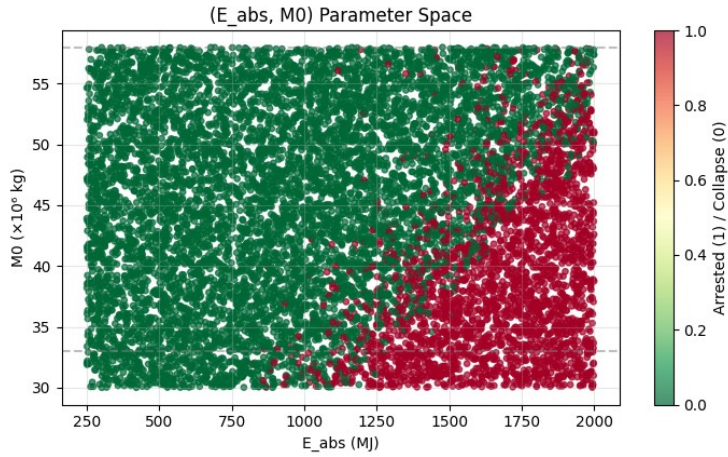


Figure 2. Heatmap of arrest probability across the (E_{abs}, M_0) parameter plane.

- **Arrest frequency:** 2,444 trials (24.4 %) produced arrest before reaching the base of the lower structure. The large majority of arrests occurred early — 57.5 % within the first 5 floors and over 99 % within the first 20 floors.
- **Strong bimodality:** The outcome distribution is sharply bimodal. Either the progression arrests relatively quickly or it becomes self-sustaining runaway collapse. Once the falling mass has accreted more than approximately 10–15 stories, kinetic energy overwhelms even the highest plausible dissipation values within the model.
- **Sensitivity ranking** (Pearson correlation with total floors crushed):

Parameter	Pearson $ r $	Interpretation
E_{abs}	0.612	Dominant driver by a wide margin
M_0	0.363	Heavier upper block increases energy margin
Floor mass	0.312	More accreted mass per story
CV_E	0.082	Minor effect near threshold only
f_{shed}	0.069	Relevant only very near threshold
λ	0.047	Negligible
α	0.036	Initial conditions swamped by later dynamics

Energy dissipation capacity per story (E_{abs}) alone accounts for the overwhelming majority of outcome variance. Most other parameters debated in the wider literature (shedding, tilt proxies, initiation details) are second-order in this formulation.

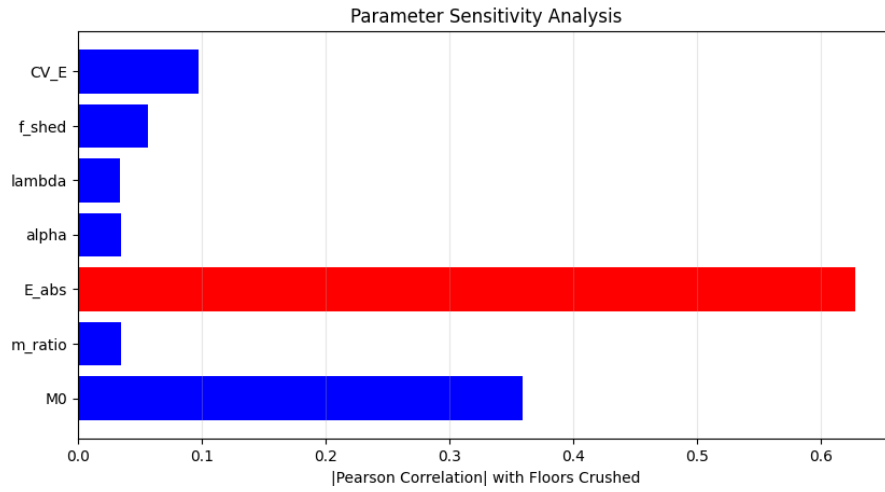


Figure 3. Sensitivity ranking (Pearson $|r|$ with number of floors crushed).

The Threshold Structure

At the corrected upper-block mass ($M_0 \approx 33 \times 10^6 \text{ kg}$), the analytical arrest threshold lies at approximately **1,370 MJ**. The transition from near-certain arrest to near-certain runaway is extremely sharp. Arrest probability exceeds 95 % above $\sim 1,550 \text{ MJ}$ and falls below 5 % below $\sim 1,200 \text{ MJ}$.

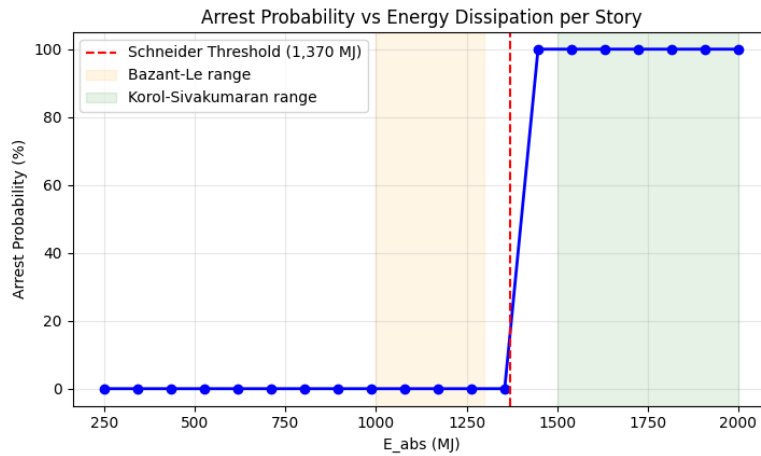


Figure 4. Arrest probability as a function of E_{abs} (other parameters at nominal values), with annotated literature ranges. The step-like transition is clearly visible.

This sharpness has direct implications for the existing scientific debate:

- The Bazant-Le (2016) rescaled range ($\approx 1,000\text{--}1,300 \text{ MJ}$) lies entirely below the threshold $\rightarrow$ runaway collapse.
- The raw Korol & Sivakumaran (2014) experimental range ($1,500\text{--}2,000 \text{ MJ}$) lies entirely above the threshold $\rightarrow$ arrest expected.

The central technical disagreement in the literature therefore maps *precisely* onto which side of a quantifiable, model-derived boundary the actual WTC columns occupied under dynamic, damaged, and partially heated conditions. That boundary question remains empirically unresolved.

Interpretation and Limitations

The 24.4 % arrest fraction should **not** be read as a posterior probability for the actual event. It is an illustrative measure of sensitivity under uniform sampling across unresolved literature ranges. A properly Bayesian analysis incorporating video timing, seismic duration, debris characteristics, and other observables would narrow the posterior — but no such synthesis has been published by the mainstream literature.

The simulation inherits all limitations of the underlying 1D discrete-floor model: fully inelastic impacts, simplified mass accretion, no explicit tilt or 3D load redistribution, idealized tapering, etc. Three-dimensional effects could either accelerate collapse (via misalignment and loss of lateral support) or retard it (via alternative load paths). The ensemble therefore provides informative sensitivity bounds rather than a complete physical prediction.

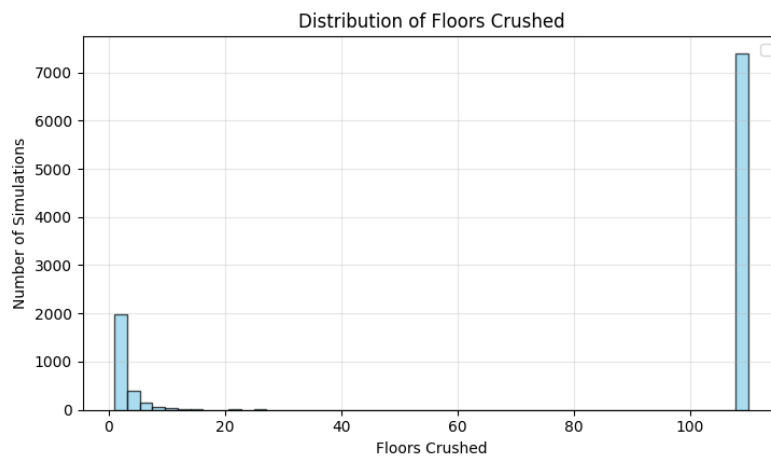


Figure 5. Histogram of floors crushed across all trials, illustrating strong bimodality.

Nevertheless, the central methodological conclusion is robust: **qualitative outcome is not stable** across the parameter uncertainty that the published record has left unresolved. A model in which total collapse versus arrest depends so sensitively on a single disputed quantity (effective dynamic dissipation capacity) has not yet demonstrated the robustness required for claims of generic inevitability.

This ensemble analysis concretizes the epistemic diagnosis developed throughout this inventory: plausibility under selected parameters has been shown. Robustness across realistic variance has not.

The literature disagreement is not interpretive but mechanistic: different papers place the actual WTC columns on opposite sides of a deterministic bifurcation.

↳ *This ensemble tests parameter uncertainty within a single model form — the Bažant-Zhou/Schneider discrete-floor structure. It does not test uncertainty in the model’s form itself; §6.9 examines that separately. A model can be robust to parameter variation within its own form while remaining one of several mechanically distinct, competing forms, none validated against the others.*

Code Availability and Reproducibility

All Monte Carlo simulations were implemented in Python (code available at <https://github.com/Akareyon/ProgressiveCollapseSensitivityEnsemble>). Parameter ranges are drawn exclusively from the peer-reviewed literature as cited in the code documentation. Results are fully

reproducible: running the ensemble with *seed*=42 produces identical arrest probabilities and sensitivity rankings. The 10,000-trial results are provided in supplementary data (ensemble_results.json).

↳ *The ensemble analysis (§4.4) addresses robustness: whether progression is generic or fine-tuned. The kinematic constraint (Section 4.3.2) addresses a prior question: whether the official collision model is internally consistent with the claimed kinematic outcome. These are distinct questions. Both remain unresolved in the published literature.*

A Temporal Instability Beyond the Ensemble: The Late Jolt

The ensemble above treats E_{abs} as an unknown constant, sampled once per story and held fixed for the duration of a single collapse. A separate analysis asks a different, more exacting question of the same equation: not “which constant value of E_{abs} is compatible with arrest or runaway,” but “what value of E_{abs} does the actual, most complete video record require at each point in the collapse, if the equation of motion is Bažant’s own?”

Schneider extends the video-kinematic tracking of the North Tower’s descent well beyond the roughly three seconds Bažant’s own analyses cover — out to over nine seconds, using three independent video sources cross-synchronized frame by frame — and fits Bažant’s own Crush-Down equation to the fuller record. The required E_{abs} is not stable across the event. For the first 4.6 seconds, matching the observed motion requires no more than roughly 250 MJ per story. Between 4.6 and 7.7 seconds, the same fit requires a minimum of roughly 1,950–2,050 MJ per story — an increase Schneider describes as occurring “by almost an order of magnitude.” After 7.7 seconds, the required value falls back to its original low level.

This is not merely a value inside the disputed range the parameter table above already lists; it sits above the upper end of it. Bažant and Le’s own 2016 estimate of the columns’ maximum physical dissipation capacity is 1,000–1,300 MJ per story — itself a downward rescaling, by a factor of 2/3, of Korol and Sivakumaran’s raw experimental measurement of 1,500–2,000 MJ. The minimum dissipation the video record demands during the 4.6–7.7 second window exceeds even Bažant and Le’s own later, more considered ceiling. Schneider states the implication directly: “If the maximal possible dissipation of energy per storey is on a scale below 2000 MJ... then this implies that the principal assumption is wrong” — that the collapse, over this specific interval, was not gravity-driven as the model describes it. Notably, the required value sits close to the raw, uncorrected Korol-Sivakumaran figure; the tension traces specifically to Bažant and Le’s own downward rescaling of that data, not to the underlying experimental measurement.

Two further details sharpen rather than soften the finding. The required dissipation, had it persisted roughly one additional second, would have arrested the collapse outright under the same equation. And whatever produced the rise also reversed it: the fit shows the required dissipation returning to its original low value after 7.7 seconds, in the same structure. Schneider proposes no mechanism for either transition: “We do not speculate how it was possible that the resisting structure rises and decays by an order of magnitude, but do point out that this was the case.”

↳ *This finding is not independently peer-reviewed — the same evidentiary tier as other non-peer-reviewed sources cited throughout this document, and it should be weighed accordingly. It does not establish what caused the fluctuation, and it is not offered as evidence of an alternative mechanism. It establishes something narrower: that Bažant’s own governing equation, fit to the fullest available video record and checked against Bažant’s own later, self-authored parameter revision, does not reconcile with itself over a specific three-second interval — using no assumption external to the official mechanism.*

4.5 When Naive Models Normally Work

In ordinary engineering practice, naive models like the above:

- estimate pile-driving penetration
- evaluate impact cratering depth
- approximate collision outcomes
- assess preliminary arrest conditions

They succeed because most structural systems operate far from instability. In those regions:

- responses are quasi-linear
- local resistance approximates global behavior
- connection variability does not dominate

Upper bounds set by naive diagnostics are not just pedagogical; they are routinely conservative and reliable.

When naive bounds **fail spectacularly** in real systems, the failure itself is informative.

4.6 Critical Regimes, Propagation Metaphors, and Spatiotemporal Coherence Constraints

There exist domains in structural dynamics where naive engineering models lose predictive power:

- Euler buckling and snap-through instability
- dynamic fracture and branching crack fronts
- percolation and cascading failure
- avalanche release in metastable systems
- phase-transition analogs in continua
- progressive collapse near instability thresholds

These are **critical regimes**. In such regimes:

- global behavior is controlled by mode selection rather than local stress alone,
- small parametric changes produce large qualitative jumps in outcome,
- nonlinearity amplifies heterogeneity, and
- conservation relations alone do not determine trajectory.

When an explanation invokes crushing fronts, compaction waves, adiabatic heating, or progressive collapse analogies, it is asserting that the system entered such a critical regime. That is a strong claim. It carries a burden: **the regime must be shown to be generically reachable and stable under realistic variance**, not merely possible under tuned assumptions.

4.6.1 Propagation Metaphors as Heuristic Proxies for Critical Regimes

Descriptions of progressive collapse frequently invoke metaphors such as dominoes, chain reactions, avalanches, or “houses of cards.” These metaphors are not explanations; they are informal references to specific classes of critical-regime dynamics:

- **Dominoes** require pre-aligned geometry, low dissipation, and negligible reserve capacity.
- **Chain reactions** require local events that amplify energy or reduce resistance at each step.

- **Avalanches** occur in metastable systems already near a critical slope, with friction tuned close to failure.
- **Houses of cards** are deliberately unstable configurations with minimal redundancy.

In each case, propagation depends not merely on initiation, but on the system residing within a **narrow critical regime**. Outside that regime, disturbances arrest, dissipate, or localize. Engineered steel structures differ fundamentally from these analog systems: they are designed with redundancy, multiple load paths, and energy dissipation mechanisms that oppose synchronized failure. Metaphors therefore shift the burden of proof rather than discharge it. They rephrase the question: *Does resistance decrease faster than destabilizing forces accumulate across many sequential engagements?* That must be demonstrated, not assumed from analogy.

4.6.2 Spatiotemporal Coherence as an Additional Constraint

Reduced-order progressive-collapse models often assume an approximately planar collapse front that remains coherent across the building footprint. This assumption introduces a further, often implicit constraint: **near-synchronous engagement and failure of load-bearing elements across plan dimensions of order 60–70 m**.

As descent velocities increase, the decisive deformation phase for effective loss of axial resistance (e.g., a few centimeters of plastic shortening or buckling) can occur over milliseconds. Meanwhile, stress redistribution and kinematic information propagate at finite wave speeds ($\sim 5\text{--}6$ km/s in steel/composite framing), yielding transit times across the footprint of ~ 0.01 s. When the collapse front advances rapidly, local failure completion can therefore occur on timescales comparable to or shorter than full-plan wave transit. This increases sensitivity to heterogeneity:

- phase lags between load paths,
- spatial gradients in damage and temperature,
- rotational modes and eccentric loading,
- local hang-ups and partial engagement

all become more likely unless a broader, slower coordinating mechanism averages the load across the footprint. Highly localized, short-duration impulses favor loss of planarity and symmetry. Conversely, more distributed, ductile deformation modes accommodate asynchrony but tend to reduce localization and suppress idealized front propagation.

This coherence requirement further narrows the region of parameter space in which one-dimensional collapse-front models are expected to apply. It is an additional constraint on the basin of attraction for sustained, symmetric, near-vertical progression.

4.6.3 Synthesis: A Narrow Basin with Multiple Simultaneous Constraints

Taken together, these considerations indicate that progressive collapse propagation—if it occurs as described—must satisfy **multiple coupled constraints simultaneously**:

- entry into a critical regime where simple bounds fail,
- resistance reduction outpacing demand growth across sequential stages,
- localization of dissipation into a thin collapse front, and
- maintenance of spatiotemporal coherence across the building footprint despite heterogeneity.

Each constraint individually narrows the basin of attraction. Their conjunction narrows it further. Predictions of sustained, symmetric collapse therefore represent a **limiting case within a restricted parameter regime**, not a generic consequence of gravity and fire.

Demonstrating inevitability under these conditions requires showing that this multi-constraint regime is **broadly accessible and stable under realistic variance**, rather than being a narrow corridor requiring coordinated assumptions about localization, dissipation, accretion efficiency, and coherence. Absent such demonstration, claims of inevitability overstate the evidentiary basis of current models.

4.7 Porosity, Load Paths, and Zeroth-Order Intuition

High volumetric porosity (“mostly air”) is sometimes cited as evidence that the towers offered limited resistance to downward motion. This intuition treats porosity as a proxy for structural weakness. This intuition is non-diagnostic.

In materials science, porosity generally reduces *material* strength, often approximated by relations of the form:

$$\sigma \approx \sigma_0 (1 - \Phi)^n$$

where Φ is porosity and n depends on material and microstructure. Such relations describe homogenized continua. They do not directly characterize the behavior of engineered structures, where resistance is governed by load paths, geometry, and force redistribution rather than bulk density.

Many highly porous systems — trusses, lattices, honeycombs, bone, and wood — achieve substantial load-bearing capacity by organizing material along stretching-dominated paths. In these systems, porosity reflects optimization for **specific strength**, not an absence of resistance.

The relevance of porosity to collapse dynamics therefore cannot be inferred from volume fraction alone. The same structural porosity that reduces self-weight also enables axial reserve capacity, redundancy, and redistribution — features that ordinarily act against progressive failure.

References to “mostly air” thus function as zeroth-order intuitions rather than mechanisms. They do not specify how engineered load paths lose resistance, how energy dissipation is bypassed, or why arrest would be generically suppressed once descent begins. Porosity explains nothing without quantified load-path analysis. Density alone is not mechanism.

4.8 Scale Arguments and the Square–Cube Misconception

Appeals to the square–cube law are frequently used to claim that large buildings are “naturally” prone to progressive collapse. This reflects a fundamental error: engineering design explicitly defeats naive geometric scaling.

4.8.1 The Square–Cube Law in Unengineered Systems

For geometrically similar, unengineered structures scaled by factor k :

- **Volume (and mass)** scale as $\propto k^3$
- **Cross-sectional area (and nominal strength)** scale as $\propto k^2$
- **Stress under self-weight** therefore scales as $\propto k^3/k^2 = k$

This relationship explains why unengineered systems—biological skeletons, geological formations, or homogeneous solids—fail at large scales. A structure scaled up by a factor of ten experiences approximately ten times the self-weight stress in its supporting members if geometric similarity is preserved.

4.8.2 Why Engineered Structures Do Not Follow Square–Cube Scaling

Buildings are not scaled biological organisms or homogeneous solids. They are deliberately engineered systems whose geometry, member sizing, load paths, and safety margins are selected to satisfy codified performance requirements:

- **Member cross-sections** are sized to satisfy load combinations rather than geometric similarity
- **Design load combinations and robustness requirements** become more demanding with building height (e.g., wind, seismic, P– Δ effects)
- **Allowable stresses and demand–capacity ratios** are constrained by code
- **Redundant load paths and continuity** are introduced explicitly to mitigate disproportionate collapse

Accordingly, mass-to-strength ratios follow **engineering scaling**, not square–cube scaling. Tall buildings do not approach gravitational self-failure by virtue of size.

4.8.3 Empirical Confirmation

Empirically, large engineered structures do not exhibit self-weight instability:

- Tall towers and long-span bridges do not buckle under gravity as a class
- Large buildings routinely remain stable under wind, seismic, and fire loading without global collapse
- Very tall structures (e.g., supertall towers) demonstrate that engineered scaling defeats naive square–cube limitations

The WTC towers were designed under the same engineering paradigm: member sizes and load paths were selected to carry design loads with substantial margins relative to self-weight.



Figure 6. South Tower in early stages of construction. Credit: Leffler

4.8.4 Implications for Progressive Collapse

Square–cube scaling does not explain progressive collapse in engineered structures, for the reasons already established in §4.8.2–4.8.3. What it also fails to address is a distinct question: global robustness and resistance to localized dynamic collapse fronts are separate properties. The latter is governed by post-peak material behavior and localization phenomena, not overall structural capacity. Square–cube considerations address static capacity; progressive collapse is governed by dynamic processes with fundamentally different scaling properties. The relevant question is therefore dynamic demand versus dissipation capacity, not static scale effects.

4.8.5 The Size Effect Law: A Precise Domain Boundary

A more sophisticated version of the scale argument invokes not naive square–cube stress scaling but the quasi-brittle size effect law — Bažant’s own established contribution to fracture mechanics — to argue that the towers’ unusual scale explains anomalous completeness of collapse. This argument has a stated, precise domain, not an open one.

The size effect law distinguishes failure modes by the slope of nominal strength against size on a log–log plot. For “elasticity with allowable stress, or elastoplasticity with any failure criterion... expressed in terms of the stress or strain components,” geometrically similar structures “fail at the same nominal stress”

regardless of size — a slope of 0, “the basic, reference case, in which we say there is no size effect.” [BAZANT-S34](#) Quasi-brittle fracture (slope $-1/6$ to $-1/4$, transitioning toward the LEFM asymptote of $-1/2$ at large scale) arises specifically in “materials... characterized by the existence of a large fracture process zone with distributed cracking damage”: concrete, rock, ceramics, fiber composites, sea ice, wood, and similarly heterogeneous materials. Weibull statistical size effect, a separate mechanism again, “has been enormously successful in applications to metal structures embrittled by fatigue” — where failure initiates from a still-microscopic flaw prior to large-scale plastic deformation.

Neither mechanism describes the failure mode the crush-down model assigns to steel columns. F_b — the column-buckling resistance term (§6.11.2) — is explicitly a plastic, three-hinge mechanism: ductile yielding and large-scale plastic deformation, not brittle fracture from a small flaw. This is precisely the reference case the law itself identifies as showing no size effect. Concrete floor-slab comminution (§4.10.1, §5.6) sits inside the size effect law’s actual domain, and its use there is unproblematic. Extending the same apparatus to the steel columns’ failure mode would apply a theory outside the domain its own literature states for it.

4.9 Dynamic Scaling and Adverse Growth of Demands

Dynamic processes scale differently from static capacity:

- **Kinetic energy:** ($E_k \propto m v^2$)
- **Momentum:** ($p \propto m v$)
- **Dissipation capacity:** scales primarily with material strength, geometry, and deformation mechanisms

As structures increase in size, gravitationally attainable velocities remain bounded by height, while participating mass increases. This produces adverse scaling in dynamic demands even when static stability improves with size.

The relevant analytical question is therefore not:

“Do large structures fail more readily because they are large?”

but:

“Does scale-aware dynamic analysis predict the observed collapse behavior across plausible parameter variation, or only under narrowly tuned assumptions about localization, dissipation, and mass participation?”

Dynamic scaling requires dimensionless severity parameters comparing gravitational energy release to localized dissipation capacity. The interpretation of such parameters depends critically on how dissipation is modeled and where plastic work is assumed to occur.

We define the **severity parameter**

$$\Gamma \equiv E_g / E_d = mgh / (F_c h) = mg / F_c$$

where m is the mass of the falling block, g is gravitational acceleration, h is story height, F_c is the mean crush resistance of one story, E_g is the gravitational potential energy released over one story, and E_d is the plastic dissipation capacity assigned to one story. Γ compares the gravitational energy released by a one-story drop to the plastic energy dissipation capacity attributed to a single story.

↳ **Notation note:** capital Γ is used to avoid collision with the strain-rate $\dot{\gamma}$ introduced later in Exhibit B’s rate-dependent material model. E_g/E_d , rather than Bažant’s own W_g/W_p , are used deliberately: those symbols are not only borrowed but are themselves defined inconsistently across Bažant’s own papers (§6.11.3), and this document’s E_g/E_d are not identical to either of his definitions.

4.9.1 Interpretation of Γ

The condition $\Gamma > 1$ does not characterize the static stability of the intact structure. Static stability is governed by force equilibrium ($F_c > mg$) rather than by energy dissipation requirements.

Instead, $\Gamma > 1$ indicates that, under the modeling assumption that dissipation is localized to a single story, that story alone cannot absorb the energy associated with a one-story drop of the falling mass. Γ therefore characterizes **arrest capacity under a specific collapse mechanism**, not structural adequacy under service or design loads.

4.9.2 Localization of Dissipation

The assignment $E_d = F_c h$ reflects the assumption that most plastic work occurs in a thin collapse front approximately one story thick. This localization is motivated by strain-softening behavior and post-peak instability in compression members, which tend to concentrate deformation once a story begins to lose load-carrying capacity — the same localization assumption examined for its parameter sensitivity in §4.10 (see especially ξ_{out} , §4.10.2, which quantifies energy escaping this front).

If dissipation is instead distributed over multiple stories,

$$E_d^{(eff)} = \sum_i F_{c,i} h \quad (i = 1 \text{ to } N),$$

then the effective severity parameter becomes

$$\Gamma^{(eff)} = mgh / \sum_i F_{c,i} h,$$

which can be substantially smaller than the single-story value. The magnitude of Γ therefore depends on the assumed degree of localization in the collapse mechanism.

The localization assumption is not left implicit in the primary literature. A published closure gives an explicit mechanical argument for it: once the plastic or fracturing response is triggered at a story, the local wave speed in that story drops toward zero, so “no wave attaining the material strength can penetrate beyond the crushing or plastic front. Only harmless elastic waves can.” ^{D25} On this account, stress capable of triggering further damage cannot propagate ahead of the front by construction, which is offered as the physical reason dissipation should be expected to concentrate there rather than distribute.

This is a genuine mechanical argument, not a bare assumption, and engaging with it directly is more useful than treating localization as unmotivated. It does not, however, resolve the sensitivity question raised throughout this section: even granting that damage-capable stress cannot outrun the front, the argument establishes only that the front is where damage *initiates* — it does not by itself fix what fraction of the total energy budget is dissipated there versus lost to lateral ejection, fracture, or deformation of material already at or behind the front (ξ_{out} , §4.10.2), nor does it address the debris-geometry tension below, which concerns where *mass* ends up after the front has passed, not where stress waves can travel in advance of it.

4.9.3 Initial Γ and the Onset of Collapse

Evaluating Γ for the initial falling mass prior to any accretion yields

$$\Gamma_0 \equiv m_0 g / F_c.$$

A value $\Gamma_0 > 1$ does not imply that the intact structure is unstable or “pre-failed.” It indicates that, within the assumed localized dissipation mechanism, a single intact story does not possess sufficient plastic energy capacity to arrest a one-story drop of the initial falling mass. This is a statement about **arrest under dynamic impact within a localized failure mode**, not about the stability of the undamaged structure.

This is also where the effective resistance factor R (§4.10.3) enters directly: if F_c in the expression above is itself a degraded value, $F_c = R \cdot F_{c,nominal}$, then $\Gamma_0 = m_0 g / (R \cdot F_{c,nominal})$, and the entire question of whether Γ_0 exceeds unity can hinge on an assumption about R that — per §6.7.1 — is independently measured for local static column loss but not for this loading regime.

4.9.4 Mass Accretion and Height Dependence of Γ

If floor mass is assumed to accrete to the falling block as collapse progresses,

$$m(z) = m_0 + \mu z,$$

then the severity parameter becomes height-dependent:

$$\Gamma(z) = m(z)g / F_c.$$

Under this assumption, Γ increases with collapse progression, reinforcing the tendency toward runaway collapse once a localized collapse front is established. This height dependence follows directly from the accretion assumption and does not arise from geometric scale effects; it is the same mass-accretion mechanism examined for its own sensitivity in §4.10.4.

4.10 Localized Dissipation, Deformation Limits, and Parameter Sensitivity

Reduced-order progressive-collapse models embed assumptions about where dissipation occurs and how rapidly axial resistance degrades once instability is triggered. Several parameters jointly control whether downward propagation is mechanically self-sustaining. The most influential are the deformation limits governing post-peak behavior, the degree of energy loss outside the collapse front, the effective resistance reduction, and the efficiency of mass accretion.

4.10.1 Deformation Limits and Post-Peak Softening

Two parameters encode assumptions about how much deformation a story can sustain before effective loss of axial load-bearing capacity:

- u_c/h : the characteristic deformation ratio associated with near-complete loss of axial resistance under sustained gravity load;
- θ_a : an effective buckling or rotation angle associated with snap-through instability.

Representative parameterizations use values on the order of:

$$u_c/h \sim 10^{-2}, \theta_a \sim 5^\circ\text{--}7^\circ$$

corresponding to absolute deformations of only a few centimeters over a story height of several meters. These values reflect the assumption of rapid post-peak softening and limited deformation capacity prior to near-complete loss of axial resistance.

Real steel frame systems exhibit a range of post-peak behaviors depending on member slenderness, bracing, connection ductility, continuity, catenary action, and load redistribution. Larger deformation capacity increases effective plastic dissipation capacity and produces spatially distributed dissipation rather than extreme localization.

The compaction ratio also generates an independently checkable prediction the model itself doesn't state explicitly: applying $\lambda \approx 0.15\text{--}0.22$ across the full crushed height implies a final compacted pile on the order of 60–90 m — roughly 15–20 stories. Post-collapse imagery and LIDAR survey data of the site instead show a debris field comparable in height to the remaining perimeter “trident” column stubs (on the order of a few stories), with depth below street level in places.

This is not merely a matter of imprecise compaction ratios: a substantial fraction of material was evidently ejected laterally as dust and debris rather than remaining within the footprint the crush-down/crush-up model assumes — itself consistent with ξ_{out} (§4.10.2) being far from negligible.

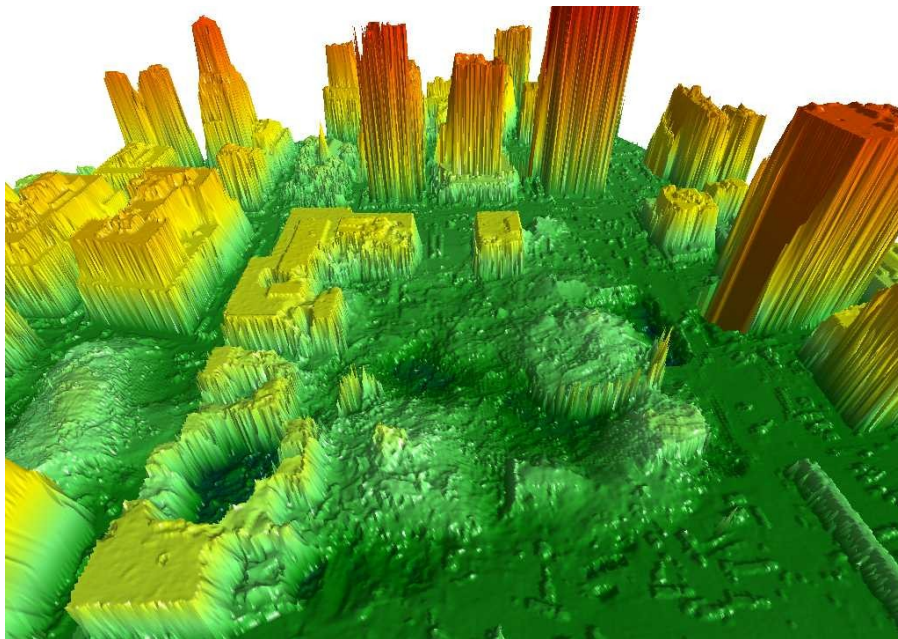


Figure 7. Digital surface model created by LIDAR mapping. Credit: NOAA NGS

This tension has, in fact, been directly addressed in the primary literature. A published discussion raised essentially this calculation, projecting an 83 m compacted-debris height plus a 53 m intact upper block from the model's own stated parameters — far exceeding the observed debris field. Bažant and Le's closure answered with a specific, quantitative resolution: invoking the model's own mass-shedding fraction ($\kappa_{out} \approx 0.2$), it states that the tall, narrow pile predicted immediately on impact “exists only for a split second... [before it] immediately begins to spread rapidly outside the tower footprint,” and calculates — holding rubble density constant — a lateral spread of roughly 60 m beyond the footprint at a slope of about 20°, a figure the closure states “agrees well with the typical slope of rubble piles seen in the demolitions of buildings.” [D27](#)

This resolves the qualitative objection but replaces it with a specific, checkable quantitative one: does the actual debris field extend roughly 60 m beyond the tower footprints at roughly a 20° slope, as this closure's own calculation predicts? That comparison — footprint-plus-60m modeled extent against surveyed or LIDAR debris-field geometry — is, to our knowledge, not published, and would settle the specific claim rather than the general assumption.

4.10.2 Energy Partitioning Outside the Collapse Front (ξ_{out})

The parameter ξ_{out} represents the fraction of gravitational energy release and kinetic energy dissipated outside the localized collapse front, including through:

- lateral ejection and fragmentation of debris,
- fracture and tearing of connections,
- deformation of the falling block,
- vibration and wave radiation into surrounding structure.

Larger values of ξ_{out} reduce the net energy available to drive crushing at the collapse front. In the limiting case ($\xi_{out} \rightarrow 1$), localized crushing cannot be energetically sustained. Small values of ξ_{out} concentrate dissipation within the collapse front, enabling continued downward propagation.

4.10.3 Effective Resistance Reduction (R)

The factor R scales the effective mean crush resistance to account for degradation due to dynamic effects, damage accumulation, imperfections, and loss of load-path effectiveness during collapse. Values $R < 1$ reduce the effective dissipation capacity (W_p) and increase the severity parameter Γ . Conversely, engagement of alternative load paths, catenary action, and strain-rate strengthening in steel would increase the effective resistance relative to conservative estimates.

↳ **Notation note:** R is this document's own synthesis quantity, applied to whole-story effective resistance. Bažant's own literature uses β for a related but narrower quantity — a reduction factor applied specifically to the column-buckling force component, $F_b = \beta F_p$ (§6.11.2) — not identical in scope, though conceptually the same kind of adjustment.

4.10.4 Mass Accretion Efficiency

Mass accretion governs the growth of the falling mass and therefore the growth of dynamic demand with collapse progression. High accretion efficiency increases momentum and gravitational energy release per story, reinforcing runaway behavior. Partial loss of mass through fragmentation or lateral ejection reduces demand growth and weakens the tendency toward sustained propagation.

4.10.5 Coupled Influence on Propagation

The condition for sustained downward propagation is:

$$(1 - \xi_{out})E_g(z) \gtrsim RE_d(u_c/h, \theta_a),$$

with $E_g(z)$ increasing with accretion and E_d governed by deformation limits and post-peak resistance. The parameters u_c/h , θ_a , ξ_{out} , R , and accretion efficiency therefore act multiplicatively on the demand–capacity balance. This is the coupled form of the severity parameter Γ defined in §4.9: $\Gamma_{coupled} \equiv (1 - \xi_{out})E_g(z) / [RE_d(u_c/h, \theta_a)]$, with propagation favored as $\Gamma_{coupled} \gtrsim 1$.

Robust conclusions regarding collapse propagation require demonstration that predicted behavior persists across a broad and physically plausible parameter range, rather than arising from a narrow band of low-ductility, high-localization, high-accretion assumptions.

Summary Table: Key Parameters and Their Directional Effects

Parameter	Physical Meaning	If Increased	Effect on Collapse Propensity
u_c/h	Deformation capacity before loss of axial resistance	More ductility	↓ Decreases propensity (more dissipation, less localization)
θ_a	Buckling/rotation angle at instability	Greater post-peak tolerance	↓ Decreases propensity
ξ_{out}	Energy dissipated outside collapse front	More external dissipation	↓ Decreases propensity
R	Effective resistance factor	Higher effective resistance	↓ Decreases propensity
Accretion efficiency	Fraction of floor mass captured	Less accretion	↓ Decreases propensity
Localization thickness	Width of collapse front	Thicker front (multi-story)	↓ Decreases propensity

↳ These parameters combine multiplicatively into the severity parameter Γ (§4.9, §4.10.5). $\Gamma \gtrsim 1$ indicates propagation is not self-arresting under the stated assumptions; the table above shows the direction in which each parameter pushes Γ .

4.11 Synthesis: Scale, Dynamics, and Localization

The preceding sections separate three distinct issues that are often conflated:

1. **Geometric scale effects (§4.8):** Naive square–cube scaling applies to unengineered systems and does not govern the stability of engineered structures. Building size alone does not imply gravitational self-instability or an inherent predisposition to collapse.
2. **Dynamic demand scaling (§4.9):** Dynamic collapse processes can exhibit adverse scaling because kinetic energy and momentum scale with participating mass, while attainable velocities under gravity remain bounded. This establishes that scale effects, if present, arise from dynamics rather than geometry.
3. **Localization of dissipation (§4.10):** The severity parameter Γ compares gravitational energy release to plastic dissipation capacity under the specific assumption that dissipation localizes into a thin collapse front. The magnitude and interpretation of Γ therefore depend on the degree of localization assumed.

Taken together, these distinctions clarify that conclusions regarding runaway progressive collapse are **conditional** on the assumed collapse mechanism. In particular:

- Square–cube arguments do not explain progressive collapse in engineered structures.
- Dynamic scaling produces adverse demand–capacity ratios when kinetic energy growth exceeds dissipation capacity growth.

- Whether these ratios exceed unity in practice depends on the extent to which plastic dissipation localizes versus distributes across multiple stories and load paths.

Accordingly, assessments of collapse propagation must be framed in terms of **mechanism-specific dynamic demand and dissipation**, rather than appeals to size or static capacity alone.

Chapter 5 — Energy Scales, Impact Comparisons, and Diagnostic Tensions

Naive bounds establish what gravity alone cannot easily achieve. Energy-scale comparisons add precision: they quantify how much energy was applied, where it went, and why structurally similar inputs produced drastically different outcomes. This chapter examines those contrasts.

5.1 Why Compare Energy Scales at All

Energy comparisons do not explain collapse mechanics. They perform a narrower function:

they test internal coherence of claimed mechanisms

If two events differ drastically in consequence while similar or larger energy inputs are involved, explanation must identify:

- *where* energy went,
- *how* it was dissipated,
- *why* qualitatively different structural responses occurred.

Without this, accounts become narrative rather than mechanistic.

5.2 Horizontal Impact: Aircraft Strikes as a Structural Disturbance

The aircraft impacts delivered kinetic energies on the order of gigajoules. The loading was:

- high-velocity,
- localized,
- largely horizontal,
- accompanied by fragmentation and fuel dispersal.

Observed global structural response:

- transient lateral sway,
- damped oscillation within design expectations,
- no immediate loss of global stability.

Despite intense local damage and fires, the structures remained standing for extended periods.

The contrast is not energy magnitude but failure topology. Horizontal impact did not trigger a failure pathway with positive feedback; vertical progressive failure did.

This establishes an empirical fact:

Large energy input alone is insufficient to cause immediate global collapse in tall framed-tube structures.

Any adequate account of the subsequent global collapse must therefore identify **the specific mechanism by which later, smaller incremental energy inputs produced complete failure.**

5.3 Quantitative Energy Asymmetry

The horizontal–vertical contrast can be quantified using energy estimates from NIST and Bažant’s models.

Aircraft impact:

- **Mass:** ~150,000 kg
- **Velocity:** WTC1: ~470 mph (~210 m/s), **WTC2: 542 mph** ± 24 mph (~242 m/s)
- **KE** = $\frac{1}{2}mv^2 \approx 3.3 \text{ GJ}$ / **4.4 GJ**

Observed response:

- Transient sway amplitude ~1/3 of design envelope [NCSTAR-1, p. 182](#)
- Damped oscillation with period nearly equal to undamaged structure
- Global stability maintained for **56/102** minutes

First story drop (Bažant’s model):

- Upper block mass: 58,000 tonnes
- Drop height: 3.7 m (one story)
- Gravitational energy: $PE = mgh \approx 2.1 \text{ GJ}$ [BAZANT-405](#)

Observed outcome:

- Onset of sustained downward motion
- Descent through ~85 undamaged stories
- Total structural destruction

Diagnostic observation:

A larger energy input ([3.3, 4.4] GJ) delivered horizontally—where the structure is designed to flex—was absorbed without global failure.

A smaller energy input (2.1 GJ) delivered vertically—where the structure has massive reserve capacity—allegedly initiated total collapse.

This asymmetry identifies that:

- **Loading direction** and **failure topology** dominate over energy magnitude alone
- Horizontal impact did not couple into a self-reinforcing failure mode
- Vertical progression did

Gravity provides sufficient total energy. The question is mechanism: by what pathway do vertical increments couple into coordinated collapse while larger horizontal inputs do not?

5.4 The Horizontal–Vertical Outcome Contrast

The asymmetry established in §5.3 is a diagnostic tension, not merely an observation. It requires that:

- **loading mode,**
- **time history,**
- **failure topology,**
- **momentum transfer pathways**

dominate outcomes.

A coherent explanation must therefore establish, not presuppose:

- that the system in fact transitioned from demonstrated robustness to a self-sustaining runaway,
- why dissipation pathways credited with preventing collapse earlier would cease to operate later,
- why expected asymmetries (tilt, hang-up, partial collapse) would be suppressed rather than realized.

Merely invoking “gravity did the rest” leaves the mechanism unspecified — and presupposing the transition it is meant to explain leaves the claim unexamined

The mechanism that could reconcile the two outcomes is a self-sustaining, escalating cascade — the vertical case entering a regime the lateral case never approached. That the towers entered such a regime is the claim under examination throughout this document (§3, §4.6, §6.5.1), not a premise available to explain the contrast away.

5.5 Momentum Transfer and Dynamic Amplification

Dynamic amplification is often cited as the missing ingredient.

The core claim:

- once motion begins, momentum grows,
- each impact raises local demand,
- resistance falls behind energy input,
- runaway progression follows.

This is plausible in abstract form, but subject to a prior kinematic constraint: under NIST’s own collision model, the theoretical ceiling on average acceleration is $g/3 \approx 3.27 \text{ m/s}^2$ (§4.3.2). Dynamic amplification must therefore either operate within that ceiling, or invoke additional mechanisms not present in the official collision model. It becomes sufficient only when four additional statements are demonstrated:

1. **progression remains axial without enforced symmetry**
2. **falling mass remains coherent under violent interaction**
3. **rising resistance does not overtake increasing mass**
4. **arrest probability remains negligible across realistic variance**

Each is empirical. None are guaranteed by amplification alone.

5.6 Dissipation Pathways and the Accounting Problem

Total gravitational energy during collapse is large.

What matters is partitioning:

- plastic deformation,
- fracture and comminution,
- ejection of material,
- heating of air and dust,
- seismic coupling,
- acoustic release.

Statements such as ‘gravity supplied sufficient energy’ are non-diagnostic.

The relevant question is:

Does the proposed mechanism partition energy in a way that produces the specific observed pattern of acceleration, symmetry, and material pulverization?

That requires:

- modeling of crack propagation and comminution,
- structural breakup dynamics,
- air–dust interaction,
- momentum exchange in non-rigid bodies.

Where modeling exists, it is typically calibrated, not variance-tested.

A sensitivity analysis was performed... [but] a larger variation of the energy dissipation capacity of columns (i.e. more than 50% increase) was recognized to cause noticeable deviations from the video record... Therefore, the proper way of using these data together with this video record is to exploit them for calibrating the energy dissipation per column, restricted, of course, to the realistic range of uncertainties... [BAZANT-585](#)

A concrete instance of this pattern

An independent estimate placed total dissipated energy at roughly two orders of magnitude above the reduced-order model’s implied figure, driven primarily by a much larger assumed comminution energy per unit mass. The published response rejects this on two grounds — a specific, checkable objection to the comminution-energy figure used, and a more general objection that the estimate appeared in “online postings and an online journal not subjected to mechanics reviewing.” The first is a substantive engagement on the merits; the second is a venue-based dismissal independent of whether the number itself is right. [D28](#)

The comminution-energy figure is not an ad hoc estimate; it draws on an extensive, independently published body of work on fracture energetics of quasi-brittle materials (§4.8.5). What has not been shown, to our knowledge, is that the specific per-unit-mass value applied to the WTC case has been checked against that broader published parameter range by an independent party — the open question concerns a single calibrated application, not the underlying theory.

The dispute illustrates §5.6’s point directly: energy partitioning between competing dissipation channels remains a live, unresolved disagreement between calibrated point estimates, not a settled accounting.

5.7 The Ordering of Difficulty

Lateral robustness is the tested, designed-for capacity; axial failure of a few stories looks like it should be the easier case to arrest. The energy asymmetry of §5.3 reverses this ordering, and the reversal only resolves if the vertical case escalated into a self-sustaining cascade the lateral case never had to enter — the load-sharing switch identified at §6.5.1, treated there as an assumption, not a demonstrated transition. Absent that escalation, the ordering has no resolution: a smaller energy input defeated the same structure a larger one did not.

5.8 Epistemic Limits of Energy Comparisons

They:

- require energy-partitioning explanations,
- expose tensions between robustness and inevitability claims,
- demand explicit mechanisms for mode selection,
- highlight the non-triviality of the observed progression.

The key point:

The joint outcome of rapid, near-axial, global collapse with extensive comminution under heterogeneous conditions remains physically demanding. Energy magnitudes alone do not resolve that demand.

Chapter 6 — Models of Inevitability: Calibration, Validation, and Circularity Risks

6.1 What Inevitability Requires

A claim of inevitability is stronger than a claim of plausibility.

- *Plausible* means: an account is internally consistent with known physical laws, given its stated input assumptions. This is a weak bar. Any outcome admits a consistent account under sufficiently permissive assumptions; the substantive question is always whether those assumptions are themselves warranted (§6.2–§6.4), not whether the account coheres once they are granted.
- *Inevitable* means: across realistic variability, the same outcome occurs with high probability.

To justify inevitability, an explanation must show that:

1. the observed outcome occupies a **large basin of attraction**, and
2. nearby states in parameter space converge to that same outcome.

A model that reproduces one observed case is therefore insufficient. Inevitability is demonstrated not by fit, but by **robust recurrence under variance**.

6.2 Calibration, Validation, and Their Epistemic Roles

All serious models of unique historical events are calibrated. That is normal scientific practice.

What is at issue here is not calibration itself, but the **confounding of calibration with validation**.

Calibration:

- adjust uncertain parameters,
- match observed behavior,
- identify ranges compatible with data.

Validation:

- specify the model before observing new data,
- predict outcomes not used in calibration,
- survive independent tests.

When the same dataset is used to:

- tune free parameters and
- claim the success of the tuned model,

calibration and validation collapse into a single step. That is where circularity risk arises.

6.3 Characteristic properties of “proof-of-collapse” models

Models explicitly developed to show that collapse progression must occur share certain structural features:

- constrained degrees of freedom (often one-dimensional vertical motion),
- rigid or quasi-rigid upper blocks,
- homogenization of discrete structural elements,
- simplified load–displacement laws,
- parameters selected to favor progression,
- symmetry enforced or inherited from boundary conditions.

None of these features are illegitimate. They are recognizable as **idealizations**.

What follows from them, however, is crucial:

A model built to show that collapse proceeds cannot, by construction, discover arrest dominated by unsymmetrical, multimode behavior it excludes.

It evaluates sufficiency under the assumptions it encodes. It does not evaluate inevitability under real-world variability.

6.4 Circularity in Proof-of-Collapse Modeling

A characteristic reasoning loop is common across such analyses:

1. collapse is observed,
2. a model is designed to demonstrate collapse progression,
3. free parameters are chosen or tuned to match observed timing and behavior,
4. the resulting match is cited as evidence that collapse was inevitable.

Formally, the same empirical record plays two roles:

- constraint for parameter selection, and
- test for explanatory success.

It is a **structural epistemic limitation**. It produces strong **retrodiction**, not independent prediction.

6.5 Limits imposed by boundary choices

When:

- lateral motion is suppressed,
- block breakup is neglected,
- multiple failure fronts are excluded,
- local connection variability is smeared into continuum resistance,

the resulting model space excludes:

- toppling,
- partial collapse,
- asymmetric arrest,
- competing descent modes.

Therefore, when such models report that collapse progresses rather than arrests, the result is:

a statement about what happens in the reduced model space, not about the full dynamics of real structures under variance.

Inevitability claims derived from such restricted models rely on extending their conclusions beyond the domains they actually explore.

6.5.1 The Load-Sharing Switch

The “local connection variability... smeared into continuum resistance” identified above is not a generic simplification. It is one specific, locatable substitution, and it is what actually produces the appearance of a transition between two severity conditions this document has cited throughout without yet separating them formally.

Call Γ_{static} the condition governing the intact, as-built structure: $F_c^{sys} > mg$, where F_c^{sys} is the story’s *system* capacity — the redundant, code-compliant, safety-factored resistance of a connected frame, the quantity every statement about the towers’ decades of service life and design margin (§3.3) actually describes. Call $\Gamma_{effective}$ the condition the crush-down model evaluates during collapse: $E_g > E_d$, i.e. $mg > R \cdot F_c^{iso}$ (§4.9.3, §4.10.3), where F_c^{iso} is computed from the standard three-hinge plastic buckling formula for a single, isolated column — the textbook calculation for a member with no lateral support from its neighbors.

Applied to the actual structure, the comparison this framework needs has three terms, not two — and only one of the two steps between them is licensed by damage.

F_c^{sys} is what the standing tower’s decades of service life describe: connected-system capacity, columns sharing load through floor diaphragms and the hat truss, the quantity every statement about pre-collapse robustness actually credits.

F_c^{iso} is the isolated three-hinge capacity, stripped of that connection. This is what the crush-down model assigns to a story once the descending mass reaches it — and it is what the model assigns to every one of the eighty-plus stories below the fire zone, none of which were struck, heated, or damaged before the front arrived. Nothing happens to these floors between being credited with F_c^{sys} and being recomputed at F_c^{iso} . Only their position relative to the crush front changes.

$R \cdot F_c^{iso}$ is the further-reduced capacity assigned to the one floor actually struck and heated. This step, unlike the one before it, answers to something real: R (§4.10.3) is the model’s own accounting for fire and impact damage, applied on top of the isolated baseline.

The chain runs $F_c^{sys} \rightarrow F_c^{iso} \rightarrow R \cdot F_c^{iso}$. The second step is damage. The first is not. It is applied to every undamaged floor the front will reach, on no basis but which side of the front that floor is currently on.

These are not the same quantity observed before and after a physical transition. They are two different quantities, computed by two different methods, and nothing in the crush-down formalism derives one from the other. The reduction to a 1D model — $F_c(z)$, a function of height alone — has no representation for a column at height z being helped by the columns beside it, or by the floor diaphragm and hat truss connecting them; that capacity was never in the model to lose. The one justification offered for this reduction, in a published closure, is that once plastic or fracturing response is triggered at a story, local wave speed drops toward zero, so “no wave attaining the material strength can penetrate beyond the crushing or plastic front” [D25](#) (§4.9.2). That establishes only that damage cannot outrun the front — that stories below remain undamaged until reached. It does not establish that stories still intact and adjacent to the front cannot share load with it while both remain connected. Those are different claims, and only the first has been argued.

The parameter that could, in principle, carry the correction the other way — restoring some fraction of system capacity to the isolated-member formula — is R (§4.10.3), and R is used in exactly one direction. Every stated value and uncertainty range for it across the corpus (2/3 revised to 0.73; 0.24 revised to 0.26; §6.11.2) is a further *reduction* below the isolated three-hinge capacity, never a correction *upward* toward the connected-system capacity Γ_{static} already credits the structure with. The one empirically measured analog available points the other way: alternate-load-path testing found most instrumented members exceeding individual demand-to-capacity limits while the structure still did not progress to collapse (§6.7.1) — system capacity, in a tested, connected frame, exceeding isolated-member capacity, not the reverse.

This chain also governs the model’s initiation criterion, not only its propagation. Downward crush-down is stated to begin from rest — no velocity, no momentum contribution needed — whenever $mz_0g > F_c(z_0)$ at the critical floor. [BAZANT-466](#) $F_c(z_0)$ here is $R \cdot F_c^{iso}$, the third term above, applied to the one floor damage actually reached. The criterion never asks whether that floor’s undamaged neighbors — untouched by fire on either side — could carry redistributed load the way a tested, connected frame does (§6.7.1, entire columns removed, structure held): that comparison would require F_c^{sys} , the first term, and the model makes it nowhere. It compares the third term to weight, at rest, and on a “yes” proceeds to use the second term — undamaged, isolated capacity — for every floor that follows.

*The same substitution reappears, from a different angle, in Exhibit B’s **Anti-Newtonian Solid**: a structure robust under service load but offering negligible resistance once dynamic descent begins is exactly what a system-capacity-to-isolated-capacity switch would produce, restated as a material property instead of a structural one.*

What this yields is not a physical account of how a stable structure enters a supercritical regime. It is a definitional one: Γ_{static} is computed on the system, $\Gamma_{effective}$ is computed on the isolated member, and the space between them — where the claim of an entered critical regime (§4.6) would need to be earned — is never

modeled. It is assumed away by which capacity formula is selected upstream of the equation, not derived within it.

This is what Chapter 5's energy contrast (§5.4, §5.7) depends on for any resolution at all. Without it, a smaller vertical input defeated the same structure a larger lateral one did not, with no accounting given.

6.5.1a The gap is measured, not assumed

The model's engaged-story resistance is fixed by its own literature at $F_c^{iso} \approx 500\text{--}2,000\text{ MJ per story}$ (E_d , §4.4). The connected-system capacity of the same stories is measured, not estimated, in NIST's own baseline structural analysis (NCSTAR 1-2A, Tables 5-1, 5-4, 5-8): under original design gravity and wind loads, the 31,086 exterior-wall columns spanning floors 9–106 carry a mean demand-to-capacity ratio of 0.76, with 1.1% of members exceeding 1.0; core columns sit at a mean DCR of 0.83–0.86 across every load case examined, original and refined alike. Even under NIST's own refined, higher wind-load estimate — a load case designed to be conservative, not representative of September 11 conditions — mean exterior-column DCR rises only to 1.09–1.10, with 72–74% of members exceeding the code-allowable stress level, not the yield or ultimate strength (DCRs here are computed to AISC (1989) allowable stress design, where 1.0 denotes roughly $0.6F_y$ for tension, lower for compression-governed members). NIST's own simplified calculation separately finds the intact floor/connection system could carry roughly eleven additional floors under gradual loading and six under sudden loading before exceeding assumed capacity. [NIST-FAQ](#)

These figures certify the load-bearing and load-transfer performance of the connected system that stood for three decades, under gravity and wind — not under aircraft impact, which this report does not analyze. They do not supply an ultimate dynamic capacity for post-initiation stories under multi-story accreting impact, and this section does not convert a static, dimensionless DCR into an energy-per-story quantity comparable to F_c^{iso} ; the two are not shown to be commensurable here, and no numeric ratio between them is claimed.

What the two descriptions state, in their own terms, does not require that conversion to be in tension. NIST's own baseline analysis certifies a connected story, under its design loads, carrying demand at or near the code-allowable level, with the overwhelming majority of members below it — a system with margin to spare, measured as a system. The crush-down model assigns the same physical stories an isolated-member crush resistance, computed with no credit for that connection, at the moment the front engages them. One description certifies comfortable systemic margin. The other assumes its near-total absence.

The model's own literature does state a conversion — column fracturing and end-restraint flexibility, entering as a fitted reduction factor [BAZANT-585](#) [BAZANT-EPN](#)[^] — but states it as a calibration, not a demonstration: the factor's magnitude is set by matching the video record, and the only measured analogs of connected-story response under member removal [SEZEN](#) indicate restoration of system capacity, not its reduction to isolated-member values (§6.5.1), with the reduction applied in exactly one direction. Neither this document nor the model's literature shows the measured margin disappearing at the front; the literature asserts the reduction, calibrates it, and moves on. This is not a claim that arrest occurred. It is a statement that the transition between two descriptions of the same stories is asserted where it is not demonstrated, in the stories' own terms, without needing to force those terms into shared units.

6.5.2 The Snap-Through Cascade and the Localization–Kinematics Tension

§4.3.2 established the $g/3$ asymptote for the pure stationary-target, perfectly inelastic accretion limit and identified the resulting kinematic constraint on any explanation of near-free-fall descent. §4.6 established the broader methodological point: when a progressive-collapse explanation invokes snap-through, crushing

fronts, or other critical-regime dynamics, the transition into that regime and its persistence under realistic variance become questions requiring demonstration rather than assumption.

Bažant's own formulation supplies an important refinement here. It does not simply ignore snap-through; it replaces the unstable softening response of individual failing stories with a nonsoftening, energetically equivalent characterization of that snap-through. The relevant issue is therefore not whether the mathematical treatment contains an instability, but what that energetic equivalence establishes — and what remains outside its scope.

Bažant's own framing

Rather than homogenizing the tower's stiffness in the conventional way — which he argues would produce an ill-posed strain-softening continuum requiring nonlocal regularization (Bažant & Verdure, 2007) — Bažant constructs an energetically equivalent resistance F_c for each failing story. He describes the construction explicitly as nonstandard homogenization aimed at energy dissipation rather than stiffness, analogous to the energetically equivalent transition in van der Waals theory and to the crack-band treatment of distributed softening:

“It corresponds to nonstandard homogenization, in which the aim is not homogenized stiffness but homogenized energy dissipation — this approach is analogous to the energetically equivalent transition in the van der Waals theory of gas–liquid phase changes, and the energy equivalence is also analogous to the crack band model for softening distributed damage.” (Bažant & Verdure, 2007)

The phase-transition language should be understood precisely. Bažant is not proposing an additional thermodynamic phase transition that must first be discovered in the tower. He is using an established energetic-equivalence analogy to replace an unstable softening branch with a tractable nonsoftening representation. The construction is part of the mathematical model itself, and there is no need to treat it as rhetorical overreach in order to identify the remaining evidentiary problem.

What the equivalence guarantees, and what it explicitly does not

Bažant states the scope of the construction precisely:

“Although the dynamic $u(t)$ history for the replacement F_c is not the same as for the actual $F(u)$, the final values of displacement u and velocity $\dot{u}$ at the end of crushing of a story are exactly the same.” (Bažant & Verdure, 2007)

He further states that, provided F_c is not large enough to arrest the motion, the replacement has no effect on the overall change in velocity from the beginning to the end of column crushing. The construction deliberately sacrifices the detailed transient $u(t)$ history while preserving the relevant endpoint state and net velocity change across the modeled crushing interval. This is a coherent reduced-order construction, and the difference between the true transient trajectory and the equivalent one is not, by itself, a defect. The important boundary lies one step earlier.

Moving the kinematic issue to the correct boundary

Bažant's equivalence maps an already specified initial state into a final state after the story has been crushed. It therefore answers an *intra-story* question: given the velocity with which the crushing event begins, does the energetically equivalent F_c produce the correct final velocity and displacement? According to the stated construction, yes.

The g/3 diagnostic concerns a different question: *what mechanical state does the next portion of the structure occupy immediately before it is engaged, and how does its participation alter the momentum transfer at that boundary?*

This is not a question the full Bažant formulation simply omits. His continuum model explicitly specifies an accreting mass based on the initial mass density of the undisturbed tower and includes both compaction and mass loss at the crushing front; in the crush-down phase, the compacted zone and upper structure are treated as a moving system with momentum modified by compaction and the fraction κ_{out} ejected from the cascade. The elementary g/3 result is recovered only in the special stationary-accretion limit in which those additional effects are suppressed.

The distinction is therefore between two closure problems: **intra-story energetic equivalence** — whether F_c correctly represents the crushing event once its initial state has been established — and **inter-story dynamical closure** — whether the full set of accretion, compaction, mass-loss, resistance, and velocity assumptions produces the observed motion when one crushing event hands off to the next. The first is what the Maxwell-line-type construction addresses. The second is what the g/3 diagnostic exposes. This does not mean Bažant's full formulation is identical to the elementary g/3 model — it is not — but his additional terms must supply a quantitatively adequate departure from that limit. The burden is therefore shifted from the elementary ceiling itself to the magnitude and physical justification of the mechanisms used to depart from it.

The localization–kinematics tension

Bažant's localization argument addresses the spatial side of this problem. As discussed in §4.9.2 and §6.5.1, once plastic or fracturing response is triggered in the damaged region, the relevant local wave speed falls toward zero, so a stress wave capable of producing comparable damage cannot outrun the crushing or plastic front — only elastic disturbances can propagate ahead. This is a genuine mechanical rationale for localization of *damage-producing* deformation, not an assumption asserted without physical basis.

But localization of damage is not equivalent to localization of all mechanically relevant motion. The argument constrains the propagation of damage-capable stress ahead of the front. It does not, by itself, establish that the structure ahead of the front is kinematically identical to an undeformed, stationary target in every degree of freedom relevant to the subsequent collision.

The formalism needs the lower structure to remain sufficiently intact to justify localization of damage, while the observed kinematics potentially require it to be sufficiently mechanically participating that it cannot be treated as a stationary collision target.

If the lower structure is effectively stationary at the instant it is engaged, the momentum-transfer penalty identified in §4.3.2 remains part of the collision mechanism. If it is already moving or deforming sufficiently to alter that collision, the stationary-target boundary condition is incomplete and the mechanism producing that prior participation must be specified and quantified. The localization argument alone does not choose between these alternatives.

It is worth being precise about what Bažant’s own governing equations specify on this point, since the disjunction above should not be read as agnostic between two mechanisms already latent in the published model. They are not. The crush-down equation derives the kinetic-energy accounting for each newly accreted slice of mass explicitly as being accelerated *from rest* to the velocity of the moving block: “its kinetic energy is increased by the kinetic energy due to accelerating every infinitesimal slice... of mass... to velocity v ” (Bažant & Verdure, 2007). Neither the crush-down nor the crush-up equation of motion contains a term crediting engaged mass with velocity prior to accretion. The published mechanism therefore reconciles observed kinematics through exactly two disclosed channels — the magnitude of F_c relative to gravitational demand (the severity condition of §4.9) and the shedding fraction κ_{out} — and nothing else. A prior-participation mechanism, if invoked at all, is not a feature the equations already contain; it would be a third, undisclosed ingredient with no corresponding term in the model as published. Recognizing this sharpens rather than weakens the diagnostic: the reconciliation the published model actually attempts is fully accounted for by F_c and κ_{out} , which is what the remainder of this section evaluates.

That distinction matters because §4.6 has already identified spatiotemporal coherence as an independent constraint on any reduced-order cascade model. A coherent front extending across the building footprint is itself a nontrivial dynamical condition in a heterogeneous frame with multiple load paths.

Critical-regime consequence and the persistence burden

If the cascade departs from the stationary-accretion limit sufficiently to reconcile the kinematics — whether through mass participation, compaction, or (were it invoked) prior dynamical participation — the mechanism is no longer adequately characterized as passive sequential crushing. It becomes a critical, self-sustaining cascade in which the mechanical state of material ahead of the active damage front matters to the propagation law. That is precisely the class of problem for which the Inventory’s critical-regime criterion (§4.6) applies: the transition into the regime, its mode selection, and its persistence must be demonstrated rather than inferred from the existence of a mathematically self-consistent post-transition solution.

The evidentiary burden therefore scales with the claimed extent of the cascade. A mechanism invoked to explain one local snap-through need not automatically explain sustained participation across dozens of subsequently engaged stories. If the same dynamical departure from stationary accretion is required repeatedly, its persistence — not merely its existence at one story — becomes part of what must be demonstrated.

The reconciliation is not free

The mechanisms capable of modifying the simple accretion picture are not interchangeable, and none is free.

Mass shedding changes the mass retained by the descending system and therefore changes both the momentum-transfer problem and the energy available to subsequent propagation. But the effects run in opposite directions. Reduced retained mass can lessen the mass that subsequently has to be accelerated, while the same shedding removes mass and energy from the downward cascade. The Inventory already treats κ_{out} as a mass-shedding parameter and ξ_{out} as a separate outside-front dissipation parameter; they should not be conflated. More importantly, the propagation model assigns mass retention a destabilizing role: greater accretion increases momentum and gravitational demand, whereas fragmentation and lateral ejection reduce demand growth and weaken sustained propagation. Mass shedding therefore cannot simply be invoked as a one-directional solution to the $g/3$ problem; its net effect must be evaluated in the coupled propagation condition:

$$(1-\xi_{out})E_g(z) \gtrsim RE_d(u_c/h, \theta_a)$$

Because the $g/3$ asymptote is independent of the magnitude of accreted density and depends only on the functional form of accretion, uniform shedding does not alter the ceiling itself; it delays the approach to it. The operative question is therefore whether the specific κ_{out} value keeps the accreted-to-initial mass ratio small enough across the roughly 80–110 floors actually traversed, not whether shedding occurs at all.

Pre-engagement or prior deformation poses a different requirement, and one with no corresponding term in Bažant’s published equations. If material below the nominal front is invoked as already participating sufficiently to alter the collision, its cause, timing, spatial extent, and energetic consequences must be specified, especially given that such participation would need to recur repeatedly over the many stories below the initiation zone. Conversely, merely asserting that the lower structure is somehow “pre-engaged” without showing that this participation materially changes the collision does not close the kinematic problem either. Neither extreme — assuming stationarity without checking, or asserting participation without demonstrating it — discharges the burden by itself.

Mathematical closure vs. physical closure

This distinction places Bažant’s snap-through construction in its proper methodological position. The energetic homogenization is mathematically coherent — an explicit reduced-order representation of an unstable story response, whose scope Bažant states clearly. There is no need to challenge that construction to identify the remaining evidentiary problem, which is global physical closure.

A locally energy-equivalent crushing law does not by itself demonstrate that the real structure remains within the dynamical regime required for repeated application of that law. The full cascade additionally requires sufficiently localized dissipation; sufficiently coherent mass participation; sufficiently favorable momentum transfer; sufficiently rapid effective resistance reduction; sufficiently limited competing dissipation and mass loss; and sufficiently strong spatiotemporal coherence to maintain the assumed propagation mode. These constraints are already identified separately throughout the Inventory. The ensemble analysis (§4.4) supplies the next methodological layer: even within the same discrete-floor model form, the qualitative outcome is sensitive to unresolved parameter choices, with arrest and runaway occupying different sides of a sharp model-derived boundary. The ensemble explicitly does not test uncertainty in the model form itself.

Thus this section does not establish that the cascade is impossible, nor that Bažant’s snap-through homogenization is mathematically defective. It establishes a narrower and more useful point:

The snap-through construction closes the energetic representation of an already-engaged crushing event. The full continuum formulation also supplies an inter-story dynamical prescription — reconciling observed kinematics through F_c magnitude and κ_{out} alone, on its own published terms — but the $g/3$ diagnostic identifies the momentum-transfer penalty against which those terms must be quantitatively assessed. If the required departure ultimately depends on some further, undisclosed mechanism, then the transition into and persistence of that mechanism across the remaining structure become part of the evidentiary burden already identified in §4.6 and tested for parameter sensitivity in the ensemble of §4.4.

The resulting burden is therefore not to invent an alternative collapse mechanism. It is to demonstrate that the published mechanism supplies, from independently justified physical assumptions, a sufficiently large departure from the stationary-target accretion limit while simultaneously satisfying the energy-partition, resistance, mass-retention, and coherence constraints required for continued propagation.

If that demonstration succeeds across a broad and physically plausible parameter range, the cascade mechanism becomes robust in the sense required by §6.1. If it succeeds only within a narrow combination of

effective resistance, compaction, mass participation, shedding, deformation limits, and coherence assumptions, then the mathematics establishes a possible propagation regime, not by itself the demonstrated inevitability of that regime.

Mathematical closure is not the same as physical closure. A reduced model can be internally exact with respect to the quantity it was constructed to preserve while remaining empirically underdetermined with respect to the coupled physical conditions that determine whether its reduced mechanism adequately describes the real structure.

That is the unresolved question here.

§4.4 supplies an independent, empirical complement to this theoretical tension: fit to the fullest available video record, the required E_{abs} itself is not stable across the collapse, rising by nearly an order of magnitude for roughly three seconds before falling back. Neither finding identifies what the lower structure was doing during that window. Together they establish that the question — kinematic here, energy-budget there — is live from two independent directions.

6.6 The missing quantity: arrest frequency under variance

The relevant unanswered question is simple:

Across realistic variability of input parameters, how often does the modeled structure arrest, partially collapse, or fail asymmetrically?

This requires:

- large ensemble runs,
- stochastic parameter distributions,
- no symmetry constraints,
- open reporting of arrest rates and divergence cases.

What currently exists in the public record largely consists of:

- single trajectories,
- parameter sweeps guided by observed outcomes,
- high-fidelity reconstructions tuned to match video evidence.

These demonstrate that collapse *can* be represented in models. They do not establish that the observed collapses occupy a **broad, stable basin of attraction**.

6.7 Calibration transparency and sensitivity

A nontrivial proportion of model parameters affecting progression are:

- not directly measured for the WTC structures,
- inferred from analogous systems,
- adjusted within physically “reasonable” bounds,
- sensitive to assumptions about insulation loss, connection behavior, and heat histories.

Two requirements follow if inevitability is to be asserted:

1. parameter choices and sensitivity must be fully transparent, and
2. ensemble statistics must be reported across the ranges of uncertainty.

When transparency or ensemble reporting is absent, claims of robustness rest on **unexamined fragility**.

6.7.1 A Worked Example: The Measured and the Unmeasured Mobilization Factor

The abstract point of §6.7 — that key coupling parameters are inferred rather than measured — can be made concrete with a single quantity: R , the effective resistance factor already defined in §4.10.3, which scales mean crush resistance downward to account for dynamic effects, damage accumulation, and loss of load-path effectiveness during collapse. It is important to distinguish two adjacent bodies of evidence that bear on R , because they are frequently conflated.

What has been independently measured

A mature, decades-old research program — predating the WTC collapse and driven by post-Ronan Point and post-Oklahoma City design requirements, not by it — has repeatedly removed columns from real and large-scale steel and composite frames and measured the dynamic response. [LEYENDECKER](#) [ELLINGWOOD](#) [STAROSSEK](#) [SEZEN](#)

This “alternate load path” literature yields a genuine, independently derived dynamic amplification factor — typically clustering near 1.5–2×, consistent with the design factor long used in GSA/DoD guidance — and it generally documents substantial reserve capacity: catenary action in beams, large connection rotations, and load redistribution well beyond simple flexural predictions. In at least one full-scale test, most instrumented members exceeded standard demand-to-capacity limits once a second column was removed, yet the structure did not progress to collapse. [SEZEN](#)

This is real, independent, non-circular evidence, and it should be credited as such. For the question it actually answers — does a code-compliant steel frame reliably bridge over the sudden loss of one or two columns — the tested record leans toward robustness, not fragility.

What has not been measured

That is a different question from the one the crush-down models need answered. The alternate-load-path tests load an otherwise *intact* structure with a *static or moderately dynamic* loss of one support. The crush-down R concerns something else entirely: how much of a floor’s capacity is mobilized when it is struck by an *already-falling, multi-story accreting mass*, at velocities and impact energies well outside the alternate-load-path envelope, during an *already-propagating* failure rather than a single local one. No published test of a steel high-rise subassembly under that specific loading regime — a falling stacked mass impacting intact stories in sequence — appears in the record; the closest experimental analog is a pancake-collapse impact test on single-story, 1:3-scale reinforced-concrete frame–shear-wall models. [WU](#)

6.8 What the existing modeling record does show

Crediting it fairly: the record shows that progressive gravity-driven collapse is mechanically plausible, that dynamic amplification matters, and that calibrated simulations can visually match large portions of the observed event. These are genuine advances in collapse modeling, not merely negative findings.

What remains missing is precisely the ensemble evidence identified in §6.6: whether progression is generic rather than finely conditioned, whether symmetry is naturally selected rather than imposed, whether arrest is

rare rather than pruned away by modeling choices, and whether the observed outcome recurs broadly across uncertainty. Those, not single-trajectory matches, are the conditions inevitability claims require.

6.9 Model Plurality and the Limits of “Generic” Inevitability Claims

Claims of post-initiation inevitability are further constrained by the existence of **multiple peer-reviewed mechanical frameworks** that model the collapse sequence using different “generic” assumptions.

Representative examples include:

- **Bažant / NIST:** one-dimensional rigid-block, gravity-driven progressive collapse [BAZANT-405](#), [NCSTAR-1](#), [BAZANT-466](#), [BAZANT-476](#), [BAZANT-499](#), [BAZANT-585](#), [BAZANT-625](#), [BAZANT-EPN](#),
- **Seffen:** propagating instability models treating collapse as a wave-like failure mode; [SEFFEN](#)
- **Usmani:** global fire–structure interaction emphasizing thermally induced system-level forces. [USMANI](#)

These differences are not minor parametric adjustments. They reflect fundamentally different physical pictures:

Seffen treats collapse as a propagating instability requiring specific wave conditions—still “generic” in his view, but with stricter mathematical constraints than simple energy comparison. Usmani emphasizes that initiation mode (thermal expansion laterally compromising the core) determines progression character, challenging the decoupling of initiation from dynamics.

These approaches differ in how they treat energy dissipation, symmetry, resistance evolution, and the coupling between initiation and progression. All are mechanically conventional; none invoke non-standard physics.

A separate, more elementary body of work — the naive collision-model treatments discussed in §4.3.2, including Szuladziński’s — establishes a ceiling rather than a competing physical picture: any account claiming kinematic outcomes approaching free fall must reckon with the bound imposed by inelastic collision alone ($a_{max} \approx g/3$), independently derived at least four times. That ceiling is not itself one of the “distinct physical pictures” tallied above; it is a constraint every one of them, including Bažant/NIST, Seffen, and Usmani, must satisfy or explicitly exceed.

The relevance of this plurality is methodological rather than adjudicative. When multiple, mathematically distinct models—each employing simplified but defensible assumptions—can account for aspects of the observed outcome, the observation does not uniquely select any one of them as a proof of inevitability. Instead, the outcome is **underdetermined** by the available modeling record.

Accordingly, the existence of competing ‘generic’ models demonstrates that calibrated models establish plausibility, while inevitability claims require demonstrated robustness independent of modeling choices.

6.10 The appropriate methodological conclusion

Returning to the plausibility/inevitability distinction of §6.1: the current modeling record justifies “can collapse under these conditions,” but it does not yet justify “had to collapse in this way.” That difference is not rhetorical. It is the boundary between explanation and epistemic closure.

§6.5.1 locates where, specifically, that boundary is crossed in this model: not at a physical transition the governing equations describe, but at the point where system capacity is silently replaced by isolated-member

capacity in the resistance term itself. The model does not independently demonstrate that the real structure enters and remains in the critical, self-sustaining regime represented by its reduced resistance law.

6.11 The Burden of Proving “Must” and “Inevitable”

The modeling literature does not conceal its aim: to establish that progressive collapse was inevitable once initiated, driven by gravity alone. This intent is stated plainly and repeatedly, framing the analyses as demonstrations of necessity rather than explorations of possibility. From Bažant and Zhou (2002):

“Since we are not attempting to model the details of the real failure mechanism but seek only to prove that the towers must have collapsed and do so in the way seen...”

In the accompanying discussion:

“The wedging was not considered in the analysis because the stated aim was to prove that the towers must have collapsed. For that purpose, the most optimistic assumptions about the structure resistance had to be made, and the assumption of wedging would not be of that kind.”

NIST builds on this foundation, limiting detailed simulation to pre-initiation events while deeming progression self-evident:

“The specific objectives were: 1. Determine why and how WTC 1 and WTC 2 collapsed following the initial impacts of the aircraft and why and how WTC 7 collapsed.” — (*NCSTAR 1*, p. xxxix)

Yet:

“The focus of the investigation was on the sequence of events from the instant of aircraft impact to the initiation of collapse for each tower. For brevity in this report, this sequence is referred to as the ‘probable collapse sequence,’ although it includes little analysis of the structural behaviour of the tower after the conditions for collapse initiation were reached and collapse became inevitable.” —(*footnote on p. xxxvii*)

And again:

“The focus of the Investigation was on the sequence of events from the instant of aircraft impact to the initiation of collapse for each tower. For brevity in this report, this sequence is referred to as the “probable collapse sequence,” although it does not actually include the structural behavior of the tower after the conditions for collapse initiation were reached and collapse became inevitable.” —(*footnote on p. 82*)

Bažant and Verdure (2007) reinforce this, tying it to a specific criterion while dismissing alternatives:

“Regardless of the load capacity of the columns, there is no way to deny the inevitability of progressive collapse driven by gravity alone if this criterion is satisfied (for the World Trade Center it is satisfied with an order-of-magnitude margin).”

“Therefore, no further analysis has been necessary to prove that the WTC towers had to fall the way they did, due to gravity alone. However, a theory describing the progressive collapse dynamics beyond the initial trigger, with the WTC as a paradigm, could nevertheless be very useful for other purposes, especially for learning from demolitions. It could also help to clear up misunderstanding (and thus to dispel the myth of planted explosives).”

“The subsequent progressive collapse was not simulated at NIST because its inevitability, once triggered by impact after column buckling, had already been proven by Bažant and Zhou’s (2002a) comparison of kinetic energy to energy absorption capability.”

“Some critics have been under the mistaken impression that collapse cannot occur if (because of safety factors used in design) the weight mg of the upper part is less than the load capacity F_0 of the floor. This led them to postulate various strange ideas (such as “fracture wave” and planted explosives). However, the criterion in Eq. (5) makes it clear that this impression is erroneous. If Eq. (5) is violated, there is (regardless of F_0) no way to deny the inevitability of progressive collapse driven only by gravity.”

“If the total (internal) energy loss during the crushing of one story (representing the energy dissipated by the complete crushing and compaction of one story, minus the loss of gravity potential during the crushing of that story) exceeds the kinetic energy impacted to that story, collapse will continue to the next story. This is the criterion of progressive collapse trigger [Eq. (5)]. If it is satisfied, there is no way to deny the inevitability of progressive collapse driven by gravity alone (regardless of by how much the combined strength of columns of one floor may exceed the weight of the part of the tower above that floor).” [BAZANT-466](#)

The logical structure is clear:

- Adopt maximally collapse-favorable assumptions (e.g., frangible columns, rigid upper block, negligible comminution energy).
- Show that under these assumptions, gravitational energy exceeds resistance.
- Conclude that collapse was therefore inevitable—“doomed,” “must have collapsed,” “had to fall” — *regardless of whether those assumptions reflect physical reality or variance.*

These statements elevate calibrated, assumption-bound models to proofs of inevitability. The lever turns here: “Inevitable” implies a dynamical necessity that holds across realistic uncertainties — not a conditional outcome under optimistic simplifications. Where models are built to prove “must” via tuning (e.g., selecting parameters like R , Γ , ξ_{out} to match descent without exploring arrest under variance), they demonstrate compatibility, not necessity. The epistemic demand remains: Show the basin for “must” is broad, or retract to “could.”

Later Bažant papers (2008–2017) refine component mechanisms and respond to select criticisms without addressing the central issue: post-initiation inevitability is still advanced without basin mapping or robustness demonstration under realistic variance.

6.11.1 Case Study: The Free-Fall Baseline Discrepancy

A further methodological concern arises within the Bažant corpus itself, and it is sharper than a simple shift of reference number. Bažant and Zhou (2002) describe the collapse as “almost free fall.” Bažant and Verdure (2007) calculate 10.8 s against a free-fall baseline explicitly defined as running “from the tower top to the ground” — 9.21 s, 17% longer, described as “not much longer.” Bažant et al. (2008) calculate 12.81 s against a baseline of 7.74 s — 65.5% longer, presented as proof that critics’ free-fall claims are “totally out of range of the free fall hypothesis.” [BAZANT-405](#) [BAZANT-466](#) [BAZANT-476](#) Reconstructing the geometry behind the 2008 figure shows that its baseline is not “tower top to ground” at all. It is the height of the visible tower top at the moment the growing, compacting debris pile beneath it first reaches the foundation — a point the paper’s own figure (Fig. 8) places at roughly 125 m above grade, consistent with the compaction ratio (≈ 0.20) used throughout this literature (0.15–0.22). The 65.5%-longer-than-free-fall statistic — offered as the proof that settles the free-fall question — therefore covers only the phase ending while roughly thirty stories of visible structure are still standing above the debris pile. Whatever follows is governed by a separate equation (the “crush-up” phase) and is excluded from the figure entirely. The same 2008 paper elsewhere reports a third free-fall figure — 8.61 s, the calculated free-fall time for fragments ejected from the North Tower’s 96th-floor initiation zone to reach the ground — and explicitly states this is “not the same” quantity as 7.74 s — without ever telling the reader, in the surrounding prose, what physical interval the comparison figure actually spans or why the two should differ. Three numerically distinct “free fall” quantities for the same event circulate across two papers without a single explicit reconciliation. Backing out the baseline the retained percentage implies — $12.40/1.655 \approx 7.49$ s — gives a figure close to, but not identical with, 2008’s own stated 7.74 s baseline, a gap of about three percent left unaddressed either way.

A fourth data point extends the same pattern. Le and Bažant (2022) report a refined model prediction of 12.40 s (North Tower), distinct from both 2008’s own model prediction (12.81 s) and the seismically-derived figure both papers cite (12.59 s) — yet retain the identical “65.5%... longer than free fall” figure from 2008 without restating what baseline the new percentage is computed against. A statistic can survive a revision to the number it was computed from only if the comparison baseline was revised by the same proportion; neither paper states that it was.

The contradiction with NIST is compounded by this same scope question, not merely left standing unresolved. NIST states the upper building section “came down essentially in free fall,” without specifying whether this describes the complete visible descent or some sub-phase of it. Bažant (2008) states the observed motion is “totally out of range of the free fall hypothesis” and offers the calculated 65.5% excess as proof — but that figure, as shown above, applies only to the crush-down phase, ending while the tower top is still roughly 125 m above the ground. If NIST’s “essentially in free fall” characterization is understood to describe the full visible collapse — the natural public reading, and the one load-bearing in popular science communication — then Bažant’s rebuttal and NIST’s claim are not demonstrably describing the same physical interval. Both statements are cited as components of a single official account. Neither specifies its own scope precisely enough to be checked against the other, let alone acknowledges the other’s existence.

6.11.2 Case Study: The Drift of β Between 2008 and 2017

A second, independent instance of the same pattern — a pivotal parameter revised between papers without a stated basis sufficient for a reader to assess the revision — appears in the resistance term itself, rather than in the kinematic bookkeeping examined above.

Bažant’s model decomposes total crushing resistance into four components, $F_c = F_b + F_s + F_a + F_e$ (buckling/fracture, comminution, air ejection, lateral ejection), with the buckling term further written $F_b =$

βF_p , where F_p is the standard three-hinge column-buckling force and β a reduction factor applied to it. [BAZANT-EPN](#)

↳ This β is Bažant’s own notation, distinct from both Němec’s mass-retention β (§4.3.2) and this document’s own resistance factor R (§4.10.3).

In the 2017 revision, Bažant’s own value for this parameter changes:

“the reduction factor β of the column resisting force was increased from 2/3 to 0.73 for normal-strength columns, and from 0.24 to 0.26 for high-strength columns” [BAZANT-EPN](#)

relative to the earlier analysis (Bažant et al. 2008, this document’s BAZANT-476). The stated basis for the revision is “recent information” — a phrase that does not, within the passage itself, name a source, a test, or a methodology that would let a reader independently evaluate whether the new values are better constrained than the old ones, or simply better fitted.

Two observations follow, echoing §6.11.1:

- the revision is real and disclosed — this is not a hidden change, and Bažant reports it openly as a correction to his own earlier figures. Crediting that transparency is appropriate.
- the revision is nonetheless under-specified in a specific way: both the 2008 point estimates and the 2017 revised estimates fall inside the same uncertainty range 2008 itself discloses ($\beta \in [0.5, 0.8]$ normal-strength, $[0.1, 0.3]$ high-strength) — so this is not a case of a value moving outside previously claimed bounds, but of the central estimate shifting within them, without the paper stating what specifically justified the new central value over the old one. This matters because β in 2008 was itself calibrated by fitting the model’s output against the same video and seismic records the model is later checked against (the paper is explicit about this: “[the fit] gives $\beta = 2/3$... reasonably matches the video and seismic records”), which is precisely the calibration-validation conflation examined in §6.2 and §6.4. A second, undocumented re-fit in 2017 repeats the same structure: adjust a free parameter within its own prior bounds until it matches, again, without an independent test.

This is the same structure identified throughout Chapter 6: a single-point calibration, revised without a disclosed method for the revision, standing in for the ensemble-level robustness check that would show whether the conclusion survives across the range the parameter has actually occupied across the literature’s own papers.

6.11.3 Case Study: The Redefinition of W_g Between 2002 and 2007

A third instance of the same pattern touches the demand-side term every arrest criterion in this framework is compared against: the gravitational-energy quantity itself.

Bažant and Zhou (2002) define it as the release of potential energy over *two* story-heights, matched to their specific plastic-hinge mechanism: “the total release of gravitational potential energy is $W_g = mg \cdot 2h \approx 2 \times 2.1 \text{ GN m} = 4.2 \text{ GN m}$.” This is the figure behind the paper’s headline result, $W_g/W_p \approx 8.4$.

Bažant and Verdure (2007) redefine the same symbol as “ $W_g = gm(z)u_f$ = loss of gravity when the upper part of the tower is moved down by distance u_f ,” where $u_f = (1-\lambda)h$ — the *compacted* displacement of a *single* story, strictly less than one full story-height once $\lambda > 0$, and evaluated at a general coordinate z rather than tied to the specific two-hinge geometry of the 2002 mechanism.

Both quantities are called W_g . Neither paper flags the change, states that the earlier definition has been superseded, or reconciles the two — the 2007 paper cites the 2002 result approvingly (“the tower was doomed once the top part of the tower dropped through the height of one story or even 0.5 m”) without noting that its own W_g no longer means what the number it is endorsing was computed from.

This does not undermine either individual calculation on its own terms — each is internally consistent with the mechanism it accompanies. What it shows is that W_g is not a fixed, transferable quantity across this literature the way its constant reuse across six papers might suggest; it is redefined to fit whichever collapse-front geometry each paper is modeling, silently. A reader citing “Bažant’s W_g ” without specifying which paper’s W_g is citing one of at least two materially different quantities.

6.11.4 Case Study: The Hinge-Count Assumption and Dynamic Buckling Mode Order

Bažant’s plastic-hinge energy dissipation calculation — the basis for his reported ratio $W_g/W_p \approx 8.4$ — rests on a stated bound: “the sum of the rotation angles $\theta_i \dots$ on one column line, $\sum \theta_i$, cannot exceed 2π . This upper-bound value, which is independent of the number of floors spanned by the buckle, is used in the present calculations since... it represents the most optimistic hypothesis, maximizing the plastic energy dissipation.” [BAZANT-405](#)

The claim that this bound is independent of hinge count carries a hidden premise: that a single hinge can absorb its full share of the total rotation without fracturing first. Real steel connections cannot absorb unbounded rotation before fracture; some per-hinge limit exists.

A separate critique of the same calculation argues the actual mechanism should involve more hinges than the single-hinge case the 8.4 figure is computed from, recalculating the ratio to 2.8 on that basis. [CHEREPANOV-2008](#) That specific recalculation is the source’s own and has not been independently reproduced here. The mechanism it invokes — “the greater is the dynamic load, the higher is the mode of buckling” — is independently established, apart from any claim about this collapse specifically: experimental and numerical work on columns struck axially at increasing impact velocity finds “the higher the impact velocity is, the shorter the buckled wavelength becomes.” [CUJ](#) A shorter buckled wavelength means more buckle segments — more hinges — over the same column length.

If per-hinge rotation capacity is fracture-limited, and impact velocities in the relevant range push toward shorter wavelengths and more hinges, then the achievable total rotation before fracture-limited failure exceeds what a single-hinge calculation credits, and the gap grows with hinge count. Bažant’s 2π bound, applied “independent of the number of floors spanned by the buckle,” implicitly treats hinge rotation capacity as unlimited by fracture. Whether that assumption is in fact the most optimistic one available — the specific claim made for it — is therefore itself conditional on a fracture-capacity premise the paper does not state or test.

This does not establish what the actual hinge count or achievable rotation was during the collapse; that would require destructive testing and debris analysis to a standard this document has already found wanting (Chapter 7). What it establishes is narrower and does not depend on that evidence: the claim that the hinge-count assumption is the most optimistic hypothesis available is itself an assumption, and independent, non-WTC literature on dynamic buckling mode order gives a specific, mechanistic reason to treat it as contestable rather than settled. This is a fourth instance of the pattern examined above — a quantity presented as maximally favorable to the “must collapse” conclusion resting, on inspection, on an unstated premise — arrived at from outside the Bažant corpus, pointing at the same structure §6.5.1 identifies from within it.

6.11.5 Case Study: The Uncorrected Misprint in the Original Paper’s Headline Impact Figure

Bažant and Zhou (2002) offer two separate estimates of the dynamic overload the falling upper section’s impact imposes on the structure below: a spring-model estimate (Eq. 1, $\approx 31\times$) and an elastic-wave estimate (Eq. 2, $\approx 64.5\times$), the latter stated to hold “only for a small fraction of a second at the moment of impact,” with the spring estimate said to become “appropriate” once the wave has propagated through the structure. Eq. 2 requires a cross-sectional area term A , defined in the original paper as “cross section area of building.”

Bažant and Verdure (2007), in a parenthetical aside unrelated to its main argument, states: “note a misprint in Eq. (2) of Bažant and Zhou 2002a: A should be the combined cross section area of all columns, which means that Eq. (1), rather than (2), is decisive.” [BAZANT-466](#)

This is a correction to the original paper’s own most widely cited numerical claim, made five years later by one of the same authors, in a single sentence, with no further discussion and no restated value. No subsequent paper in this corpus — 2008, 2011, 2016, 2017, or 2022 — restates, revisits, or references this correction; the $64.5\times$ figure it disavows continues to circulate in the wider literature, including in critical responses to it, without either side noting that the paper’s own follow-up work does not rely on it. A number withdrawn once, in passing, by its own author, and never mentioned again by anyone building on the paper that contains it, has functionally not been withdrawn at all for a reader working only from the original.

6.11.6 The Compute Defense

“...that’s in this day and age what computers do”

The exhibit commonly begins with the later literature’s invocations of simulation. It should begin here, on the record, before any of it existed: [EASTDAY](#)

New York authorities’ decision to ship the twin towers’ scrap to recyclers has raised the anger of victims’ families and some engineers who believe the massive girders should be further examined to help determine how the towers collapsed.

But New York Mayor Mike Bloomberg insisted there are better ways to study the tragedy of September 11.

“If you want to take a look at the construction methods and the design, that’s in this day and age what computers do,” said Bloomberg, a former engineering major. “Just looking at a piece of metal generally doesn’t tell you anything.”

Three features of this statement carry diagnostic weight.

First, chronology. The statement was published January 24, 2002 — 135 days after the event, three weeks into the speaker’s first term. At that date there existed no FEMA building-performance study, no NIST investigation, no finite-element model of the towers at any refinement level, calibrated or otherwise. The compute defense was therefore not a conclusion drawn from the study of the collapse; it was the premise on which the physical study of the collapse was foreclosed. What was traded away was the primary, unrepeatable record — steel whose failure modes, if examined, could never be un-examined — and what it was traded against was an instrument that did not yet exist, whose adequacy would thereafter be assessed by its own producers.

Second, the burden inversion. If “that’s in this day and age what computers do” is adequate to supersede metallurgical examination, then the computational analyses inherit, in full, the evidential burden vacated by the steel. What they cannot then be treated as is corroborated by the absence of physical findings, because that absence is the product of the very argument invoked to produce it. An analysis population exempted, by prior disposal of the cross-checking evidence, from confrontation with the artifact it purports to simulate is insulated in the precise sense diagnosed at §III-2.5.1 — except that here the insulation is not a by-product of the analysis but a precondition of it, arranged before the analysis began. The contemporaneous objection was stated on the record and overruled; the record disposed of is therefore an adjudicated loss, not an inadvertent one.

Third — and this is the finding — the statement’s later career inverts its own direction. In January 2002, the instruments were invoked to render the steel dispensable: nothing can be learned from a piece of metal, so its recycling forfeits nothing. In the later literature catalogued in this section, the same instruments are invoked to render conclusions unimpeachable in the opposite direction: initiation cannot be resolved more finely, and dissenting analysis cannot be adjudicated, because the steel is gone. The two positions are jointly untenable. If computation suffices, then its failure to yield finer conclusions indicts the analysis, not the absent artifact. If computation cannot suffice without the artifact, then the disposal was not evidentially neutral, and “that’s what computers do” was the burden of proof being destroyed rather than met. Either branch is costly; the only unavailable option is the one taken — the defense asserted at the point of destruction and withdrawn at the point of evaluation, drawn in whichever direction each moment’s argumentative need requires. The structure is that of the load-sharing substitution diagnosed at §6.5.1, performed at the level of the evidentiary base itself: what the instrument was invoked to establish (redundancy of the artifact) is silently exchanged for what it is later credited with establishing (conclusions in the absence of the artifact), with neither proposition stated in its own terms.

“Computationally Infeasible”

The claim that high-fidelity global progression modeling of the towers was computationally infeasible appears in two institutions’ retrospectives: NIST’s FAQ (“the propagation of the collapse was readily explained without the same complexity of modeling” — FAQ 10, 2011 revision) and the crush-down corpus (“a detailed numerical modeling of the entire collapse process ... would have been computationally infeasible” [BAZANT-625](#), 2022). The claim is auditable against the practice of its own authors, in four dated instances:

1. **June 2002.** Lu & Jiang [LU-JIANG](#) publish an LS-DYNA simulation of the WTC collapse with explicit fracture-strain parameter sweeps (0.5%, 1%, 5%), eighteen months after the events, obtaining arrest at ~100 m below the impact zone for the middle value. Whatever hardware this required, it existed in 2002, outside NIST.
2. **2005.** NIST declares the tower progression analysis outside the executed scope (NCSTAR 1, footnotes pp. xxxvii, 82), while its contractual state of the art included a complete global model of the towers (ANSYS initiation model; see NIST 2009 withholding finding [NIST-SAFETY](#)).
3. **2008.** NIST executes a full progressive-collapse simulation — for WTC 7. The computation declared unnecessary for 110 stories is performed for 47, by the same institution, three years later.
4. **2008 and 2012.** Isobe et al. execute element-by-element progressive-collapse simulations of the towers [ISOBE](#), obtaining collapse speeds below free fall under intact splices — i.e., the computation “shown unnecessary” in 2022 was performed, twice, within the decade it was defended.

The verb mood shifts over the sequence: 2005’s “would have been infeasible” (NIST) becomes 2008’s performed-anyway (WTC 7) becomes 2022’s “shown unnecessary” (Le & Bazant, 2022). Three grammatical

moods, one unchanged conclusion. The 2022 formulation is the strongest: it no longer even claims infeasibility, only irrelevance — cf. §1.6’s observation that burden language migrates when justification is retired but conclusion is not.

Consumer-hardware replications (discrete-element visualizations [KOSTACK](#)) are noted here only for the direction of the barrier, not as validation: a 2002 academic group, a 2008 federal team, and a 2025 hobbyist all cleared the 2005 declared boundary. Whatever the barrier was, it was not computation.

Chapter 7 — Evidentiary Gaps and Epistemic Asymmetry

7.1 Method as Evidence

Scientific inference does not rely only on what is measured. It also relies on **what is preserved, tested, and made available for replication**.

Where evidence is fragile, method is decisive.

Three questions are therefore legitimate:

- What physical evidence was preserved?
- What discriminating tests were performed?
- What opportunities for resolution were foregone?

They do not assign motive. They evaluate **epistemic hygiene**.

7.2 The evidence that no longer exists

Large fractions of:

- recovered structural steel,
- connection assemblies,
- fire-affected members,

were **recycled before full forensic cataloguing and testing**. [KNOWLES](#), [KOHN](#), [MANNING](#), [EASTDAY](#)

This limited later possibilities to:

- map failure modes member-by-member,
 - conduct systematic metallurgical testing,
 - perform comparative residue analysis,
 - physically reassemble critical regions.
-

7.3 Untested alternative signatures

Standard investigative protocols in structural failures and fire/explosion analysis normally include: [NFPA-921](#), [PETROSKI](#)

- residue testing where multiple mechanisms are conceivable,
- preservation of fracture surfaces,
- archiving of critical members,
- chain-of-custody documentation for later scrutiny.

In this case, testing regimes that could have **distinguished among mechanisms** were often not undertaken on the grounds that certain outcomes were already deemed unnecessary to consider.

This converts conclusion into premise:

When alternative mechanisms are excluded **before** discriminating tests, the result reflects assumption rather than test outcome.

7.4 Asymmetry in doubt

A recurrent pattern emerges:

- uncertainty counts **against** unapproved hypotheses,
- uncertainty counts **for** the accepted one.

Examples include:

- absence of residue → cited as absence of alternative mechanism while testing for residues was limited or foregone;
- non-reproduction → cited as lack of expertise while independent reproduction was not materially enabled;
- evidence loss → treated as administratively necessary while its epistemic cost was not acknowledged.

The asymmetry is not subtle:

doubt is applied asymmetrically.

One methodological approach is asked to meet **experimental standards under missing evidence**, another approach is allowed to rest on **plausibility and authority** despite the same evidentiary gaps.

That asymmetry is a methodological flaw. Ultimate correctness does not retroactively validate asymmetric standards.

7.5 Lost opportunities for closure

The following would have materially increased epistemic clarity:

- comprehensive preservation of structural steel,
- systematic destructive testing and microanalysis,
- standardized residue screening,
- public access to modeling inputs and ensembles,
- independent reproduction programs,
- explicit falsifiability criteria.

Most of these no longer exist as possibilities. They were foreclosed not by physics, but by procedural choices.

The result is not mystery. It is **avoidable opacity**.

7.6 What follows — and what does not

This chapter supports **one** conclusion only:

Where evidence is disposed of or untested, uncertainty increases, and claims of finality weaken.

The conclusion follows directly:

In the presence of evidentiary gaps created by investigative choices, strong claims of inevitability should be regarded as unearned.

Chapter 8 — A Positive Research Program

A constructive research program follows directly from the preceding analysis. The purpose is to make resolution possible. Historical uniqueness does not prevent validation; it shifts it toward component phenomena and adjacent reference classes. It aims to determine whether the observed collapses are:

- robust consequences of impact and fire, or
- outcomes that occupy a narrow, fine-tuned region of behavior.

If inevitability is correct, this program will find it. If it is not, this program will show the limits of current models.

The program contains four components.

8.1 Transparency and Reproducibility

Required steps:

- publication of model inputs, source code, and resistance curves
- assumption documentation
- publication of parameter provenance and sensitivities
- release of arrest statistics rather than only successful runs
- independent reproduction of key simulations by unaffiliated groups

Outcome:

- if results reproduce across groups → confidence increases
- if they do not → sensitivity and fragility must be explained

This is the minimum standard in any other mature quantitative field.

8.2 Ensemble Modeling Across Variance

Single “hero simulations” are non-diagnostic with respect to robustness.

What matters is **frequency** of the observed outcome.

Research target:

Distribute input uncertainties realistically and run large ensembles.

Variables include:

- damage distribution
- insulation loss uncertainty
- fire evolution variability
- connection strength distributions
- initial asymmetries and eccentricities

Measured quantities:

- probability of global collapse
- probability of arrest
- probability of asymmetric/tipping outcomes
- dispersion of collapse duration and symmetry metrics

Interpretation:

- high recurrence $\Rightarrow$ robustness
- low recurrence $\Rightarrow$ fine-tuning

8.3 Outcome-Space Mapping

The observed collapse corresponds to a location in multidimensional parameter space defined by duration, symmetry, completeness, and fragmentation.

The task is to **map neighborhoods** in that space.

Program goals:

- identify basins of attraction
- identify boundaries of arrest versus progression
- identify competing dynamical modes
- determine where the observed outcome lies in that topology

Key diagnostic question:

Is the observed collapse inside a wide basin, or on a narrow ridge?

8.4 Laboratory and Physical Modeling

Physical testing is expensive, but it is worth noting that part of this program already exists — just not aimed at the regime that matters most here.

Already available. Alternate-load-path testing — removing one or two columns from real or large-scale steel frames and measuring the dynamic response — is a mature, independently funded research program with decades of data (§6.7.1). It constrains dynamic amplification factors and connection rotation capacity for *local, static-to-moderately-dynamic* loss of support. This work should be used, not re-derived.

Not yet available. No comparable physical test exists for the specific regime the crush-down models depend on: resistance mobilized in *intact* stories struck by an *already-falling, multi-story accreting mass*. Candidates that would close this gap include:

- scaled connection failure tests under dynamic loading, extended to impact energies representative of multi-story accretion rather than single-column loss
- column buckling and fracture energy characterization at those energies
- stacked-mass progressive collapse analogs with variance — i.e., physical drop tests of an accreting mass onto instrumented, intact multi-story steel subassemblies, the direct analog of the crush-down R
- granular comminution and energy partitioning studies

Desired outcomes:

- independent validation of resistance curves used in models
- empirical arrest probabilities
- constraints on comminution energy budgets

The goal is not to recreate the towers.

The goal is to **constrain the parameter space in which models operate** — and, specifically, to replace the one load-bearing parameter that is currently supplied by assumption with one supplied by measurement.

8.5 Resolution Criteria

The program is explicitly falsifiable.

Resolution in favor of robustness

The inevitability claim would be strengthened if:

- ensemble models with realistic variance
- open inputs and independent reproduction
- empirically constrained resistance functions

show that:

- rapid, near-axial, non-arresting progression
- arises frequently and stably
- across broad parameter ranges

This would justify the term “**robust consequence.**”

Resolution in favor of fine-tuning

Conversely, inevitability would weaken if:

- outcomes require narrow tuning,
- collapse arrests under modest variance,
- symmetry requires constraints,
- parameter sensitivity is extreme.

That would justify the term “**critical regime near threshold.**”

8.6 What This Program Does

It does one thing:

articulate the experimental and modeling pathway by which the question of robustness versus fine-tuning can be settled to scientific standards.

8.7 A Precedent Already in the Record

Part of the ensemble program called for above has, in fact, already been run once. Lu and Jiang (2002) modeled the WTC collapse in LS-DYNA and varied a single parameter — the fracture plastic strain of the steel — across three values. At 0.5%, the simulation produced total chain collapse. At 1%, propagation arrested roughly 100 m below the impact zone. At 5%, damage remained local, with no chain collapse at all. [LU-JIANG](#)

This is a published, peer-adjacent demonstration that a plausible, narrow-range material-parameter change produces arrest rather than total collapse in a full simulation of this event — precisely the “sketch of a plausible-arrest scenario” this document’s argument (§I, §6.6) identifies as missing from the published record. It is not missing; it exists, from 2002, and does not appear to have been cited or engaged by the later literature this document otherwise tracks.

Closing

The refusal to pretend to know is not a destination.

It is the beginning of a **program of disciplined inquiry**.

The path forward is clear:

- transparency,
- ensembles,
- empirical constraints,
- outcome-space mapping.

If inevitability is true, this program will reveal it. If it is not, this program will reveal that too.

Either way, science wins.

Coda — On Epicycles of Inevitability

The central claim of this document is narrow:

- collapse occurred,
- collapse is physically possible,
- **inevitability has not been demonstrated.**

What is at issue here is **not what happened**, but **how we claim to know**.

The history of science is littered with confident systems built on:

- calibration mistaken for validation,
- theory-saving patches,
- appeals to consensus in lieu of experiment,
- destruction or neglect of contradictory evidence.

Their names are familiar: **epicycles, miasma, phlogiston** — not because this case is claimed to share their fate, but because they share a diagnosis: confident systems that mistook internal consistency for demonstrated truth. The analogy is about method, not about the outcome..

The lesson is not cynicism. It is discipline.

Claims of physical inevitability must earn their status through transparency, reproducibility, robustness across variance, and internal kinematic consistency — the requirement that stated mechanisms produce stated outcomes under their own premises.

Where replication is difficult, variability high, sensitivity strong, and evidence incomplete, closure is premature.

Refusing to pretend to know is not weakness. It is the core strength of scientific reasoning.

The correct position in such domains is neither credulity nor dogmatism, but disciplined openness:

| The standards by which science judges inevitability have not yet been met.

Until they are, inquiry remains legitimate.

And that is not a scandal.

It is science.

Appendix: Derivations and Method Notes

This appendix expands the arguments in the main document with explicit formulations, assumptions, and diagnostic implications. Its purpose is strictly methodological:

- to show how each claim cashes out mathematically or physically
- to separate assumptions from results
- to identify where robustness has and has not been demonstrated

No additional hypotheses are introduced. Where data are uncertain or contested, this is stated directly.

Concordance with cited sources

This document's parameters (Γ , R , ξ_{out}) are its own synthesis and are not identical to any single source's notation. Correspondences below are noted at first use of each; “—” means the source does not use a directly equivalent named symbol in the material quoted or cited in this document.

This document	Meaning	Bažant & Zhou 2002	Bažant & Verdure 2007	Bažant et al. 2008	Schneider 2017
W_g (Bažant — unstable across his own papers, see	gravitational energy released	$mg \cdot 2h$, two story- heights (2002)	$gm(z)(1-\lambda)h$, compacted single	—	—

This document	Meaning	Bažant & Zhou 2002	Bažant & Verdure 2007	Bažant et al. 2008	Schneider 2017
§6.11.3)			story (2007)		
W_p (Bažant)	plastic dissipation capacity, per story	matched to 2002's two-story mechanism	matched to 2007's compacted-story mechanism	—	arrest threshold
E_g (this document)	gravitational energy released, per story — this document's own quantity, distinct from either Bažant definition above	—	—	—	—
E_d (this document)	plastic dissipation capacity, per story — this document's own quantity	—	—	—	—
—	floor load capacity	—	F_0	—	—
—	criterion: $K < W_c$ triggers/prevents collapse	—	K, W_c	—	re-derived, shown non-optimal
compaction ratio (§4.4: λ)	fraction of story height lost to compaction	—	$\lambda = (1 - \kappa_{out})\mu/\mu_c$ (introduced)	value ≈ 0.20 implied by Fig. 8 reconstruction (§6.11)	cited, same range
β (Němec)	mass fraction hitting the structure below, per collision (§4.3.2)	—	—	—	—
κ_{out} (Bažant)	mass shedding fraction — not the same quantity as this document's ξ_{out}	—	κ_{out} (mass fraction, inside $\lambda = (1 - \kappa_{out})\mu/\mu_c$)	κ_{out} (retained)	—
Γ	this document's severity ratio, $\Gamma \equiv E_g/E_d$	(no named ratio; comparison stated in prose)	(no named ratio)	—	—
R	this document's resistance-degradation factor	—	—	—	—
ξ_{out}	this document's outside-front	—	—	—	—

This document	Meaning	Bažant & Zhou 2002	Bažant & Verdure 2007	Bažant et al. 2008	Schneider 2017
	dissipation fraction				
F_c (§4.9–§4.10) = F_c^{iso} (§6.5.1)	mean crush resistance of one story, entering W_p/R /the crush- down criterion — isolated three-hinge column capacity	—	—	—	—
F_c^{sys} (this document, §6.5.1)	system (redundant, connected-frame) resistance of a story — governs Γ_{static} , never enters the crush-down criterion	—	—	—	—
Γ_{static} (this document, §6.5.1)	$F_c^{\text{sys}} > mg$, the as- built stability condition	—	—	—	—
$\Gamma_{\text{effective}}$ (this document, §6.5.1)	$W_g > W_p$, i.e. $mg >$ $R \cdot F_c^{\text{iso}}$, the condition actually evaluated during collapse	—	—	—	—

Notation

For clarity, a small number of symbols are used consistently throughout this appendix:

- **O** – observed joint outcome vector of the collapse
- **T** – total fraction of structural loss
- **S** – axial symmetry metric (normalized centroid deviation)
- **V** – descent velocity normalized to free-fall
- **C** – comminution metric (inverse characteristic fragment size)
- **m, g, h** – mass, gravitational acceleration, characteristic drop height
- **M_n** – accreted falling mass after *n* collisions (*M*(*x*) in continuum form, as a function of drop distance *x*)
- **v_n, v_n'** – velocity immediately before and immediately after the (*n*+1)-th inelastic collision
- **a** – instantaneous or asymptotic acceleration of the falling mass (distinct from *V*, which is descent velocity normalized to free-fall)
- **N** – number of floors in the structure
- **E_g** – gravitational potential energy
- **P_s** – static capacity of a structural element or story
- **P_d** – dynamic demand under impact or progression loading

- F_{mob} — the net resisting force transmitted by the structure below the falling mass at a given instant, inferred via d’Alembert’s principle from observed acceleration (§4.3.3); distinct from R (§4.10.3), the model’s own dimensionless resistance-degradation factor

Numerical values are **order-of-magnitude indicators**. The argument concerns **structure**, not precision.

Note 1 — The Replication Problem (Physical Difficulty)

The joint outcome is defined from observable quantities — total descent time, acceleration profile, symmetry of descent, extent of structural loss, degree of comminution — approximated by a vector $O = (T, S, V, C)$: fraction of structural loss, axial symmetry, descent rate normalized to free fall, and an inverse fragment-size comminution metric. The diagnostic feature is not any single component but the covariance of high T , S , V , and C together; many structural failure modes produce one or two of these, and the conjunction is rare.

A minimal replication test formalizes this directly: construct a multi-story gravity-loaded structure, remove or isolate the top fraction, impose downward impact, and record the fraction of trials reproducing this joint outcome, $P_{\text{obs}} = N_{\text{observed-like}} / N_{\text{trials}}$, across realistic variance. This is precisely the quantity the ensemble in §4.4 estimates for the discrete-floor model; robust inevitability requires P_{obs} high across variance, not merely nonzero at a single calibrated point.

↳ *Difficulty is empirical; it does not decide mechanism, but it constrains claims of inevitability. A clearly falsifiable research program.*

Standard Error Sources

Three sources of measurement uncertainty affect confidence bounds: (i) video kinematic reconstruction (frame rate, occlusion), (ii) model idealization (dimensionality reduction, boundary conditions), and (iii) structural heterogeneity (material variance, undocumented damage). Claims of inevitability must demonstrate robustness across these uncertainties.

Note 2 — Sequential Failure, Reserve Capacity, and Fine-Tuning

2.1 The Sequential Failure Constraint

Required conditions per floor:

1. sufficient static capacity with safety margin
2. insufficient dynamic capacity under impact

Let

- P_s — static capacity
- P_d — dynamic demand

Then sequential progression requires:

$$P_s > P_{\text{service}} \text{ AND } P_d > P_s$$

This is a mutual constraint. Deviations in either direction lead to:

- premature global failure
- arrest

- asymmetric collapse

↳ *Necessary condition for idealized progression models.*

2.2 Default System Response Without Tuning

Structural frames exhibit:

- redundancy
- load redistribution
- local hinge formation
- disorder in strength and damage

Network and fiber-bundle analogs show: [NEWMAN, STROGATZ](#)

- local load sharing → localization and arrest
- equal load sharing → global progression

Therefore, any model that yields smooth progression must justify the **coordinating mechanism** that converts a heterogeneous real frame into an effectively equal-load-sharing system.

↳ *Qualitative but widely supported in progressive collapse mechanics.*

2.3 Vertical Reserve Capacity and Basins of Attraction

Given the axial reserve capacity established in §3.3 [KRAUTHAMMER](#), the natural basin of attraction for a damaged tall structure includes partial collapse, tilt and rotation, load redistribution, and arrest. Rapid, symmetric axial progression occupies a narrower basin requiring suppression of all of these stabilizing modes across many stories simultaneously.

↳ *Design-logic inference. Robustness requires demonstration.*

Note 3 — Naive Upper Bounds

3.1 Function of Naive Models

Zeroth-order models provide **conservative ceilings** on what gravity alone can achieve. If an event exceeds such ceilings, added mechanisms must be identified and justified.

3.2 Inelastic Momentum Exchange

Single-step inelastic collision, in the notation used throughout this section (§3.4 below generalizes this to a full accretion sequence):

$$\mathbf{v}_n' = [M_n / (M_n + m)] \cdot \mathbf{v}_n$$

where M_n is the falling mass before the collision, m is the mass of the floor engaged, v_n is the velocity immediately before impact, and v_n' is the velocity immediately after. Arrest occurs when dissipative demand exceeds kinetic energy.

3.3 Breakdown of Naive Models

When naive models cease to be predictive, the system is in a **critical regime** requiring continuum treatment:

- propagating crushing fronts
- dynamic fracture
- compaction waves

The requirement then becomes:

┆ demonstrate entry into a critical regime and robustness within it.

3.4 Derivation of the Naive Collision Ceiling (§4.3.2)

The $g/3$ acceleration ceiling and the one-third energy-dissipation figure cited in §4.3.2 both follow from the same elementary setup: a sequence of perfectly inelastic collisions between an accreting falling mass and a chain of equal, stationary floor masses, under gravity, with no resistance term.

Setup. Let the falling mass after n collisions be M_n , with each floor contributing equal mass m . Between collisions, the mass falls freely through one story height h , gaining velocity according to standard kinematics. At each collision, momentum is conserved:

$$M_n v_n = (M_n + m) v_n'$$

where v_n is the velocity immediately before the $(n+1)$ -th collision and v_n' is the velocity immediately after. Kinetic energy is not conserved in this step — the difference is dissipated as collision loss, which is the source of the one-third figure below.

Steady-state acceleration bound. In the continuum limit (many stories, m small relative to M_n), the discrete recursion above can be treated as a continuous equation of motion for the accreting mass $M(x)$ at position x :

$$d/dx [M(x)^2 v(x)^2] = 2 M(x)^2 g$$

Solving for the terminal (large- x) behavior of this equation — the regime in which the mass has accreted enough that early conditions no longer matter — yields a constant asymptotic acceleration of

$$a = g/3$$

independent of the specific mass-accretion profile, provided mass is added continuously and each addition is perfectly inelastic. This is the result reported by Némec et al.; structurally it is the standard “steady accretion under gravity with inelastic capture” form. It is the same equation as the classic falling-chain problem — a uniform chain drawn by its own weight from a stationary heap through a hole at the table edge — for which the pair of values $g/3$ (nonconservative, inelastic link-by-link transfer) and $g/2$ (conservative transfer) is independently documented, with the association between value and transfer regime depending on the configuration, and with the $g/3$ asymptote confirmed experimentally in a mechanical apparatus unrelated to this subject ($a/g = 0.328 \pm 0.004$). [WONG-CAYLEY](#) The equivalent formalism appears in the variable-mass-systems literature as the “inverse rocket problem.” [MUNGAN](#)

↳ *A related but distinct problem — a chain already in free fall striking a stationary plate, rather than a body accreting mass as it falls — illustrates that idealized momentum-based ceilings are not always conservative once real material behavior is measured. Classical theory for that problem predicts a reaction force of three times the static weight of the deposited material at any instant; measured experiments found the chain falling faster than free fall and generating forces up to roughly six times the static weight, from a tension effect at*

the landing point absent from the idealized model. [HAMM-GEMINARD](#) That is not the accretion problem analyzed above — the $g/3$ ceiling has been separately confirmed experimentally on its own terms — but it is a documented instance of a real system exceeding an idealized momentum bound once material-specific effects are included, noted here for completeness rather than as a qualification of the $g/3$ result.

Energy-dissipation form of the same result. The same physics can be read off in energy terms rather than acceleration terms. At each inelastic collision, the fraction of kinetic energy lost is

$$\Delta KE / KE = m / (M_n + m)$$

Summed over a full-height collapse with many equal-mass floors, this collisional loss alone — before any column resistance, fracture, or comminution energy is added — removes an asymptotically constant fraction of the gravitational energy being released. In the energy domain the same differentiated equation of motion (with $ma = mg - \mu v^2$, $d(KE)/dt = mgv - \frac{1}{2}\mu v^3$) gives the instantaneous dissipated fraction as $\frac{1}{2}\mu v^3/mgv = v^2/(2yg) = f/(2g)$, which at the stable fixed point $f = 2g/3^*$ is exactly one-third — the same conservation law read out in the energy domain, not an independent confirmation. The one-third is an asymptotic value, reached together with the $g/3$ asymptote: early in the descent f is small and the fraction is lower, so the exact total over a finite descent with a sub-free-fall initiation condition ($\alpha < 1$) falls slightly below one-third, of order 0.30–0.32 depending on initiation. “Approximately one-third” is fair for the bulk of the descent; an Overview citing the figure should not present it as an exact finite-descent total.

This is the result reported by Szuladziński: not a separate finding, but the energy-domain expression of the same conservation law that produces the $g/3$ acceleration ceiling above.

Why the two figures agree. $g/3$ (acceleration) and one-third (energy fraction dissipated) are two readouts of one conservation law, not two independent confirmations. A model that conserves momentum through perfectly inelastic, resistance-free accretion cannot exceed either bound; violating one implies violating the other, since both are derived from the same momentum-conservation step applied to the same collision sequence. Reporting both together, as §4.3.2 does, is a consistency check on the arithmetic — not two separate pieces of evidence.

Numerical instantiation (Kosciuk & Robbin; Hoffman). Setting $g = 9.81 \text{ m/s}^2$ (or 32 ft/s^2) and integrating the discrete recursion directly — rather than taking the continuum limit above — over $N = 110$ floors of height $h \approx 3.7 \text{ m}$ (12.4 ft) reproduces the same regime numerically: a minimum descent time of $\approx 9.2 \text{ s}$ at the (physically unreachable) pure free-fall limit, rising to $\approx 15.8 \text{ s}$ once floor-by-floor inelastic collision losses are included with zero additional structural resistance. Both figures are consistent with, and numerically bounded by, the closed-form $g/3$ result above: the asymptotic acceleration envelope alone gives $t = \sqrt{6y/g} \approx 16.0 \text{ s}$ for the full 417 m descent, with the directly integrated 15.8 s sitting slightly below it for the expected reason that the early phase runs above $g/3$.

↳ This derivation establishes a ceiling under the most collapse-favorable assumptions available (no column resistance, no fracture energy, no lateral mass loss). It does not model the actual collapse, and it does not by itself indicate what mechanism, if any, is required to explain observed kinematics at or beyond this ceiling.

Note 5 — A Bounded Reconstruction: The Average Column Implied by W_p

5.1 The energy-equivalent average, not a static capacity

Bažant’s own formulation defines crush resistance as an average over a load-displacement curve: $F_c = (1/h)\int_0^h F(u) du$ (§4.9). Dividing W_p by story height and column count reconstructs the average of an already-

varying, already-degrading resistance — not a claim that any column sustained a fixed force throughout crushing.

5.2 The arithmetic

Using this document’s own figures — $W_p \approx 0.5 \text{ GN}\cdot\text{m}$ (§6.11.3), $h \approx 3.7 \text{ m}$, $N = 287$ columns per floor — Bažant’s own relation gives:

$$F_c = W_p/h \approx 135 \text{ MN}, F_{c,\text{avg}} \approx 470 \text{ kN/column}$$

And via his own hinge-rotation assumption, $W_p \approx N \cdot M_p \cdot 2\pi$ ($\Sigma\theta_i = 2\pi$, §6.11.4):

$$M_{p,\text{avg}} \approx W_p/(287 \cdot 2\pi) \approx 277 \text{ kN}\cdot\text{m}$$

5.3 Checking against a verified population, not a single column

Bažant’s own paper specifies the perimeter column geometry at the aircraft impact level: “almost square, with the width of 356 mm... at the aircraft impact level the web was 10 mm thick.” [BAZANT-476](#) For this geometry, $Z_p \approx 1.80 \times 10^6 \text{ mm}^3$.

NIST’s own steel-specification report gives the complete, verified grade distribution for perimeter columns across the fire-affected floors (WTC 1, floors 92–100): 978 documented column instances across twelve specified grades, from 45 to 100 ksi. [NCSTAR-1-3A](#) The population-weighted mean yield strength is **64.2 ksi (443 MPa)**.

Applying this verified population average to Bažant’s own stated impact-level geometry gives:

$$M_p \approx 443 \text{ MPa} \times 1.80 \times 10^6 \text{ mm}^3 \approx 796 \text{ kN}\cdot\text{m}$$

a factor of roughly **2.9×** above the 277–280 kN·m average implied by the 0.5/0.51 GJ figures.

5.4 Pivot to the actual Floor-96 NIST column inventory

The 10 mm thickness is the impact-level figure. The actual Floor-96 perimeter schedule is dominated by type 120 (6.4 mm plates) with smaller numbers of types 121–123 (up to 11.1 mm). [NCSTAR 1-5G Tables B-1 & B-3](#) The resulting weighted-average plastic modulus is only $\approx 1.30 \times 10^6 \text{ mm}^3$.

Using the same population-weighted 64.2 ksi grade, the ideal perimeter contribution is $\approx 0.87 \text{ GJ}$. Adding the complete 47-member core schedule for Floor 96, ⁵(#NCSTAR-1-5G) evaluated on the weak axis with the predominantly 36–42 ksi grades, contributes a further $\approx 0.22 \text{ GJ}$. The total ideal plastic-hinge energy for the cold 96th story is therefore:

$$W_{\text{ideal}} \approx 1.09 \text{ GJ}.$$

5.5 Bažant’s own answer and the quantitative test

Bažant’s model has a stated response to exactly this gap: the reduction factor β is steel-grade dependent, “reasonably expected to lie within the range [0.5, 0.8] for normal structural steel... but in the range [0.1, 0.3] for the high-strength steel,” on the stated grounds that “high-strength steel has much lower ductility,” fracturing earlier and dissipating comparable energy despite higher initial capacity. [BAZANT-476](#) Matching the 0.51 GJ reference to the NIST-derived ideal energy of 1.09 GJ requires:

5 NCSTAR 1-5G Table B-6

$$\beta \approx 0.47\text{--}0.58$$

(depending on whether the compaction factor $1-\lambda \approx 0.8$ is applied). This interval lies inside the range Bažant himself adopted for normal-strength steel.

Final Note

The appendix is not advocacy. It is method. It keeps a clean distinction between:

- what is measured
- what is assumed
- what is derived
- what remains to be shown

That distinction is the central object of this document.

Overview

Claim	Basis Cited	What Was Demonstrated	What Was Not Demonstrated
Progressive collapse was inevitable once initiation began	Bažant 2002–2017; NCSTAR 1 “doomed”/“global collapse ensued”	Calibrated models showing plausibility for selected parameter sets	Ensemble robustness, basin size, arrest frequency under variance
Symmetry of descent was natural	Tube/core system + gravity; visual observation	Post-hoc narrative coherence	Mechanism for suppression of asymmetry, tilt, and partial collapse
Near-free-fall descent compatible with gravity-driven collapse	Video measurements; “essentially in free fall” (NCSTAR 1; FAQ 31)	Kinematic description, unspecific in scope	Reconciliation with the $g/3$ accretion ceiling: $\sim 1.8\text{--}2.3\times$ discrepancy using NIST’s own 11 s / 9 s figures; $\sim 3\times$ only under a sustained $a \approx g$ reading of “essentially in free fall,” whose scope NIST does not state
Fire + impact exhausted reserve capacity	NIST fire simulations and structural analysis	Initiation plausibility	Systematic testing of alternative damage/insulation scenarios; partial-failure regimes
Modeling captures progression mechanics adequately	Closed-source FE models; idealized 1-D crushing models	Case-calibrated trajectories	Independent reproduction, open inputs, sensitivity mapping, out-of-sample validation
Alternative-mechanism adjudication	NIST FAQ 28–29; Request-for-Correction response	No positive findings for explosives in the tested record; no corroborating evidence <i>for</i>	Systematic residue screening (explicitly not performed); alternative-hypothesis modeling at equal effort.

... It does not make any difference how beautiful your guess is, it does not make any difference how smart you are, who made the guess, or what his name is — if it disagrees with experiment, it is wrong.

— [Richard Feynman](#)

Chronology of Closure

(The Refusal to Pretend to Know: Pt. III)

Preface: Purpose and Limits

This document examines how explanations of the collapse of the World Trade Center Twin Towers developed, changed, and were socially stabilized over time. Rather than seeking to prove what happened, it asks a more foundational question:

How did explanatory closure develop—through what balance of methodological demonstration, practical constraint, and professional convergence?

The focus is chronological and procedural. We track:

- shifts in technical explanation
- expansion and contraction of uncertainty claims
- treatment of disagreement within scientific and professional communities
- separation between what was modeled and what was asserted
- how public rhetoric converged faster than technical understanding

Analytic Scope:

This analysis is restricted to procedural chronology of explanatory development, symmetry of hypothesis testing, consistency of evidentiary standards, temporal relationship between demonstration and closure, and discourse mechanisms handling dissent. Event reconstruction, mechanism adjudication, and attribution of intent lie outside this analysis.

What it does evaluate is **process**:

- whether competing hypotheses were tested symmetrically
- whether evidentiary standards were applied consistently
- whether closure followed demonstration or preceded it
- whether dissent was addressed methodologically or socially

The inquiry matters because closure is never merely academic. When a question as basic as structural collapse behavior is declared settled without transparent resolution of its open problems, the cost is not only technical. The cost is **erosion of public trust in scientific method itself**.

Accordingly, this document should be read alongside *Epistemic Limits* and the *Dialectic Inventory*. Together, these documents establish that closure in science must arise from method, not decree.

Chapter 1 — Chronology of Explanatory Claims

This section summarizes the evolution of explanatory claims regarding the collapse of the World Trade Center Twin Towers. It focuses on how interpretations were offered, revised, and consolidated over time.

1. Pre-Event Professional Expectations

Before September 11, 2001, professional expectations within structural engineering generally treated large high-rise steel structures as robust against global failure from localized damage.

- Statements following the 1993 bombing described the towers as capable of absorbing aircraft impact loads without global collapse.
- Design engineers noted large safety margins, extensive redundancy, and the high capacity of the framed-tube system.
- In early 2001, interviews characterized an aircraft strike as causing localized damage analogous to perforation of a mesh, without expectation of total failure.

These pre-event expectations formed the baseline from which later surprise was measured.

2. Immediate Characterizations on September 11

During and shortly after the collapses, rapid explanations were offered in real time. These included:

- assertions that intense fires caused progressive structural weakening
- suggestions of “pancake-type” collapse mechanisms
- references to prior impact damage as an initiating factor

These early statements were provisional. They shaped initial public framing despite being formed without detailed structural information.

3. First Days: Competing Hypotheses

In the week following the event, multiple explanatory narratives circulated simultaneously:

- **steel melting due to jet-fuel fires**
- **truss-sagging leading to inward bowing of perimeter columns**
- **progressive floor collapse sequences**
- **inevitability claims based on gravitational progression models**

At this stage, explanations differed not only in mechanism but in emphasis: some focused on temperature thresholds, some on redistribution of loads, and others on dynamic progression after initiation. No single narrative dominated.

4. Institutional Investigations

The institutional investigation period formalized the inquiry process.

- Initial FEMA findings summarized observable damage patterns and explored multiple possible mechanisms.
- NIST later initiated a multi-year investigation emphasizing aircraft impact, fire progression, structural response, and collapse initiation.
- The stated scope explicitly prioritized the sequence up to initiation; post-initiation propagation was generally treated as inevitable once global instability began.

The gradual movement from open hypotheses to a dominant explanatory narrative occurred over this phase.

5. Post-Investigation Consolidation

Following release of the final reports:

- public communications increasingly presented “impact + fire + gravity” as a settled explanation
- much earlier terminology (e.g., steel melting, pancake collapse) was deemphasized or abandoned
- later analytical papers framed further debate as unnecessary or superfluous

Over time, the multiplicity of early hypotheses gave way to a single dominant narrative structure. That narrative did not reflect the diversity of initial viewpoints.

Chapter 2 — Narrative Closure and Treatment of Dissent

1. From Tentative Statements to Certainty Claims

Early statements were conditional, qualified, and framed as provisional interpretations. Over time, similar classes of claims were increasingly presented as settled, with less explicit emphasis on uncertainties, contested assumptions, or model dependence.

The published record shows a relative increase in rhetorical certainty, even though the underlying mechanical models retained substantial idealizations and parameter uncertainties.

It is not always clear from the literature whether this reflects changes in the authors’ internal confidence, external communication demands, or genuine tightening of the models.

2. Changing Stories Without Retrospective Accounting

Throughout the chronology, explanations changed in material ways:

- from “steel melting” to “loss of strength without melting”
- from “pancaking floors” to “column failure driven by inward bowing”
- from “possible collapse” to “inevitable collapse after initiation”

These changes occurred without systematic public documentation of which earlier claims were withdrawn, which were retained, and why revisions were made. The record therefore displays evolution without clear epistemic bookkeeping.

3 The Domino Metaphor and Its Limits

Analogies to “domino effects” or “house-of-cards” failure are common in public-facing explanations, but they rarely survive into detailed mechanical treatments, where strong uniformity requirements and controlled boundary conditions become explicit. [HEVESHS](#)

This suggests a gap between the intuitive metaphors used for reassurance and the more complex mechanics required when collapse progression is modeled quantitatively.

4. Consensus Claims and the Appearance of Uniformity

Over time, media and institutional communication increasingly used formulations such as:

- “experts agree that...”
- “it is now understood that...”

even where the technical literature exhibited:

- multiple competing models
- unresolved quantitative disagreements
- continued debate over assumptions and parameter choices

This produced an appearance of consensus that exceeded the documented level of expert agreement on details.

5 Treatment of Skepticism and Methodological Critique

Method-focused critiques — those questioning modeling assumptions, calibration practices, or the scope of investigations — were often reframed as:

- challenges to institutional authority
- endorsement of specific alternative hypotheses
- “conspiracy” positions

Rather than being addressed within normal scientific discourse, many were moved into social or political categories. The net result was reduced space for ordinary methodological scrutiny.

The treatment of competing hypothesis classes has not only shaped conclusions, but also the standards applied to them. Two recurrent reasoning patterns deserve explicit attention.

5.1 Epistemic Drift Through Contrast

In the absence of detailed, publicly testable mechanical demonstrations, an informational vacuum formed. As highly speculative alternatives (directed-energy devices, mini-nuclear charges, “exotic” demolitions) filled the vacuum, their weaknesses — real or perceived — sometimes functioned rhetorically as a contrast class: the implausibility of these proposals was invoked as indirect support for the official narrative, even where mechanical questions about the baseline model remained underdetermined.

In such cases, the weakness of alternatives substituted, in practice, for a fully demonstrated strength of the primary hypothesis.

5.2 A Category Error in Burden-Shifting

A parallel shift occurred in burden-shifting. Mechanical objections—regarding synchronization of failure, comminution, near-axial descent, and the suppression of lateral modes—were increasingly met with sociological answers rather than mechanical ones: too many people would have to be involved; secrecy would be impossible; logistics would be prohibitive. [BAZANT-EPN](#) This is a category error. In a mechanical problem, the burden is mechanical: quantified forces, constraints, energy dissipation, load paths, and robustness under variability. Redirecting the discussion to plausibility of conspiracy does not resolve questions of structural sufficiency. In its most compressed form this inversion yielded the argument that because enormous quantities of energetic material would be required, this is evidence that none were required. The extremity of an alternative was taken as confirmation of the primary hypothesis without the primary hypothesis being demonstrated. Rhetorical closure was achieved; mechanical closure was not.

5.3 The Europhysics Incident: Institutional Response to Published Critique

Before examining the institutional response, the source itself should be characterized plainly. The article’s authors are, by their own stated bios, Steven Jones (physicist), Robert Korol (civil engineer), Anthony Szamboti (mechanical engineer, publishing chiefly in the *Journal of 9/11 Studies*), and Ted Walter, identified in the article as director of strategy and development for Architects & Engineers for 9/11 Truth (AE911Truth). The article’s own stated conclusion is that “the evidence points overwhelmingly to the conclusion that all three buildings were destroyed by controlled demolition” — a conclusion outside this document’s stated scope (§I). This document takes no position on it, and cites the episode for a narrower question: whether the article’s specific mechanical and energy-based claims were engaged on their merits or answered in social and procedural terms (§5.1–5.2 above). That the authors held, and disclosed, a firm prior conclusion does not settle whether their specific calculations were correct — and it does not excuse a response addressed to affiliation and procedure rather than to the calculations. Neither fact is being used here to prejudge the other.

The following reconstructs the institutional response to a published critique, not to adjudicate the critique’s mechanical claims, but to examine whether those claims were engaged, restated, or bypassed at the institutional level.

In 2016, *Europhysics News* published an invited feature article [JONES](#) examining energy budgets, descent rates, and symmetry in the WTC collapses. The institutional responses that followed compress into a single episode several patterns analyzed throughout this chapter.

NIST’s Position

NIST declined engagement: [EPN](#)

“Since our study...ended in 2008, there has been no new evidence presented that would change our findings and conclusions, and therefore, **nothing new that we can contribute to the discussion.**”

The response restated conclusions, directed readers to existing FAQs, and cited downstream code adoption as upstream validation:

“This is an **extraordinarily rapid pace** in the code making and approval process—a **solid affirmation** by the ICC that the work done by the NIST WTC investigation team was **of the highest quality**.”

Observation: Consequence is offered in place of engagement: downstream code adoption is cited as validating the investigation’s quality, while the investigation’s quality is what downstream adoption was supposed to test. The specific mechanical critiques raised in the article are not addressed in either direction.

Editorial Reaction

EPN acknowledged the article “contained some speculations and had a rather controversial conclusion” and expressed being “shocked that the published article is being used to support conspiracy theories.”

Despite justifying publication as an attempt to “trigger an open discussion leading to an undisputable truth,” the editors implemented procedural changes:

“In future, prospective authors will be asked to provide an abstract...to allow the editors to **better assess the content** of the invited articles.”

Observation: Article invited → published to trigger discussion → discussion occurred → journal expressed shock → procedures changed to prevent recurrence.

The sequence is the observation: an article was invited, published as an attempt to trigger discussion, and — once discussion occurred — met with expressions of surprise and a policy change designed to prevent similar publication in the future. Whatever the editors’ intentions, the practical effect was to treat open discussion as acceptable only up to the point of actually occurring.

Asymmetric Standards

EPN disclosed: “EPN does not have a formal review/rejection policy for invited contributions.”

Context:

- NIST’s reports: internal review only
- Bažant’s initial papers: rapid publication (first draft September 13, 2001)
- Europhysics critique: invited publication

Outcome:

- Official explanations: authoritative
- Published critique: procedural failure requiring policy correction

Neither NIST nor EPN engaged the article’s mechanical arguments: energy partitioning, arrest conditions, symmetry mechanisms, comminution calculations.

Both responses operated in social register: institutional authority, conspiracy contamination, procedural propriety.

The “New Evidence” Frame

NIST’s statement that “no new evidence” exists to change findings treats methodological gaps as resolved questions.

If “evidence” means physical samples or observational data, the statement is accurate but uninformative—the evidence was singular and largely destroyed.

If “evidence” includes ensemble modeling, sensitivity analysis, arrest probabilities, transparent inputs, or independent replication, the statement inverts burden: these were never provided, so their absence cannot constitute lack of “new” evidence.

The framing converts **unfulfilled requirements** into **settled matters**.

Pattern Integration

The incident instantiates mechanisms from 5.1 and 5.2:

Epistemic drift through contrast: Publication → conspiracy theory association → contamination → article discredited → official narrative strengthened by contrast, without being defended on merits.

Category error: Mechanical questions (energy, symmetry, arrest) → sociological responses (conspiracy implausibility) → mechanical questions unaddressed → rhetorical closure without mechanical resolution.

Diagnostic Value

The episode demonstrates:

1. Social framing preempts technical engagement

Neither response addressed specific calculations. Both invoked authority (code adoption), contamination (conspiracy theories), and procedure (editorial policy).

2. Discussion invited only under conditional closure

EPN trajectory: invite article → publish to trigger discussion → discussion occurs → express shock → prevent recurrence.

Mechanism revealed: “open discussion” was acceptable only if it arrived at predetermined conclusions.

3. The contamination catch-22

- Critique published → “conspiracy theory” association
- Critique unpublished → ignored as fringe
- Methodological questions → social stigma
- Social stigma → justifies non-engagement
- Non-engagement → feeds speculation
- Speculation → justifies closure

If accurate, this reading would make the pattern self-reinforcing: closure creates the very conditions later cited to justify closure. Whether that dynamic generalizes beyond this one incident is a separate question this document does not settle.

A second, independently documented instance of the same structure exists in the record. A Discussion paper submitted specifically to correct stated errors in a published Bažant-authored paper was rejected by the journal’s editors as “out of scope” — a characterization its authors publicly disputed on the grounds that error correction cannot, by the nature of a Discussion paper, be out of a journal’s scope. [WYNDHAM](#) Whether that specific rejection was correctly decided is not adjudicated here; what is notable is the shape of the event,

matching the pattern already examined above — a correction submitted, procedural grounds cited for non-engagement, no record of the substantive correction being addressed elsewhere.

Implication

If peer-reviewed methodological critique triggers institutional retreat, procedural barriers, and contamination claims rather than technical rebuttal, normal error-correction mechanisms are compromised.

The incident does not establish that official explanations are incorrect. It shows one case in which methodological critique was answered in social terms — authority, contamination, procedure — rather than in the mechanical terms it was framed in. Generalizing beyond this single episode would require more cases than are examined here.

When critique becomes impossible without contamination, and contamination justifies non-engagement, the result is not scientific closure but **epistemic quarantine**.

5.4 A Case Study in Technical Discourse

The Europhysics incident demonstrates these patterns at the institutional level. The peer-reviewed technical literature exhibits similar dynamics at the level of individual researcher exchanges. This section examines representative exchanges not to adjudicate their technical merits, but to observe **patterns of discourse** that shaped the closure process.

The Critics

Between 2002 and 2013, several credentialed engineers and scientists published technical critiques of the Bažant collapse models:

- **Gregory Szuladziński** — structural engineer; temporal analysis showing momentum-exchange constraints
- **Robert Korol & K.S. Sivakumaran** — structural engineering professors; energy absorption studies
- **Anthony Szamboti** — mechanical engineer; detailed mathematical critiques
- **Crockett Grabbe** — PhD physicist; analysis of motion characteristics
- **Gennady Cherepanov** — Academician, Russian Academy of Sciences; fracture mechanics analysis
- **James Gourley** — chemical engineer and attorney; methodological questions

These were not anonymous internet commentators. They were published professionals raising specific technical objections in peer-reviewed venues or formal discussion papers.

Pattern 1: Credential Dismissal

Despite professional credentials, critics were frequently characterized as lacking appropriate expertise.

Bažant's responses employed terms such as: - “lay critics” - “mistaken impression” - “erroneous” - “no meaningful calculations”

Example:

Korol and Sivakumaran—both structural engineering professors with publication records in steel structures—raised questions about energy dissipation and arrest conditions.

Their work was not addressed on technical merits in subsequent Bažant papers. Szuladziński's 2012 temporal analysis, showing that momentum exchange alone imposes significant arrest tendency, was similarly unaddressed in later publications.

Observation:

When substantive technical critiques go unaddressed while their authors are dismissed as unqualified, the discourse departs from normal scientific exchange.

A further instance: a closure response opens its reply to Gourley by noting that “the interdisciplinary interests of Gourley, a chemical engineer with a doctorate in jurisprudence, are appreciated,” before stating that “none of the discussor’s criticisms is scientifically correct” — crediting the critic’s background before dismissing the entirety of his content in the same sentence. The same closure separately describes a co-author of an independently circulated critique as “S. E. Jones, Associate Professor of Physics at Brigham Young University and a cold fusion specialist” — a description accurate on its face but selected for its association value rather than for its bearing on the physics argument under discussion. [D25](#)

Pattern 2: Strawman Characterization

In several responses, critics are described as holding positions—“collapse cannot occur if $F_0 > mg$,” “steel must melt,” or “free fall throughout”—that are stronger or cruder than the claims made in their own papers, where the focus is often on robustness, energy partitioning, or localized weakening requirements.

Example from Bažant & Verdure (2007):

“Some critics have been under the mistaken impression that collapse cannot occur if... the weight mg of the upper part is less than the load capacity F_0 of the floor. This led them to postulate various **strange ideas** (such as “fracture wave” and planted explosives).”

The characterization as “strange ideas” frames technical alternatives as beyond consideration while misrepresenting the underlying objection: not that $F_0 > mg$ prevents collapse, but that claims of inevitability require demonstrating robustness across variance, not just showing compatibility under selected parameters.

Observation:

Engaging a weakened version of critique rather than the strongest version is a departure from rigorous discourse norms.

Pattern 3: The Category Error in Response

Technical objections were sometimes met with non-technical deflections.

The structure:

1. Critic raises mechanical question (e.g., arrest conditions, energy budget)
2. Response shifts to logistical or sociological domain
3. Mechanical question remains unaddressed

Example from Bažant & Verdure (2007):

After presenting energy-based inevitability criterion:

“If Eq. 5 is violated, there is (regardless of F_0) no way to deny the inevitability of progressive collapse driven only by gravity.”

But then:

“This could also help to clear up misunderstanding (and thus to dispel the myth of planted explosives).”

Followed by discussion of how conventional demolition would require extensive preparation and “many thousands of pounds” of explosives, which would be “difficult to organize.”

The logical problem:

- Mechanical question: Does the model demonstrate robustness across realistic variance?
- Response: Alternative mechanism would be logistically difficult
- **These do not address each other**

Logistical difficulty of alternative A does not validate model B. Each must stand on its own evidence.

Observation:

When mechanical questions receive sociological answers, discourse has shifted from physics to rhetoric.

Pattern 4: Argument from Incredulity

The pattern recurs:

1. Identify what an alternative would require
2. Express that requirement as extreme or implausible
3. Treat this as evidence for preferred explanation

The form:

- “Demolition would require X tons of material”
- “X tons is difficult to place covertly”
- **Therefore progressive collapse is validated**

The fallacy:

Structurally, these arguments resemble affirming the consequent: from “if demolition, then large quantities of material and difficult logistics” and “logistics would be difficult,” they tend to infer “therefore demolition did not occur,” and then treat the gravity-driven scenario as effectively validated.

The conclusion doesn’t follow. Difficulty of one mechanism doesn’t validate another. Both require independent demonstration.

Observation:

Arguments from incredulity are rhetorical, not evidential. They function to frame alternatives as absurd without addressing the technical question: Does the preferred model demonstrate what it claims?

Pattern 5: Ignoring Substantive Technical Work

A concerning feature of the literature is not that these critiques went unaddressed — Korol and Sivakumaran’s work is engaged directly (§6.11.2’s rescaling), Szuladziński received formal Discussion-and-Closure treatment ^{D25}, and Le and Bažant (2011) responds to the class of argument MacQueen and Szamboti later formalized. ^{BAZANT-499} The concern is narrower: that engagement, where it occurred, was frequently selective, occasionally credential-inflected (§Pattern 1), and did not always address the strongest form of the critique (§Pattern 2) — closure by inadequate answer, not closure by silence.

This creates a form of closure by omission: unresolved questions about energy budgets and arrest conditions persist in parallel, rather than being clearly answered or explicitly bracketed as outside the model’s current scope.

What These Patterns Reveal

The discourse around progressive collapse, in this particular corpus, exhibits features that sit uneasily with strong norms of scientific exchange: some critiques are met with credential-focused framing or sociological deflections rather than direct engagement with the most challenging mechanical points.

In domains where evidence and models are exceptionally robust, one typically sees willingness to address the best available counterarguments on their own terms; here, by contrast, several of the more technically detailed critiques remain at the margins of the mainline narrative.

The Consequence for Closure

By framing technical dissent as:

- Unqualified (“lay critics”)
- Confused (“mistaken impression”)
- Absurd (“strange ideas”)
- Motivated (“conspiracy theories”)

Legitimate methodological questions were moved **outside the bounds of normal scientific discourse**.

These discursive patterns plausibly influenced how closure formed in practice: they reduced the visibility and uptake of certain methodological questions, narrowed the range of publicly “safe” objections, and contributed to an appearance of consensus that outpaced detailed resolution of key mechanical issues.

Methodological Implications

The handling of technical dissent matters beyond this case specifically because premature closure has costs: if the mechanism is not as well understood as claimed, design codes may be inadequate, risk assessment may be wrong, future events may not be preventable, and public trust in expertise erodes.

The Contrast with Other Investigations

In classic case studies such as Tacoma Narrows or the Hyatt Regency walkway, the published record highlights a trajectory in which early explanatory frameworks are revised in response to competing mechanisms, with dissenting views eventually assimilated into an improved consensus.

In the WTC progressive-collapse literature centered on Bažant’s models, by contrast, the inevitability framing appears early and persists; critiques that challenge key assumptions (e.g., symmetry, energy margins,

arrest conditions) are answered selectively or left unintegrated, and further large-scale modeling beyond the one-dimensional framework is often described as unnecessary for establishing post-initiation inevitability.

The difference in pattern—iterative remodeling in one case, versus defense and consolidation around an early global narrative in the other—is notable, even allowing for the very different political and evidential environments.

Conclusion

The treatment of technical dissent in progressive collapse literature exhibits systematic departures from norms of scientific discourse:

- Credential dismissal over substantive engagement
- Strawman characterization over steel-man response
- Sociological deflection over mechanical rebuttal
- Argument from incredulity over evidential demonstration
- Selective non-engagement with substantive critique

These patterns are diagnostic not of the truth or falsity of any particular collapse mechanism, but of tensions in the epistemic process: social dynamics and rhetorical framing are doing part of the work that, in a more fully resolved case, would be carried by convergent mechanical evidence and independently replicated modeling.

Closure here appears to have been driven by a mixture of technical argument, institutional authority, and discursive boundary-setting around acceptable questions, rather than by a fully symmetrical and exhaustively tested treatment of competing mechanical hypotheses.

↳ *These patterns do not resolve the collapse question. They identify places where reasoning has slipped from physics into preference. **The refusal to pretend to know begins by noticing those slips.***

5.5 The Unfalsifiable Comparator

A related pattern recurs whenever a comparative counter-example is raised: a building that suffered comparable fire and structural damage without producing a total, symmetric, progressive collapse. The response is consistently that the comparator differs from the WTC towers in some material respect — scale, construction material, arrest mechanism, damage profile. Disanalogies can be real; the pattern becomes diagnostic only in aggregate, when every comparator offered is distinguishable after the fact and none is treated as bearing on the claim under examination.

A published discussion raised the Windsor Tower (Madrid, 2005) comparison directly. [D28](#) The response distinguished it on core material (reinforced concrete rather than steel) and arrest mechanism (a specific transfer structure) (“such a comparison is meaningless because these two buildings have different structural elements and the cause of collapse is totally different”). Each distinction may be individually sound. Applied consistently across every comparator in turn, the effect is that no historical instance remains available to test the claim against, while the claim continues to be presented as demonstrated.

This is the same structure examined in §3–§4 above, applied to a different move: methodological questions redirected to case-by-case disanalogy rather than addressed as a class.

5.6 A Non-Sinister Account of the Same Pattern

The patterns catalogued above admit a less adversarial explanation, worth stating rather than leaving implicit. NIST’s stated objectives included determining why and how the towers collapsed — not merely revising codes after the fact. [NIST-FAQ](#) But the investigation itself, on its own account, narrowed that objective in execution: it treated initiation as the genuinely open question requiring detailed modeling, and post-initiation propagation as sufficiently explained by the energy comparison already available, a scope decision NIST states directly rather than conceals (§I-5). That stated justification is itself a data point worth weighing against the mundane reading, not just for it: “brevity” is an odd word to reach for in a document exceeding ten thousand pages, meticulous elsewhere about exactly the kind of structural detail the same justification waves past here (§I-5.5). A resource-constrained institution narrowing its scope is ordinary. A meticulous institution explaining that narrowing with a justification its own report’s length contradicts is a narrower and stranger thing, and this document does not have a mundane account of that specific incongruity to offer. A large, resource-constrained investigation concluding it had answered the harder question and treating the easier one as settled is a mundane project-management judgment, available without positing suppression — even if, as this document argues throughout, that judgment does not survive the scrutiny it was never subjected to. Separately, once outside critique of that judgment began, NIST’s reviewers had reason to expect that substantive engagement would be read as validating the critique’s broader framing regardless of the answer given — the same catch-22 §5.3 describes, operating from inside the institution as well as against it. A working mechanic’s clipped or credential-inflected responses to an increasingly adversarial correspondence (§5.4) are also consistent with ordinary busyness and declining patience under repeated challenge, not only with a considered strategy of exclusion. §III-2’s own stated principle — that a shift from tentative to certain language may reflect “internal confidence, external communication demands, or genuine tightening of the models,” not necessarily one specific cause — applies here with the same force it is given elsewhere in this document. The institutional-response pattern is real and documented; a mundane explanation for it is available and has not been ruled out.

Chapter 3 — Why Epistemic Closure Matters Beyond This Case

1. The Structural Problem of Complexity

Modern life demands judgment on questions that lie beyond individual expertise:

- vaccine safety
- climate dynamics
- financial stability
- AI governance
- structural failure

This is not a failure of education. It is a **structural feature** of knowledge in complex societies. Citizens must rely on **expert consensus**—but only if that consensus is **earned through method**, not declared by authority.

2. The Single Defensible Point

Process quality and outcome correctness can come apart. A conclusion being ultimately correct does not retroactively validate the methods used to reach it, and a conclusion being widely believed does not substitute for the demonstration that would earn it. That is the entirety of what this document argues about

closure, and it does not require examples from other fields to carry its weight — importing them risks exactly the move §III-2.5.1 and §III-2.5.2 diagnose elsewhere: a contrast class doing work that demonstration should do.

A Note on This Document's Own Exposure

The patterns catalogued in this chapter — credential framing, strawman substitution, contrast-class substitution, argument from incredulity — are not unique to one side of this dispute. A document that names “Anti-Newtonian Solid” and “Epicycles of Inevitability” is not exempt from the same scrutiny it applies to its subject. The claim made here is not that this document is free of rhetorical framing; it is that its central technical claims (Part II) are stated precisely enough to be checked, and have been revised where checking found them wanting. Whether that is sufficient is for the reader to judge, not for this document to declare of itself.

Conclusion

1. Why the Inquiry Remains Justified

Across *Epistemic Limits*, the *Dialectic Inventory*, and the present document, one conclusion is consistent:

Epistemic closure is unearned.

The replicability problem, the model-dependence problem, the evidentiary-preservation problem, and the rhetoric-of-consensus problem all point to the same modest conclusion:

further open inquiry is still warranted.

The appropriate end state is:

- open data where possible
- explicit uncertainty where required
- symmetric treatment of hypotheses
- ensemble-based evaluation of robustness
- preservation of material and computational evidence

These are not aspirational ideals. They are the minimum conditions under which claims of inevitability could be justified.

2. Forward Linkage

Together, *Epicycles of Inevitability* now form a coherent architecture:

1. **Epistemic Limits** — account of internal tensions and incoherencies in popular explanations
2. **A Dialectic Inventory** — physics and modeling difficulty of the observed phenomena
3. **Chronology of Closure** — social trajectory of explanation and its consequences

The task that remains is straightforward:

- reopen methodological questions without presuming answers

- evaluate robustness across variance rather than single tuned runs
- treat disagreement as instrument, not as stigma

Only then does the inquiry satisfy its own standard:

| physics first, method second, epistemology last — and closure never by decree.

Until that standard is met, claims of inevitability in the WTC collapse remain interpretive conveniences rather than demonstrated properties of the system.

References

↳ **A note on citation practice.** Listing a source here indicates it is drawn upon for a specific, checkable claim — a measurement, a derivation, a quoted statement — not that this document endorses or adopts that source’s broader conclusions, methodology, or institutional affiliation. This applies uniformly, regardless of whether a source originates from an official investigation, an academic journal, an advocacy-affiliated venue, or an independent researcher. Where a source’s own stated thesis or institutional affiliation is materially relevant to weighing a specific citation — typically because the source itself, rather than an isolated result drawn from it, is the object of discussion (e.g., III-2.5.3) — that context is disclosed at the point of use or in the entry itself. Its absence elsewhere indicates the citation concerns a narrow, independently checkable claim, not that no such context exists.

Bažant Corpus

Zdeněk P. Bažant, Yong Zhou: “Why Did the World Trade Center Collapse?”; SIAM News Vol 34 No 8 October 13, 2001. <https://web.archive.org/web/20070715213036/https://www.siam.org/news/news.php?id=578> / <https://web.archive.org/web/20120504023524/http://www.siam.org/pdf/news/578.pdf> <https://web.archive.org/web/20011031095744/http://www.tam.uiuc.edu/news/200109wtc/>

405: Zdeněk P. Bažant, Yong Zhou: “Why Did the World Trade Center Collapse?—Simple Analysis”; Journal of Engineering Mechanics, 128(1), January 2002, pp. 2–6; with Addendum, Journal of Engineering Mechanics, 128(3), March 2002, pp. 369–370; <http://www.civil.northwestern.edu/people/bazant/PDFs/Papers/405.pdf>

The original version with Eqs. (1) and (2) was originally submitted to ASCE on **September 13, 2001**

[...] we are not attempting to model the details of the real failure mechanism but **seek only to prove that the towers must have collapsed and do so in the way seen** [...]

The analysis shows that if prolonged heating caused the majority of columns of a single floor to lose their load carrying capacity, **the whole tower was doomed**. The structural resistance is found to be an order of magnitude less than necessary for survival, even though the most optimistic simplifying assumptions are introduced.

The 110-story towers of the World Trade Center were designed to withstand as a whole the forces caused by a horizontal impact of a large commercial aircraft (Appendix I). So why did a total collapse occur? The cause was the dynamic consequence of the prolonged heating of the steel columns to **very high temperature**.

If fracturing in the plastic hinges were considered, a still smaller (in fact much smaller) energy dissipation would be obtained. So **the collapse of the tower must be an almost free fall**. This conclusion is supported by the observation that the duration of the collapse of the tower, observed to be 9 s, was about the same as the duration of a free fall in a vacuum from the tower top (416 m above ground) to the top of the final heap of debris (about 25 m above ground), which is $t = \sqrt{2(416\text{m}-25\text{m})/g} = 8.93\text{s}$. It further follows that the brunt of vertical impact must have gone directly into the columns of the framed tube and the core and that any delay Δt of the front of collapse of the framed tube behind the front of collapsing ('pancaking') floors must have been negligible, or else the duration of the total collapse of the tower, $9\text{ s} + \Delta t$, would have been significantly longer than 9 s. However, even for a short delay Δt , the floors should have acted like a piston running down through an empty tube, which helps to explain the smoke and debris that was seen being expelled laterally from the collapsing tower.

It will nevertheless be appropriate to initiate research on materials and designs that would postpone the collapse of the building so as to extend the time available for evacuation, provide a hardened and better insulated stairwell, or even prevent collapse in the case of a less severe attack such as an off-center impact or the impact of an aircraft containing little fuel.

The main purpose of the present analysis is to prove that the whole tower must have collapsed if the fire destroyed the load capacity of the majority of columns of a single floor.

[From the Addendum, *J. Eng. Mech.* 128(3), pp. 369–370, March 2002, addressing a hypothetical impact at a higher, less massive floor:] “Instead of **one powerful jolt**, this could lead to a series of many small vertical impacts, none of them fatal.”

- **ADDENDUM:** <https://ascelibrary.org/doi/10.1061/%28ASCE%290733-9399%282002%29128%3A3%28369%29>

- **D20:** [Discussion by K. Sivakumar](#)

The **stated aim** was to **prove** that the towers **must have collapsed**,

466: Zdeněk P. Bažant, Mathieu Verdure: “Mechanics of Progressive Collapse: Learning from World Trade Center and Building Demolitions”; *Journal of Engineering Mechanics* 2007
<http://www.civil.northwestern.edu/people/bazant/PDFs/Papers/466.pdf>

Progressive collapse is a failure mode of great concern for tall buildings, and is also typical of building demolitions. The most infamous paradigm is the collapse of the World Trade Center towers.

The destruction of the World Trade Center on September 11, 2001 was not only the largest mass murder in U.S. history but also **a big surprise for the structural engineering profession**, perhaps the biggest since the collapse of the Tacoma Bridge in 1940. **No experienced structural engineer watching the attack expected the WTC towers to collapse. No skyscraper has ever before collapsed due to fire. The fact that the WTC towers did, beckons deep examination.**

[T]he press reports right after September 11, 2001 indicating temperature in excess of 800 °C, turned out to be groundless, but Bažant and Zhou's analysis did not depend on that.

the upper part of the tower fell, with little resistance, through at least one floor height, impacting the lower part of the tower. This triggered progressive collapse because the kinetic energy of the falling upper part exceeded - by an order of magnitude - the energy that could be absorbed by limited plastic deformations and fracturing in the lower part of the tower.

The kinetic energy of the top part of the tower impacting the floor below was found to be about 8.4 x larger than the plastic energy absorption capability of the underlying story, and considerably higher than that if fracturing were taken into account. This fact, along with the fact that during the progressive collapse of underlying stories the loss of gravitational potential per story is much greater than the energy dissipated per story, was sufficient for Bažant and Zhou to conclude, purely on energy grounds, that **the tower was doomed** once the top part of the tower dropped through the height of one story **or even 0.5 m**.

[N]o further analysis has been necessary to prove that the WTC towers had to fall the way they did, due to gravity alone. However, a theory describing the progressive collapse dynamics beyond the initial trigger, with the WTC as a paradigm, could nevertheless be very useful for other purposes, especially for learning from demolitions. It could also help to clear up misunderstanding (and thus to dispel the myth of planted explosives). Its formulation is the main objective of what follows.

As W_g [the loss of gravity when the upper part of the tower is moved down by one floor] was, for the WTC, greater than W_p [the area under the complete load-displacement curve] [Fig. 3] by an order of magnitude, acceleration of collapse from one story to the next was ensured. Some critics have been under the mistaken impression that collapse cannot occur if (because of safety factors used in design) the weight mg of the upper part is less than the load capacity F_0 of the floor. This led them to postulate various **strange ideas** (such as “fracture wave” and planted explosives). [...] If Eq. 5 $\{K < W_c$, where K = kinetic energy of the impacting mass $m(z)$ and W_c = net energy loss up to u_c during the crushing of one story. This is the criterion of preventing progressive collapse from starting [Fig. 4(c)]. Its violation triggers progressive collapse} is violated, there is regardless of [the load-displacement curve of a column] no way to deny the inevitability of progressive collapse driven only by gravity.

Ronan Point apartments, the Oklahoma City bombing, etc., demonstrate that only a vertical slice of building may undergo progressive collapse, whereas the remainder of the building stands. Such a collapse is truly a three-dimensional problem, much harder to analyze, but some cases might allow adapting the present one-dimensional model as an approximation.

1. If the total (internal) energy loss during the crushing of one story (representing the energy dissipated by the complete crushing and compaction of one story, minus the loss of gravity potential during the crushing of that story) exceeds the kinetic energy impacted to that story, collapse will continue to the next story. This is the criterion of progressive collapse trigger [Eq. (5)]. If it is satisfied, **there is no way to deny the inevitability of progressive collapse driven by gravity alone** (regardless of by how much the combined strength of columns of one floor may

exceed the weight of the part of the tower above that floor). What matters is energy, not the strength, nor stiffness. [...]

5. The present idealized model allows simple **inverse analysis** which can yield the crushing energy per story and other properties of the structure **from a precisely recorded history of motion** during collapse. From the crushing energy, one can infer the collapse mode, e.g., single-story or multistory buckling of columns.

6. **It is proposed to monitor the precise time history of displacements in building demolitions**—for example, by radio telemetry from sacrificial accelerometers, or high-speed optical camera—**and to engineer different modes of collapse to be monitored.** This should provide **invaluable information** on the energy absorption capability of various structural systems, needed for assessing the effects of explosions, impacts, earthquake, and terrorist acts.

- D25: [Discussions by Gregory Szuladzinski, James R. Gourley](#)

476: Zdeněk P. Bažant, Jia-Liang Le, Frank R. Greening, David B. Benson: “What Did and Did Not Cause Collapse of World Trade Center Twin Towers in New York?”; Journal of Engineering Mechanics 2008
<http://www.civil.northwestern.edu/people/bazant/PDFs/Papers/476.pdf>;

Previous analysis of progressive collapse showed that gravity alone suffices to explain the overall collapse of the World Trade Center Towers. However, **it remains to be determined whether the recent allegations of controlled demolition have any scientific merit.** The **present analysis proves that they do not.** The video record available for the first few seconds of collapse is shown to agree with the motion history calculated from the differential equation of progressive collapse but, despite uncertain values of some parameters, it is totally **out of range of the free fall hypothesis, on which these allegations rest.**

But are high steel temperatures really necessary to explain collapse? Not really. The initial speculation that very high temperatures were necessary to explain collapse must be now revised since tests revealed a strong temperature effect on the yield strength of the steel used. The tests by NIST showed that, at temperatures 150 °C, 250 °C and 350 °C, the yield strength of the steel used in the fire stories decreased by 12%, 19% and 25%, respectively. [...] Although a detailed computer analysis of columns stresses after aircraft impact is certainly possible, it would be quite tedious and demanding, and has not been carried out by NIST. Nevertheless, it can easily be explained that the stress in some surviving columns most likely exceeded 88% of their cold strength σ_0 . In that case, **any steel temperature ≥ 150 °C sufficed to trigger the viscoplastic buckling of columns.**

Merely to be convinced of the **inevitability of gravity driven progressive collapse**, further analysis is, for a structural engineer, superfluous. Further analysis is nevertheless needed to **dispel false myths**, and to acquire full understanding that would allow assessing the danger of progressive collapse in other situations.

- D27: [Discussion by Anders Björkman](#)

[discusser] presents no meaningful mechanics argument against the gravity driven progressive collapse model of our paper

499: Jia-Liang Le, Zdeněk P. Bažant: “Why the Observed Motion History of World Trade Center Towers Is Smooth”; Journal of Engineering Mechanics 2011

<http://www.civil.northwestern.edu/people/bazant/PDFs/Papers/499.pdf>

At the moment of impact, the velocity of the upper part must have decreased. The fact that no velocity decrease can be discerned in the videos of the early motion of the tower top has been recently exploited to claim that the collapse explanation generally accepted within the structural mechanics community was invalid. This claim is here shown to be groundless. Calculations show that the velocity drop is far too small to be perceptible in amateur video records and is much smaller than the inevitable error of such video records. [...] The collapse of the World Trade Center WTC towers has been explained as a gravity-driven process triggered by the collapse of a critical story heated by fire (Bažant and Zhou 2002; Bažant and Verdure 2007; Bažant et al. 2008; Bažant and Le 2008). **All the objections** of the proponents of the controlled demolition hypothesis **have been shown invalid.**

- **D28:** [Discussion by Crockett Grabbe](#)

discusser's [...] objections against the analysis of gravity-driven progressive collapse of the World Trade Center (WTC) towers by [sic] have **no scientific merit**

585: Jia-Liang Le, Zdeněk P. Bažant: “Mechanics of Collapse of WTC Towers Clarified by Recent Column Buckling Tests of Korol and Sivakumaran”; Int. J. Str. Stab. Dyn. 2016

<http://cee.northwestern.edu/people/bazant/PDFs/Papers/585.pdf>

[...] Korol and Sivakumaran [...] revealed that, under the simplifying assumptions of rigid end supports and unlimited ductility (or no fracturing) of unheated columns, the energy dissipation of the WTC columns would have been at maximum 3.5-times as large as that calculated by the plastic hinge mechanism normally considered for small-deflection buckling. [...] The **proper conclusion** from Korol and Sivakumaran's tests, based on close matching of the video record, is that the fracturing of unheated columns and the flexibility of their end restraints must have significantly reduced the energy dissipation in columns calculated under the assumptions of no fracture and no end restraint flexibility.

Previous studies led to a rigorous mathematical model which showed that a gravity driven collapse of the twin towers of the World Trade Center (WTC) in New York on September 11, 2011 was **inevitable.**

625: Jia-Liang Le, Zdeněk P. Bažant: “Spontaneous Collapse Mechanism of World Trade Center Twin Towers and Progressive Collapse in General”; J. Struct. Eng. 2022

<http://www.civil.northwestern.edu/people/bazant/PDFs/Papers/625.pdf>

The kinetic energy of the upper falling part was immediately (Bažant 2001) shown to be many time greater than the maximum possible energy absorption capacity of the columns of the underlying story, which made a progressive story-by-story collapse **inevitable.** [...]

For the collapse, the crucial difference from previous fires in tall buildings was that this fire was ignited simultaneously over a large volume of several stories. This caused the surface-to-volume ratio of the fire zone to be much smaller than in previous fires, which reduced the cooling rate and thus led to higher temperatures (normally the fire moves—when a neighboring place starts burning, the previous one is already cooling).

NIST’s analysis supports Bažant and Zhou’s (2002) conclusion that as soon as the top part of each tower fell through the height of at least one story, the whole tower was doomed to suffer a total collapse (NIST 2005). [...]

Of course, a detailed numerical modeling of the entire collapse process, with finite-element discretization of all columns and floors and gas flows, would have been computationally infeasible. But such a brute-force approach has actually been shown unnecessary.

The fact that the stories were numerous and that the collapse was vertical and almost simultaneous over the full tower width made possible a simplified, although realistic, one-dimensional (1D) continuum analysis. A 1D continuum model for the vertical collapse propagation of the crush front was developed by Bažant and Verdure (2007). They applied d’Alembert’s principle to formulate the dynamic equilibrium of the upper falling part, and then to derive the equations for the crush-down and crush-up phases of the collapse process.

In the subsequent studies, the model was further refined by incorporating different energy dissipation sources, such as the impact comminution of concrete slabs, fast ejection of air, and lateral ejection of debris (Bažant et al. 2008; Le and Bažant 2011, 2017a). The model was shown to match all the observations, which include video records of the motion of the upper falling part during the first few seconds of the collapse (before dust clouds obscured the view), the total collapse duration inferred from the seismic record, the particle-size distributions of the dynamically comminuted concrete floors, the explosive expansion of dust clouds, and the booms heard. This analysis dispelled all the main claims of the proponents of the controlled demolition hypothesis, which include the free-fall motion of the tower, pulverization of concrete slabs by explosives, wide spreading of concrete particles, and dust burst and booms supposedly due to explosions. The analysis showed them all to be false.

EPN: Jia-Lang Le, Zdeněk P. Bažant: “Mechanics-Based Mathematical Studies Proving Spontaneity of Post-Impact WTC Towers Collapse”; Europhysics News 48/1, January 2017.

<https://www.europhysicsnews.org/articles/ePN/abs/2017/01/ePN2017481p18/ePN2017481p18.html>

This article also presents the governing equations for the crush-down and crush-up phases, not reflected in the shorter excerpts otherwise cited from this source.

Resisting force is decomposed into four components:

$$F_c = F_b + F_s + F_a + F_e$$

where F_b is column buckling/fracture resistance, F_s the concrete-comminution energy rate, F_a air-ejection resistance, and F_e resistance from lateral debris ejection.

F_b is further expressed as $F_b = \beta F_p$, where F_p is the standard three-hinge column-buckling force and β is a reduction factor applied specifically to that component — the source of this document’s own resistance factor (§4.10.3, R after the rename), a related but narrower quantity in the original.

The article also defines the compaction ratio explicitly: $\lambda = (1 - \kappa_{out})\mu/\mu_c$, the ratio of compressed to original material height, where κ_{out} is the fraction of mass ejected outside the tower perimeter.

Notably, β 's own numerical value shifts between sources: this article revises the reduction factor from 2/3 to 0.73 for normal-strength columns and from 0.24 to 0.26 for high-strength columns, relative to the earlier analysis (Bažant et al. 2008 [BAZANT-476](#)), attributed only to unspecified “recent information.”

The cause of collapse of the World Trade Center (WTC) in New York on 9/11/2001, clarified mathematically by mechanical analysis, has been questioned by some **lay critics without any meaningful calculations**. They blame the collapse on controlled demolition, implying some sort of conspiracy. [...]

The **sonic booms heard, the rapidly expanding dust clouds and the wide ejection of debris are no surprise** (while **some critics erroneously** claimed that the booms could have been caused only by explosives). The size distribution of concrete particles, calculated from the energy of impact on floor slabs, matches the distribution of the particle sizes seen on the ground, which ranged from 0.01 mm up.

The critics claimed that such small particles could be produced only by explosives. Yet the experience from mining and tunneling [...] shows that such small particles could be produced by explosives only if the alleged authors installed about 150 tons of TNT into small holes drilled into all concrete floor slabs of each tower. The **critics do not explain how such a massive operation, requiring many workers, could have been carried out in secrecy, no one leaking it later to the public. Many workers would have been needed even for the usual demolition by explosives installed on the columns of one story**, and, to match the collapse progression as seen, the aircraft would have had to impact each tower just above the story wired by explosives.

- Zdeněk P. Bažant, Mathieu Verdure, Yong Zhou, Frank R. Greening, David B. Benson: PRESENTATION: “Mechanics of Progressive Collapse: What Did And Did Not Doom World Trade Center, And What Can We Learn?” Georgia Tech / Northwestern April 4 & May 24, 2007. <https://www.slideserve.com/sawyer/mechanics-of-progressive-collapse-what-did-and-did-not-doom-world-trade-center-and-what-can-we-learn>

Thermite cutter charges planted? — evidenced by residues of S, Cu, Si found in dust? But these must have come from gypsum wallboard, electrical wiring, galvanized sheet steel, etc. [...]

“Fracture Wave” allegedly propagated in a material pre-damaged, e.g. by explosives, led to free-fall collapse — unrealistic hypothesis, because: A uniform state on the verge of material failure cannot exist in a stable manner, because of localization instability. [...]

MAIN RESULTS: All WTC observations are explained. All lay criticisms are refuted.

Official Reports

FEMA

“World Trade Center Building Performance Study: Data Collection, Preliminary Observations, and Recommendations.”; FEMA 403 May 2002. https://people.duke.edu/~hpgavin/cee421/fema403_index.html

Construction of WTC 1 resulted in the storage of more than 4×10^{11} joules of potential energy over the 1,368-foot height of the structure. Of this, approximately 8×10^9 joules of potential energy were stored in the upper part of the structure, above the impact floors, relative to the lowest point of

impact. Once collapse initiated, much of this potential energy was rapidly converted into kinetic energy. As the large mass of the collapsing floors above accelerated and impacted on the floors below, it caused an immediate progressive series of floor failures, punching each in turn onto the floor below, accelerating as the sequence progressed. (p. 2-27)

The ability of the two towers to withstand aircraft impacts without immediate collapse was a direct function of their design and construction characteristics, as was the vulnerability of the two towers to collapse a result of the combined effects of the impacts and ensuing fires. (p. 2-38)

In the past, tall buildings have occasionally been damaged, typically by earthquakes, and experienced collapse within the damaged zones. Those structures were able to arrest collapse before they progressed to a state of total collapse. The two WTC towers were able to arrest collapse from the impact damage, but not from the resulting fires when combined with the impact effects of the aircraft attacks. Studies should be conducted to determine, given the great size and weight of the two towers, whether there are feasible design and construction features that would permit such buildings to arrest or limit a collapse, once it began. (p. 2-42)

9/11 Commission

National Commission on Terrorist Attacks Upon the United States: “9/11 Commission Report”; 2004.
<http://govinfo.library.unt.edu/911/report/index.htm>

[the] exterior walls bore most of the weight of the building. The **interior core of the buildings was a hollow steel shaft**, in which elevators and stairwells were grouped. (“Notes”, 9.1)

NIST

John Lyle, ARUP: National Institute for Standards and Technology, “Public Comments Received by NIST on DRAFT Reports”;
<http://www.nist.gov/el/disasterstudies/wtc/upload/CombinedPublicCommentsDraftTowersReports2005-2.pdf>

Simplistic Model Results

Results from five simple finite element models (of roughly the same geometric proportions as WTC1 and WTC2) demonstrate that a large hole in the side of a tube and the missing parts of the core do not significantly influence the natural frequency (see table 1).

	Model 1	Model 2	Model 3	Model 4	Model 5
	Tube & core, No holes	Large hole in side & no core	No hole with core	Large hole in side with core	Large hole in side and core missing at same level
Bending	0.1653	0.1647	0.1643	0.1636	0.1637
Torsion	0.5981	0.5889	0.5981	0.5891	0.5889

(Table 1 Variation of natural frequency in Hz)

Comment reason: The stability of the towers immediately following the impact has not really been proven by the impact analysis. Simply taking the results of the impact damaged structure and loading it statically effectively ignores and [sic] damage that may have resulted due to swaying.

Revision suggestion: Provide a more detailed analysis to take account of load path re-distribution in the core and the P-Δ effects of sway following the impact. (PDF p. 197)

NCSTAR-1

National Institute for Standards and Technology: “Federal Building and Fire Safety Investigation of the World Trade Center Disaster: Final Report of the National Construction Safety Team on the Collapses of the World Trade Center Towers (NCSTAR 1)”; September 2005 <https://www.nist.gov/el/final-reports-nist-world-trade-center-disaster-investigation>

The specific objectives were:

1. Determine why and how WTC 1 and WTC 2 collapsed following the initial impacts of the aircraft and why and how WTC 7 collapsed. [...] (NCSTAR 1, p. xxxix)

The focus of the investigation was on the sequence of events from the instant of aircraft impact to the initiation of collapse for each tower. For brevity in this report, this sequence is referred to as the ‘probable collapse sequence,’ although it **includes little analysis** of the structural behaviour of the tower after the conditions for collapse initiation were reached and **collapse became inevitable**. (NCSTAR 1, p. xxxvii (footnote))

At 9:30 a.m., EMS set up a triage desk in the lobby of WTC 2. (NCSTAR 1, p. 44)

NIST performed confirmatory tests on samples of the **236 pieces of recovered steel** to determine if the steel met the structural specifications. (NCSTAR 1, p. 68)

The focus of the Investigation was on the sequence of events from the instant of aircraft impact to the initiation of collapse for each tower. For brevity in this report, this sequence is referred to as the “probable collapse sequence,” although it **does not actually include** the structural behavior of the tower after the conditions for collapse initiation were reached and **collapse became inevitable**. (NCSTAR 1, p. 82 (footnote))

Over a period of about 18 months, **236 pieces of steel** were shipped to the NIST campus, [...] The remaining 229 samples **represented roughly 0.25 percent to 0.5 percent of the 200,000 tons of structural steel** used in the construction of the two towers. (NCSTAR 1, p. 87)

Examination of photographs showed that 16 of the exterior panels recovered from WTC 1 were exposed to fire prior to the building collapse. None of the nine recovered panels from within the fire floors of WTC 2 were directly exposed to fire. NIST used two methods to estimate the

maximum temperatures that the steel members had reached:

Observations of paint cracking due to thermal expansion. Of the more than 170 areas examined on 16 perimeter column panels, only three columns had evidence that the steel reached temperatures above 250 °C: east face, floor 98, inner web; east face, floor 92, inner web; and north face, floor 98, floor truss connector. Only two core column specimens had sufficient paint remaining to make such an analysis, and their **temperatures did not reach 250 °C**. NIST did not generalize these results, since the examined columns represented only 3 percent of the perimeter columns and 1 percent of the core columns from the fire floors.

Observations of the microstructure of the steel. High temperature excursions, such as due to a fire, can alter the basic structure of the steel and its mechanical properties. Using metallographic analysis, NIST determined that **there was no evidence that any of the samples had reached temperatures above 600 °C**. (NCSTAR 1, p.90)

6.14.2 Results of Global Analysis of WTC 1

Once the upper building section began to move downwards, the weakened structure in the impact and fire zone was not able to absorb the **tremendous energy** of the falling building section and **global collapse ensued**. (NCSTAR 1, p.145)

6.14.3 Results of Global Analysis of WTC 2

[...] As with WTC 1, once the upper building section began to move downwards, the weakened structure in the impact and fire zone was not able to absorb the tremendous energy of the falling building section and **global collapse ensued**. [...]

Since the stories below the level of collapse initiation provided little resistance to the **tremendous energy released by the falling building mass**, the building section above came down **essentially in free fall**, as seen in videos. As the stories below sequentially failed, the falling mass increased, further increasing the demand on the floors below, which were unable to arrest the moving mass. (NCSTAR 1, p.146)

NIST found no corroborating evidence for alternative hypotheses suggesting that the WTC towers were brought down by controlled demolition using explosives planted prior to September 11, 2001. [...] Instead, photographs and videos from several angles clearly show that the collapse initiated at the fire and impact floors and that the collapse progressed from the initiating floors downward, until the dust clouds obscured the view. (NCSTAR 1, pp.xxxviii, 146 & 176)

The towers **likely would not have collapsed** under the combined effects of aircraft impact and the subsequent multi-floor fires encountered on September 11 if the thermal insulation had not been widely dislodged or had been only minimally dislodged by aircraft impact. (NCSTAR 1, p.149)

6.14.7 & .8 Probable WTC [1/2] Collapse Sequence The downward movement of this structural block was more than the damaged structure could resist, and the **global collapse began**. (NCSTAR 1, pp 151 & 152)

The two aircraft hit the towers at high speed and did considerable damage to principal structural components (core columns, perimeter columns, and floors) that were directly impacted by the aircraft or associated debris. However, **the towers withstood the impacts and would have**

remained standing were it not for the dislodged insulation and the subsequent multi-floor fires. (NCSTAR 1, p. 174)

Both towers withstood the significant structural damage to the exterior walls, core columns, and floor systems due to the aircraft impact. WTC 2 was the more severely damaged building and the first to collapse. WTC 2 **displayed significant reserve capacity, as evidenced by a post-impact rooftop sway that was more than one-third of that under the hurricane force winds for which the building was designed.** The **oscillation period of this swaying was nearly equal to that calculated for the undamaged structure.** [...] The initial **jet fuel fires** themselves **lasted at most a few minutes.** (NCSTAR 1, p.182)

Jet fuel sprayed onto the surfaces of typical office workstations **burned away within a few minutes.** The jet fuel accelerated the burning of the workstation, but did not significantly affect the overall heat released.

In the simulations, none of the columns with intact insulation reached temperatures over 300 °C. Only a few isolated truss members with intact insulation were heated to temperatures over 400 °C in the WTC 1 simulations and to temperatures over 500 °C in the WTC 2 simulations. In WTC 1, **if the fires had been allowed to continue past the time of building collapse, complete burnout would likely have occurred within a short time** since the fires had already traversed around the entire floor, and most of the combustibles would already have been consumed. In WTC 2, if the fire simulation were extended for 2 hours past the time of building collapse with all windows broken, the temperatures in the truss steel on the west side of the building (where the insulation was undamaged) would likely have increased for about 40 min before **falling off rapidly** as the **combustibles were consumed.** Temperatures of 700 °C to 760 °C were reached over approximately 15 percent of the west floor area for less than 10 min. Approximately 60 percent of the floor steel had temperatures between 600 °C and 700 °C for about 15 min. Approximately 70 percent of the floor steel had temperatures that exceeded 500 °C for about 45 min. At these temperatures, the floors would be expected to sag and then recover a portion of the sag as the steel began to cool. The temperatures of the insulated exterior and core columns **would not have increased to the point where they would have experienced significant loss of strength or stiffness.** (NCSTAR 1, p. 184)

[...]he simulations were insensitive to both the amount and distribution of the jet fuel. Sensitivity studies showed that the **amount of fuel spilled in the simulation only influenced the results of the first few minutes;** the long-term behavior of the simulated fires was unaffected. ∴ NCSTAR 1-5F, p.56

Northwestern University

The study performed by Northwestern University (Bazant 2002) was a simplified approximate analysis of the overall collapse of the WTC towers which addressed the question of why a total collapse occurred. The analysis addressed the results of prolonged heating which would have caused the columns of a single floor to lose their load carrying capacity and initiated the collapse of the building. The analysis assumed loss of thermal insulation during impact, uniform temperatures of 800 °C for a uniform column size and load across a single floor, and creep buckling and loss of load carrying capacity in over half of the columns. The analysis included

evaluation of the dynamic amplification of the loads and the ability of the columns in the lower floors to dissipate the kinetic energy of the falling upper building mass through formation of plastic hinge mechanisms. The analysis found that the ratio of the kinetic energy of the upper building section dropping one floor to the deformation energy of plastic hinge rotation in the lower building columns was approximately a factor of eight.

The study by Northwestern did not address the details of impact damage, fire dynamics, or structural response of the towers. Rather, **a generalized condition was assumed of heated columns**, and the question of why there was total collapse was addressed. NIST agrees with the assessment of the tower's required structural capacity to absorb the released energy of the upper building section as it began to fall as an approximate lower bound. The likelihood of the falling building section aligning vertically with the columns below was small, given the observed tilting, so that the required capacity would be greater if interaction with the floors was also considered, as pointed out in the study. (NCSTAR 1-6, p.323)

The buildings **would likely not have collapsed** under the combined effects of aircraft impact and the subsequent jet-fuel ignited multi-floor fires, **if the insulation had not been dislodged** or had been only minimally dislodged by aircraft impact. The existing condition of the insulation prior to aircraft impact and the insulation thickness on the WTC floor system did not play a significant role in initiating collapse on September 11, 2001. (NCSTAR 1-6, p.327)

“Contemporaneous Structural Steel Specifications (NCSTAR 1-3A)”;

https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=101017

(Table 2-3: perimeter grade distribution floors 92–100, population-weighted mean 64.2 ksi)

“Mechanical Properties of Structural Steels After High Temperature Exposure (NCSTAR 1-3D)”;

https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=909059

“ $F_y=60$ ksi perimeter column flange plate from WTC 1 column 142 between floors 97–100.”

“Fire Structure Interface and Thermal Response of the World Trade Center Towers (NCSTAR 1-5G)”;

https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=101039

(Appendix B, Tables B-1, B-3, B-6: Floor-96 perimeter type schedule, plate thicknesses, and complete core-column schedule

“Structural Fire Response and Probable Collapse Sequence of the World Trade Center Towers (NCSTAR 1-6)”;

https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=101279

(Appendix B, Tables B-1, B-3, B-6: Floor-96 perimeter type schedule, plate thicknesses, and complete core-column schedule

NIST FAQ

<https://www.nist.gov/world-trade-center-investigation/study-faqs/wtc-towers-investigation>

10. Why didn't NIST fully model the collapse initiation and propagation of the WTC towers?

The first objective of the NIST WTC investigation included determining why and how WTC 1 and WTC 2 collapsed following the initial impacts of the aircraft (see NIST NCSTAR 1). Determining the sequence of events leading up to collapse initiation was critical to fulfilling this objective. Once the collapse had begun, the propagation of the collapse was readily explained without the same complexity of modeling.

28. Why didn't NIST consider a "controlled demolition" hypothesis with matching computer modeling and explanation like it did for the "pancake theory" hypothesis?

NIST's findings **do not support the "pancake theory"** of collapse, which is premised on a progressive failure of the floor systems in the WTC towers (the composite floor system—that connected the core columns and the perimeter columns—consisted of a grid of steel "trusses" integrated with a concrete slab; see diagram). Instead, the NIST investigation showed conclusively that the failure of the inwardly bowed perimeter columns initiated collapse and that the occurrence of this inward bowing required the sagging floors to remain connected to the columns and pull the columns inwards. Thus, the floors did not fail progressively to cause a pancaking phenomenon. [...]

NIST **found no corroborating evidence** for alternative hypotheses suggesting that the WTC towers were brought down by controlled demolition using explosives.

29. Did the NIST investigation look for evidence of the WTC towers being brought down by controlled demolition? Was the steel tested for explosives or thermite residues?

NIST **did not test for the residue of these compounds** in the steel.

31. How could the WTC towers collapse in only 11 seconds (WTC 1) and 9 seconds (WTC 2)—speeds that approximate that of a ball dropped from similar height in a vacuum (with no air resistance)?

[...] Since the stories below the level of collapse initiation **provided little resistance to the tremendous energy** released by the falling building mass, the building section above came down **essentially in free fall**, as seen in videos. [...]

34. Why didn't the NIST investigation consider reports of molten steel in the wreckage from the WTC towers?

The **condition of the steel in the wreckage** of the WTC towers (i.e., whether it was in a molten state or not) **was irrelevant to the investigation of the collapse** since it does not provide any conclusive information on the condition of the steel when the WTC towers were standing.

NIST Correspondence

Catherine S. Fletcher: NIST response to request for correction; September 2007.

<https://www.journalof911studies.com/volume/2007/NISTresponseToRequestForCorrectionGourleyEtal2.pdf>

[...W]e are **unable to provide a full explanation of the total collapse**. [...] NIST has stated that it found no corroborating evidence to suggest that explosives were used to bring down the buildings. NIST **did not conduct tests for explosive residue** [...] such tests would not necessarily have been conclusive

Patrick Gallagher: “Finding regarding Public Safety Information”; July 9, 2009.
<http://wayback.archive.org/web/20140415115126/http://cryptome.org/nist070709.pdf>

- [...] disclosure might jeopardize public safety [...] NIST shall not release the following information: [ANSYS collapse initiation model, Excel spreadsheets, LS-DYNA global collapse model]

Michael E. Newman: “NIST Position on the WTC investigation”;
In: “Note from the editors regarding: 15 years later: On the physics of high-rise building collapses”;
Europhysics News October 3, 2016. <https://www.europhysicsnews.org/news/670-15-years-later-on-the-physics-of-high-rise-building-collapses>

NIST firmly stands behind its investigation results, and that the body of evidence still overwhelmingly leads to the following scenarios: The WTC Towers collapsed because aircraft impact damage and debris dislodged fireproofing from critical steel components, jet fuel-initiated fires burned very hot for long duration when fed by debris and office materials, and the heat eventually weakened the exposed steel until it lost integrity and led to a global failure [...] The NIST investigation into the collapses of WTC Buildings 1, 2 and 7 was the most detailed examination of structural failure ever conducted.

Independent Analysis

Lu Xinzheng, Jiang Jianjing: “Simulation for the Collapse of WTC after Aeroplane Impact”; June 2002.
http://www.luxinzheng.net/publications/english_WTC.htm

Thomas W. Eagar, Christopher Musso: “Why Did the World Trade Center Collapse? Science, Engineering, and Speculation”, JOM December 2001. <https://studylib.net/doc/27080931/why-did-the-world-trade-center-collapse%3F-article>

when multiple members fail, the shifting loads eventually overstress the adjacent members and the collapse occurs like a row of dominoes falling down.

Frank Greening: “Energy transfer in the WTC collapse”;
https://www.researchgate.net/publication/228801720_Energy_transfer_in_the_WTC_collapse /
<https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=bcad07a4b3975c6c9c098abffb516df56bb54657>

[...] total collapse of both towers would have occurred even without the jet fuel fires.

A.S. Usmani, Y.C. Chung, J.L. Torero: “How Did The WTC Towers Collapse: A New Theory”; Fire Safety Journal, Vol 38 Issue 6 October 2003. <https://era.ed.ac.uk/server/api/core/bitstreams/6dcfefb6-7f5a-412e-94b5-6bb06114e5b5/content>

1. The analysis presented points to a compelling **fire induced collapse mechanism rather unique to the type of structure** that the WTC Twin-Towers represented
2. This analysis also shows that the collapse is initiated principally by a stability mechanism as a result of geometry changes in the structure caused by thermal expansion effects
3. Furthermore it is quite possible that the geometric changes required to precipitate collapse could result from very low temperatures not high enough to induce significant reduction in the material properties
4. It can therefore be provisionally concluded that these buildings **could have collapsed as a result of a major fire event**. This is of course assuming that any of the active fire suppression systems would either fail or be unable to control the development of the fire. This is a normal assumption when designing fire protection for buildings

K.A. Seffen: “Progressive Collapse of the World Trade Centre: a Simple Analysis”; Journal of Engineering Mechanics Volume 134, Issue 2 February 1, 2008. <https://ascelibrary.org/doi/10.1061/%28ASCE%290733-9399%282008%29134%3A2%28125%29>

The collapse behavior of the World Trade Center towers is considered formally as a propagating instability phenomenon.

- Discussion by Crockett Grabbe: March 15, 2010.
[https://ascelibrary.com/doi/10.1061/\(ASCE\)EM.1943-7889.0000025#core-collateral-c5](https://ascelibrary.com/doi/10.1061/(ASCE)EM.1943-7889.0000025#core-collateral-c5)

R. M. Korol, K. S. Sivakumaran: “Reassessing the Plastic Hinge Model for Energy Dissipation of Axially Loaded Columns”; Journal of Structures February 13, 2014.
<https://onlinelibrary.wiley.com/doi/10.1155/2014/795257>

James G. Quinittiere: “Questions on the WTC Investigation”; In: Pasman, H.J., Kirillov, I.A. (eds) “Resilience of Cities to Terrorist and other Threats. NATO Science for Peace and Security Series Series C: Environmental Security”; 2008. https://link.springer.com/chapter/10.1007/978-1-4020-8489-8_5

The NIST conclusion essentially puts the primary cause on the impact of the aircraft, while the alternative conclusion lays it at the feet of fire safety design.

G. Szuladziński: “Temporal considerations in collapse of WTC Towers”; Int. J. Structural Engineering 2012. https://www.researchgate.net/publication/264817987_Temporal_considerations_in_collapse_of_WTC_Towers

Steven Jones, Robert Korol, Anthony Szamboti, Ted Walter: “15 years later: on the physics of high-rise building collapses”; Europhysics News Vol. 47, No. 4 August 24, 2016.
<https://www.europhysicsnews.org/articles/epn/abs/2016/04/epn2016474p21/epn2016474p21.html>

Note: authors include Ted Walter, identified in the article as director of strategy and development for AE911Truth; the article's stated conclusion is that evidence "points overwhelmingly" to controlled demolition of all three WTC buildings — outside this document's stated scope (§I). Europhysics News's own editorial note describes the piece as containing "some speculation." Cited for its account of institutional response to specific claims (§III-2.5.3), not as endorsement of its thesis.

Graeme MacQueen, Tony Szamboti: "The Missing Jolt: A Simple Refutation of the NIST-Bazant Collapse Hypothesis"; Journal of 9/11 Studies January 14, 2009.

<https://journalof911studies.com/volume/2008/TheMissingJolt.pdf>

Ivan Němec, Miroslav Trcala, Jiří Vala, Adéla Vaněčková: "A Contribution to Analysis of Collapse of High-Rise Building Inspired by the Collapses of WTC1 and WTC2 — Derivation of Simple Formulas for Collapse Upper Speed and Acceleration"; Journal of Applied Mathematics and Physics Vol. 6 No 12, 2018.

<https://www.scirp.org/journal/paperinformation?paperid=89496>

Ivan Němec: "Dynamics of Collapse of a High-Rise Building: Inspired by the Collapse of the Twin Towers of the WTC"; Scholars' Press November 18, 2014 <https://www.amazon.com/Dynamics-Collapse-High-Rise-Building-Inspired/dp/3639668162>

Genady Cherepanov: "Mechanics of the WTC collapse"; International Journal of Fracture Volume 141 September 2006. <https://link.springer.com/article/10.1007/s10704-006-0081-8>

The collapse in the regime of progressive failure is shown to occur at an acceleration, which is several times less than the gravitational acceleration [...]

Genady Cherepanov: "A Critical Analysis of the Government's Theory: The Collapse of the New York Towers on September 11, 2001"; Вестник ЧГПУ им. И.Я. Яковлева, Механика Предельного Состояния, № 1, 2008.

"The dynamic instability of columns/bars occurs by higher order modes of buckling (the greater is the dynamic load, the higher is the mode of buckling)."

Note: this paper appeared in a regional Russian pedagogical-university bulletin, a different and less specialized venue than Cherepanov's peer-reviewed International Journal of Fracture papers (CHEREPANOV, above). Cited here for one narrow, checkable technical argument regarding dynamic buckling mode order (§6.11.4), independently corroborated by non-WTC literature (CUI); not as an endorsement of the paper's broader conclusion, which lies outside this document's scope (§I).

Ansgar Schneider:

"The late jolt: Re-examining the World Trade Center catastrophe"; arXiv:1712.06207 2017.

<https://arxiv.org/abs/1712.06207>

Ansgar Schneider: "Energy estimates of progressive floor collapses and the World Trade Center catastrophe"; arXiv:1712.06183 2017. <https://arxiv.org/abs/1712.06183>

Steve Kosciuk, Joel Robbin:
“A model for the collapse of the twin towers”; May 25, 2006.
<https://people.math.wisc.edu/~jwrobbin/angelic/911.pdf>

Jim Hoffman: “Modeling Aspects of the Twin Towers’ Collapse”; 2003
<https://911research.wtc7.net/wtc/models/>
<https://911research.wtc7.net/wtc/models/docs/Collapse2.java>

Physical Evidence

Scott Gabriel Knowles: “Lessons in the rubble: The world trade center and the history of disaster investigations in the United States”; History and Technology 2003.
<https://intel-writers.us/wp-content/uploads/2019/11/Lessons-in-the-rubble-The-world-trade-center-and-the-history-of-disaster-investigations-in-the-United-States.pdf>

In the key first few weeks after the September 11 attacks, the FEMA/ASCE team was not granted full access to the disaster site.

David Kohn: “Culling Through Mangled Steel”; CBS March 7, 2002.
<https://web.archive.org/web/20150112015313/http://www.cbsnews.com/news/culling-through-mangled-steel/>

Professor Abolhassan Astaneh-Asl (UC Berkeley structural engineer): “When there is a car accident and two people are killed, you keep the car until the trial is over... If a plane crashes, not only do you keep the plane, but you assemble all the pieces, take it to a hangar, and put it together. That’s only for 200, 300 people, when they die. In this case you had 3,000 people dead. You had a major machine, a major manmade structure. My wish was that we had spent whatever it takes, maybe \$50 million, \$100 million, and maybe two years, **get all this steel, carry it to a lot. Instead of recycling it, put it horizontally, and assemble it.** You have maybe 200 engineers, not just myself running around trying to figure out what’s going on. After all, **this is a crime scene** and **you have to figure out exactly what happened for this crime, and learn from it.**“

Bill Manning: “Selling Out The Investigation”; fireengineering.com January 1, 2002.
<https://web.archive.org/web/20120805233857/http://www.fireengineering.com/articles/print/volume-155/issue-1/departments/editors-opinion/elling-out-the-investigation.html>

For more than three months, structural steel from the World Trade Center has been and continues to be cut up and sold for scrap. Crucial evidence that could answer many questions about high-rise building design practices and performance under fire conditions is on the slow boat to China, perhaps never to be seen again in America until you buy your next car. [...]
Such destruction of evidence shows the astounding ignorance of government officials to the value of a **thorough, scientific investigation of the largest fire-induced collapse in world history.**
[...]

Fire Engineering has good reason to believe that the “official investigation” blessed by FEMA and run by the American Society of Civil Engineers is **a half-baked farce** that may already have been commandeered by political forces whose primary interests, to put it mildly, lie far afield of full disclosure. Except for the marginal benefit obtained from a three-day, visual walk-through of evidence sites conducted by ASCE investigation committee members-described by one close source as a “tourist trip”-**no one’s checking the evidence for anything.** [...]

Maybe we should live and work in planes. That way, if disaster strikes, we will at least be sure that a thorough investigation will help find ways to increase safety for our survivors. [...]

Based on the incident’s magnitude alone, a **full-throttle, fully resourced, forensic investigation is imperative.** More important, from a moral standpoint, for the safety of present and future generations who live and work in tall buildings-and for firefighters, always first in and last out-the **lessons about the buildings’ design and behavior in this extraordinary event must be learned and applied in the real world.** [...]

To treat the September 11 incident any differently would be the **height of stupidity and ignorance.**

The destruction and removal of evidence must stop immediately.

“WTC scrap sails for India, China”; Daily News January 2002.

<https://web.archive.org/web/20020223145930/http://origin.dailynews.lk/2002/01/22/wor02.html>

Nityanand Jayaraman & Kenny Bruno: “Trading in Disaster - World Trade Center Scrap Lands in India”; corpwatch.org <http://www.corpwatch.org/article.php?id=1608>

“India recycles Ground Zero rubble”; BBC News January 17, 2002.

https://web.archive.org/web/20040615003107/http://news.bbc.co.uk/2/hi/south_asia/1766693.stm

“Baosteel Will Recycle World Trade Center Debris”; Eastday.com January 24, 2002 (republished by china.org.cn, English edition). <http://www.china.org.cn/english/2002/Jan/25776.htm>

New York authorities’ decision to ship the twin towers’ scrap to recyclers has raised the **anger of victims’ families and some engineers** who believe the massive girders should be further examined to help determine how the towers collapsed.

But New York Mayor Mike Bloomberg insisted there are better ways to study the tragedy of September 11.

“If you want to take a look at the construction methods and the design, **that’s in this day and age what computers do,**” said Bloomberg, a former engineering major. “**Just looking at a piece of metal generally doesn’t tell you anything.**”

Asit Jolly: “September 11 steel recycled”; BBC News September 11, 2002.

https://web.archive.org/web/20040728114646/http://news.bbc.co.uk/2/hi/south_asia/2250983.stm

“Scrap Metal from World Trade Center Memorializes September 11 Around the World”; Scrapware.com September 6, 2022. <https://www.scrapware.com/blog/scrap-metal-from-world-trade-center-memorializes-september-11-around-the-world/>

Ching-Ching Ni: “A Font of Grief in New York, a Savvy Transaction in China”; Los Angeles Times January 26, 2002. <https://www.latimes.com/archives/la-xpm-2002-jan-26-mn-24932-story.html>

The families had made emotional appeals to the city of New York to stop the recycling until further clues could be found about the towers’ collapse. New York officials said the investigation could continue without the physical wreckage.

Background

“Twin Towers Structural Drawings Books”; archive.org <https://archive.org/details/fav-gerrygan>

Hans J. Herrmann, Stéphane Roux: “Statistical models for the fracture of disordered media”; North-Holland 1990. <https://books.google.de/books?id=QufMBQAAQBAJ>

William Isaac Newman, S. Leigh Phoenix: “Time-dependent fiber bundles with local load sharing”; Physical Review E January 2001. https://www.researchgate.net/publication/12025109_Time-dependent_fiber_bundles_with_local_load_sharing
https://www.researchgate.net/publication/235745896_Time-dependent_fiber_bundles_with_local_load_sharing_II_General_Weibull_fibers

Naomi Oreskes, Kristin Shrader-Frechette, Kenneth Belitz: “Verification, Validation, and Confirmation of Numerical Models in the Earth Sciences”; Science Vol 263, Issue 5147 February 4, 1994. https://people.uncw.edu/borretts/courses/bio534/readings/Oreskes1994_verification_validation_confirmation_of_numerical_models_in_the_earth_science.pdf

Steven H. Strogatz: “Exploring complex Networks”; Nature 2001. <https://www.nature.com/articles/35065725>

T. Bart Quimby: “A Beginner’s Guide to the Steel Construction Manual, 15th ed.”; 2006. <https://www.bgstructuralengineering.com/BGSCM15/Contents.htm>

Alfred Böge: “Mechanik und Festigkeitslehre”; Vieweg 1992. https://books.google.de/books/about/Mechanik_und_Festigkeitslehre.html?id=AU1iAAAAMAAJ

Richard Feynman: “Cargo Cult Science”; Caltech commencement address 1974. https://web.archive.org/web/20050304032544/http://neurotheory.columbia.edu/~ken/cargo_cult.html

There is one feature I notice that is generally missing in cargo cult science. ... It’s a kind of scientific integrity, a principle of scientific thought that corresponds to a kind of utter honesty — a kind of leaning over backwards. For example, if you’re doing an experiment, you should **report everything that you think might make it invalid — not only what you think is right about it;**

other causes that could possibly explain your results; and things you thought of that you've eliminated by some other experiment, and how they worked — to make sure the other fellow can tell they have been eliminated.

Details that could throw doubt on your interpretation must be given, if you know them. You must **do the best you can** — if you know anything at all wrong, or possibly wrong — to explain it. If you make a theory, for example, and advertise it, or put it out, then **you must also put down all the facts that disagree with it**, as well as those that agree with it. There is also a more subtle problem. When you have put a lot of ideas together to make an elaborate theory, you want to make sure, when explaining what it fits, that those things it fits are not just the things that gave you the idea for the theory; but that the finished theory makes something else come out right, in addition. In summary, the idea is to try to give all of the information to help others to judge the value of your contribution; **not just the information that leads to judgement in one particular direction or another**.

Richard Feynman: "What is Science?"; The Physics Teacher Vol 7, issue 6 1969.
http://www.fotuva.org/feynman/what_is_science.html

Science is the belief in the ignorance of experts.

Henry Petroski: "Design Paradigms: Case Histories of Error and Judgment"; Cambridge University Press May 27 1994. https://books.google.de/books/about/Design_Paradigms.html?id=C_ZroS6rY54C

So when a structure collapses or a design fails to live up to its promise, the lesson should be accessible to all, and even a jury of laypersons is presumed capable of ruling on culpability.

Ulrich BECK:
"Risk Society: Towards a New Modernity"; Sage Publications 1992.
https://books.google.de/books/about/Risk_Society.html?id=W2sDTHaSiYC

All decisions on the risks and hazards of civilization falling within the compass of knowledge production are never just questions of the substance of knowledge [...] They are *at the same time* also decisions on *who is afflicted*, the extent and type of hazard, the elements of the threat, the population concerned, delayed effect, measures to be taken, those responsible, and claims for compensation [...] This makes it clear *that the margins for scientific research become narrower and narrower as the threatening potential increases*. [...] The destructive forces scientists deal with in all fields today impose on them the inhuman law of *infallibility*. Not only is it one of the most human of all qualities to break this law, but the law itself stands in clear contradiction science's ideals of progress and critique (p. 54)

NFPA 921: "Guide for Fire and Explosion Investigations"; <https://www.nfpa.org/codes-and-standards/nfpa-921-standard-development/921>

Chapter 4: Basic Methodology

4.1 Nature of Fire Investigations. A fire or explosion investigation is a complex endeavor

involving skill, technology, knowledge, and science. The compilation of factual data, as well as an analysis of those facts, **should be accomplished objectively and truthfully**. The basic methodology of the fire investigation should rely on the use of a **systematic approach and attention to all relevant details**. The use of a systematic approach often will uncover new factual data for analysis, which **may require previous conclusions to be reevaluated**. With few exceptions, the proper methodology for a fire or explosion investigation is to first determine and establish the origin(s), then investigate the cause: circumstances, conditions, or agencies that brought the ignition source, fuel, and oxidant together.

4.2 Systematic Approach. The systematic approach recommended is that of the **scientific method**, which is used in the physical sciences. This method provides for the organizational and analytical process desirable and necessary in a successful fire investigation.

4.3 Relating Fire Investigation to the Scientific Method. The scientific method (see Figure 4.3) is a **principle of inquiry that forms a basis for legitimate scientific and engineering processes**, including fire incident investigation. It is applied using the following steps [...]

22.2.4 Exotic Accelerants. Mixtures of fuels and Class 3 or Class 4 oxidizers (see NFPA 430, Code for the Storage of Liquid and Solid Oxidizers) may produce an exceedingly hot fire and may be used to start or accelerate a fire. Thermite mixtures also produce exceedingly hot fires. Such accelerants generally leave residues that may be visually or chemically identifiable.

22.4.9.3.7 Extremism. Extremist-motivated firesetting is committed to further a social, political, or religious cause. Fires have been used as a weapon of social protest since revolutions first began. Extremist firesetters may work in groups or as in-dividuals. Also, due to planning aspects and the selection of their targets, extremist firesetters generally have a great degree of organization, as reflected in their use of more elaborate ignition or incendiary devices. Subcategories of extremist firesetting are terrorism and riot/civil disturbance. **(A) Terrorism.** The targets set by terrorists may appear to be at random; however, target locations are generally selected with some degree of political or economic significance. Political targets generally include government offices, newspapers, universities, political party headquarters, and military or law enforcement installations. Political terrorists may also target diverse properties such as animal research facilities or abortion clinics. Economic targets may include business offices, distribution facilities of utility providers (e.g., atomic generation plants), banks, or companies thought to have an adverse impact on the environment. Fires or explosions become a means of creating confusion, fear, or anarchy. The terrorist may include fire as but one of a variety of weapons, along with explosives, used in furthering his or her goal.

Quassim Cassam: "Vice Epistemology"; The Monist Vol 99, Issue 2 April 2016.
<https://academic.oup.com/monist/article/99/2/159/2563406>

Willem de Lint: "Criminology 9/11"; Globalizations Vol 17, 2020 October 2 2019.
<https://www.tandfonline.com/doi/full/10.1080/14747731.2019.1655933>

John D. Wyndham: "Peer Review in Controversial Topics—A Case Study of 9/11"; Publications, Vol. 5, No. 2, June 2017. <https://doi.org/10.3390/publications5020016>

“The Szamboti and Johns paper was rejected by ASCE editors as being ‘out of scope.’ As Szamboti and Johns have since noted, ‘It is not possible for a Discussion paper, one that simply corrects errors in a paper that is already published, to be out of scope for a journal.’”

Note: the author is a founding member and coordinator of Scientists for 9/11 Truth, and the paper states this affiliation and a controlled-demolition conclusion openly. Cited for one narrow, checkable procedural claim — a formal Discussion-paper rejection and its stated grounds — not for the paper’s broader argument or conclusions, which lie outside this document’s scope (§I).

Zdeněk P. Bažant, Er-Ping Chen: “Scaling of Structural Failure”; Applied Mechanics Reviews, Vol. 50, No. 10, October 1997. <http://www.civil.northwestern.edu/people/bazant/PDFs/Papers/S34.pdf>

“This means that, according to all these classical theories, geometrically similar structures of different sizes fail at the same nominal stress... This is the basic, reference case, in which we say that there is no size effect (on the nominal strength).”

Note: general review of Bažant’s own scaling/size-effect research program, independent of the WTC papers — not a WTC-specific source. Cited here for its explicit domain statement (§4.8.5): the quasi-brittle size effect law it reviews applies to fracture-process-zone materials (concrete, rock, ceramics, composites), not to elastoplastic failure — the category the crush-down model’s own F_b mechanism (three-hinge plastic buckling) falls under.

Popular

Mario Salvadori: “Building: The Fight Against Gravity”; Atheneum 1979.
<https://www.google.de/books/edition/Building/ek2kdx3RuLIC>

Mario Salvadori: “Why Buildings Stand Up: The Strength of Architecture”; McGraw-Hill 1982.
https://www.google.de/books/edition/Why_Buildings_Stand_Up/fUBSAAAAMAAJ

Matthys Levy, Mario Salvadori: “How Buildings Fall Down: How Structures Fail”; WW Norton 2002.
https://www.google.de/books/edition/Why_Buildings_Fall_Down/Bwd-MHINMGsC

Tom Harris: “How Building Implosions Work”; howstuffworks.com 2010.
<https://web.archive.org/web/20100805045405/https://science.howstuffworks.com/engineering/structural/building-implosion.htm>

Sometimes, though, a building is surrounded by structures that must be preserved. In this case, the blasters proceed with a true implosion, demolishing the building so that it collapses straight down into its own footprint (the total area at the base of the building). **This feat requires such skill that only a handful of demolition companies in the world will attempt it.**

Kostack Studio: “WTC Simulation 2025, Part 1 (South Tower)”; September 26, 2025.

<https://www.youtube.com/watch?v=WpHXj62Ylw0>

Jean-François Milleron, Daniel Rieber: “Method of demolition by jacking, chassis for (a) jack(s) and thrusting and bracing apparatuses”; European Patent Office March 9, 2012.

<https://patents.google.com/patent/EP2817466B1/en>

L’invention concerne un procédé de démolition d’immeuble par **vérinage**.

Hevesh5: “World Record Domino Structure DESTROYED by Squirrel!”; August 31, 2024.

<https://www.youtube.com/watch?v=Ejuf7f20okc>

“NEW World Record: 240,000 Domino Planks! (The Imaginary City)”; August 30, 2024.

<https://www.youtube.com/watch?v=qVS4vaNfPIQ>

dominoschulrecord (Michael Hoermann, Philipp Zimmermann): “Guinness world record: The tallest domino structure on earth: 10.05 meter (32.9 feet)”; August 22, 2024. <https://www.youtube.com/watch?v=Ni70KXe5jPo>

Mick West: “Towards A Replicable Physical Model Illustrating Aspects of the Collapse of The WTC Towers on 9/11”; metabunk.org March 15, 2016. <https://www.metabunk.org/threads/towards-a-replicable-physical-model-illustrating-aspects-of-the-collapse-of-the-wtc-towers-on-9-11.7396/> /

<https://www.youtube.com/watch?v=flo62pdaIMI>



Debris Piles

Kenneth Chang: "From 5000 Feet Up, Mapping Terrain For Ground Zero Workers"; The New York Times September 23, 2001. <https://www.nytimes.com/2001/09/23/nyregion/from-5000-feet-up-mapping-terrain-for-ground-zero-workers.html>

Precedents

Windsor Tower Madrid 2005. https://en.wikipedia.org/wiki/Windsor_Tower

Plasco Building Tehran 2017. https://en.wikipedia.org/wiki/Plasco_Building

One Meridian Plaza Philadelphia 1991. https://en.wikipedia.org/wiki/One_Meridian_Plaza

Ronan Point London 1968. https://en.wikipedia.org/wiki/Ronan_Point

Science of Collapse

Bruce R. Ellingwood, Donald O. Dusenberry: “Building Design for Abnormal Loads and Progressive Collapse”; Computer-Aided Civil and Infrastructure Engineering Volume 20, Issue 3 May 2005.

<https://onlinelibrary.wiley.com/doi/10.1111/j.1467-8667.2005.00387.x>

Theodor Krauthammer: “Modern Protective Structures”; CRC Press February 2008.

<https://www.taylorfrancis.com/books/mono/10.1201/9781420015423/modern-protective-structures-theodor-krauthammer-theodor-krauthammer>

C. G. Bailey, D. B. Moore: “The Structural Behaviour of Steel Frames with Composite Floorslabs Subject to Fire”; The Structural Engineer Vol 78, Issue 11 2000.

[https://www.istructe.org/journal/volumes/volume-78-\(published-in-2000\)/issue-11/the-structural-behaviour-of-steel-frames-with-comp/](https://www.istructe.org/journal/volumes/volume-78-(published-in-2000)/issue-11/the-structural-behaviour-of-steel-frames-with-comp/)

Michael Byfield, Wjesundara Mudalige, Colin Morison, Euan Stoddart: “A review of Progressive Collapse research and regulations”; Proceedings of the Institution of Civil Engineers - Structures and Buildings August 01, 2014. <https://www.emerald.com/jstbu/article-abstract/167/8/447/408815/A-review-of-progressive-collapse-research-and?redirectedFrom=fulltext> /

https://eprints.soton.ac.uk/372116/2/ICE_20SB_20Prog_20_20Collapse.pdf

Uwe Starossek: “Progressive Collapse of Structures”; ICE Publishing 2009.

<https://www.emerald.com/books/book/18999/Progressive-Collapse-of-Structures>

Edgar V Leyendecker, Bruce R Ellingwood: “Design methods for reducing the risk of progressive collapse in buildings.”; Building Science Series, NIST 1977. <https://www.nist.gov/publications/design-methods-reducing-risk-progressive-collapse-buildings>

The normal design process results in a certain amount of strength and continuity which is also available to resist progressive collapse. This has been demonstrated in the past on a number of occasions. However, since codes, standards, and building materials, as well as construction techniques change with time, it cannot be assumed that this protection will always be provided in the future. Therefore, it is considered necessary that progressive collapse provisions be included in design requirements.

H. Sezen et al.: “Progressive Collapse Testing and Analysis of a Steel Framed Building”; American Institute of Steel Construction (AISC), research report.

<https://www.aisc.org/globalassets/aisc/research-library/progressive-collapse-testing-and-analysis-of-a-steel-framed-building-sezen-et-al.pdf>

Most structural members of [the tested] building exceeded the DCR [demand-to-capacity ratio] limits once the second column was removed. However, the building did not experience [progressive collapse].

The amplification factor of 2.0 is used to account for dynamic effects, such as damping and inertia, when static analysis procedure is used.

WU Bo, HUANG Shi-xiang, ZHAO Xin-yu: “Experimental Study on Pancake Collapse of RC Frame-Shear Wall Structures”; Engineering Mechanics Vol. 29, No. 6 June 2012.

<https://www.engineeringmechanics.cn/article/exportPdf?id=5671&language=en>

In their introduction, the authors describe Bažant and Zhou’s WTC analysis as reducing the impacted lower structure to a single-degree-of-freedom spring and deriving an impact-force ratio from energy conservation alone, and characterize that approach — along with prior single-point member tests — as leaving a substantial gap relative to the impact behavior of a real, collapsing multi-body structure. Their own instrumented test found the maximum total impact load reached 7.90 times the collapsed story’s total static gravity load (self-weight of slab, beams, half of columns and shear walls, plus added mass) when concrete members struck bare steel, dropping to 3.58 times that reference load when a concrete cushion layer was present.

Note: three 1:3-scale, single-story RC frame–shear-wall models, pushed to “pancake” failure and instrumented with sixteen load cells beneath the collapse zone. Cited here for two things: (1) an independently measured load-multiplication factor for a floor-on-floor impact event, several times larger than the $\sim 1.5\text{--}2\times$ dynamic amplification found in the alternate-load-path literature (§SEZEN), and (2) because the paper explicitly frames its own motivation as an empirical check on exactly the class of reduced-order model discussed in this chapter — its authors cite Bažant and Zhou’s single-DOF treatment by name and report that real multi-body impact is measurably more complex. Not a steel high-rise, not a multi-story accreting mass, and not a direct source for β (§6.7.1) — the authors’ own stated conclusion is that more quantitative testing is needed before their correction factor generalizes, and the crush-down regime remains open on that basis.

J.M. Alexander: “An Approximate Analysis of the Collapse of Thin Cylindrical Shells under Axial Loading”; The Quarterly Journal of Mechanics and Applied Mathematics Volume 13, Issue 1 1960.

<https://academic.oup.com/qjmam/article-abstract/13/1/10/1919943>

T. Wierzbicki, W. Abramowicz: “On the Crushing Mechanics of Thin-Walled Structures”; J. Appl. Mech. Dec 1983. <https://asmedigitalcollection.asme.org/appliedmechanics/article-abstract/50/4a/727/390676/On-the-Crushing-Mechanics-of-Thin-Walled>

Daigoro Isobe: “Numerical Simulations for Investigation On True Cause of the Total Collapse of the WTC Towers”; International Symposium on Structures under Earthquake, Impact, and Blast Loading. Osaka, Japan October 10-11, 2008. <https://www.kz.tsukuba.ac.jp/%7Eisobe/187.pdf>

It is possible that the highly-rapid total collapse of the WTC towers was caused not only by the buckling and strength reduction of members due to elevated temperatures, but by the original weakness of the member joints and the destruction of those during the collision of the aircraft.

Daigoro Isobe, Le Thi Thai Thanh, Zion Sasaki: “Numerical Simulations on the Collapse Behaviors of High-Rise Towers”; International Journal of Protective Structures Vol 3 · Number 1 2012.

https://web.archive.org/web/20160116144044/http://www.us-jpri.org/en/reports/tsukuba_20150930.pdf

In general, the tower remained standing for a longer period of time due to the catenary action of the outrigger truss system only if the load paths in the tower were protected and if the member connections were strong enough. However, in these analyses, the collapse speed never reached a value as high as that of the free fall observed in the WTC collapse, which occurred while the splices between column sections still retained their tensile strength. From these results, it is evident that the high-speed collapse of the WTC towers might have been caused by an inherent weakness in their member connections in addition to the destruction directly caused by the aircraft impact.

Nada Elkady, Levingshan Augusthus Nelson, Laurence Weekes, Nirvan Makoond, Manuel Buitrago: “Progressive collapse: Past, present, future and beyond”; Structures Volume 62 April 2024.

<https://www.sciencedirect.com/science/article/pii/S2352012424002832>

Foad Kiakojouri, Valerio De Biagi, Bernardino Chiaia, Mohammad Reza Sheidaii: “Progressive collapse of framed building structures: Current knowledge and future prospects”; Engineering Structures Vol 206 March 1, 2020. <https://www.sciencedirect.com/science/article/abs/pii/S0141029619322576>

Sherif El-Tawil, Honghao Li, Sashi Kunnath: “Computational Simulation of Gravity-Induced Progressive Collapse of Steel-Frame Buildings: Current Trends and Future Research Needs”; Journal of Structural Engineering August 2014.

https://www.researchgate.net/publication/273745457_Computational_Simulation_of_Gravity-Induced_Progressive_Collapse_of_Steel-Frame_Buildings_Current_Trends_and_Future_Research_Needs

Li Shan, Floriana Petrone, Sashi Kunnath: “Robustness of RC buildings to progressive collapse: Influence of building height”; Engineering Structures Vol 183 March 15, 2019.

https://www.sciencedirect.com/science/article/abs/pii/S0141029618327895?fr=RR-2&ref=pdf_download&rr=9c113249dfc54da8

Comparison of progressive collapse in buildings of different heights, with initial column removal beginning at the corner and at the lowest floor, shows that taller buildings possess greater resilience to progressive collapse than buildings with fewer stories and same arrangement of vertical elements in the plan. [...] In conclusion, for the specific study here presented, in all buildings the collapse has a distinct local character, involving the horizontal elements directly connected to the area where columns are removed, and is controlled by fracture of the reinforcing steel. Overall, findings from the study suggest that, although taller buildings possess more robustness than shorter buildings because of their inherent redundancy, it cannot be extrapolated that collapse resistance will continue to be enhanced with even taller buildings.

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[345347816_Assessment_of_Building_Robustness_against_Disproportionate_Collapse/links/5fa48631299bf10f732892b7/Assessment-of-Building-Robustness-against-Disproportionate-Collapse.pdf](https://doi.org/10.1016/j.jcsr.2016.05.001)

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<https://repository.library.northeastern.edu/files/neu:810/fulltext.pdf>

Faber, Michael H.; Kübler, Oliver; Fontana, Mario; Knobloch, Markus: “Failure consequences and reliability acceptance criteria for exceptional building structures – A Study taking Basis in the Failure of the World Trade Center Twin Towers”; Institute of Structural Engineering, Swiss Federal Institute of Technology July 2004. [https://doi.org/https://doi.org/10.3929/ethz-a-004807495](https://doi.org/10.3929/ethz-a-004807495)

Guo-Qiang Li, Jing-Zhou Zhang, Liu-Lian Li, Bin-Hui Jiang, Tao-Chun Yang and Jian Jiang: “Progressive Collapse Resistance of Steel Framed Buildings under Extreme Events”; Advanced Steel Construction – Vol. 17 No. 3 March 13, 2021. https://ascjournal.com/down/vol17no3/Vol17no3_10.pdf

A systematic methodology based on both Monte Carlo simulation and probability density evolution method should be developed to generate a robust and sufficient large dataset for training and testing.

Jian Jiang, Qijie Zhang, Liulian Li, Wei Chen, Jihong Ye, Guo-Qiang Li: “Review on Quantitative Measures of Robustness for Building Structures Against Disproportionate Collapse”; International Journal of High-Rise Buildings Vol 9, No 2, June 2020. [https://global.ctbuh.org/resources/papers/4330-02.127~154\(%E1%84%8C%E1%85%B5%E1%84%8F%E1%85%B2%E1%84%85%E1%85%B5\).pdf](https://global.ctbuh.org/resources/papers/4330-02.127~154(%E1%84%8C%E1%85%B5%E1%84%8F%E1%85%B2%E1%84%85%E1%85%B5).pdf)

Liang Zong, Wanquan Fang, Yichi Zhang, Jian Cui: “Progressive collapse analysis on modular steel construction based on a simplified joint model”; Thin-Walled Structures Vol 198 May 2024.
<https://www.sciencedirect.com/science/article/abs/pii/S0263823124001769>

Wei-Jian Yi, Fan Yi, Yun Zhou: “Experimental Studies on Progressive Collapse Behavior of RC Frame Structures: Advances and Future Needs”; International Journal of Concrete Structures and Materials July 16th, 2021. <https://link.springer.com/article/10.1186/s40069-021-00469-6>

some demolition experiments suggested that **resistance of actual buildings** subjected to multi-column removal **could be higher than anticipated** because of the collaboration of several load-resisting mechanisms.

Shijie Cui, Hee Kiat Cheong, Hong Hao: “Dynamic Buckling of Simply Supported Columns under Axial Slamming”; Journal of Engineering Mechanics, ASCE, Vol. 125, No. 5, May 1999, pp. 513–520.
[https://doi.org/10.1061/\(ASCE\)0733-9399\(1999\)125:5\(513\)](https://doi.org/10.1061/(ASCE)0733-9399(1999)125:5(513))

“The higher the impact velocity is, the shorter the buckled wavelength becomes, and then the fracture point approaches the impact end.”

Note: independent of the WTC literature — general experimental/numerical study of column buckling under axial impact. Cited for the general relationship between impact velocity and buckle-segment (hinge) count (§4.12 / §6.11), not in connection with any WTC-specific claim.

Chun Wa Wong, Seo Ho Youn, Kosuke Yasui: “The falling chain of Hopkins, Tait, Steele and Cayley”; European Journal of Physics, 28(3), 385–400. 2007 <https://arxiv.org/abs/physics/0612250>

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Eugenio Hamm, Jean-Christophe Géminard: “The weight of a falling chain, revisited”; American Journal of Physics, 78(8), 828–833. 2010 <https://pubs.aip.org/aapt/ajp/article-abstract/78/8/828/1058354/The-weight-of-a-falling-chain-revisited>

Boyd F Edwards, Joseph W Wilder, Earl E Scime: “Dynamics of falling raindrops”; European Journal of Physics, Volume 22, Number 2. 2001 https://www.researchgate.net/publication/231005364_Dynamics_of_falling_raindrops

Pictures

Figure 1: The North Tower collapses. Credit: Photo taken by NYPD Detective and helicopter pilot Greg Semendinger. <https://www.yahoo.com/news/us/articles/former-nypd-pilot-reveals-unseen-110243230.html>

Figure 6: The South Tower during construction. Credit: Warren K. Leffler, June 1969, U.S. News & World Report Magazine Photograph Collection. <https://guides.loc.gov/us-news-and-world-report-photograph-collection>

Figure 7: Digital surface model created by LIDAR mapping. Credit: NOAA NGS. <http://www.noaanews.noaa.gov/stories/s781.htm>

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Drafting tools, including multiple large language models, assisted with language polishing (because the author is not a native speaker) and with the implementation and verification of numerical computations; the analytical derivations, parameter choices, data selection, and all interpretations remain the author's, as do any errors. All final editorial decisions, sourcing, and structural framing likewise remain the author's sole responsibility.

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URL: <https://akareyon.wordpress.com/epicycles-of-inevitability/>

PDF: <https://raw.githubusercontent.com/Akareyon/EpicyclesOfInevitability/main/EpicyclesOfInevitability.pdf>

Exhibit A

Timeline: Evolution of the scientific consensus on the collapse of the WTC Twin Towers

Before It Happened

a large cantilever beam

1964 :: Robert E. Rapp, in “The World Trade Center: An Architectural and Engineering Milestone” praises the robustness of the structure to be built for *Contemporary Steel Design*, Vol. 1, No. 4, explaining how it becomes a huge internally-braced cantilever box beam where the Vierendeel truss walls act as web members taking shear; while the walls act as flanges to absorb compressive and tensile forces. [[The American Iron Steel Institute](#)]

could still withstand

1964 Engineering News-Record <http://911research.wtc7.net/mirrors/guardian2/wtc/eng-news-record.htm>

“The World Trade Center towers would have an inherent capacity to resist unforeseen calamities”
[...]

“live loads on [the perimeter columns] can be increased more than 2,000% before failure occurs”
[...]

One “could cut away all the first story columns on one side of the building, and partway from the corners to the perpendicular sides, and the building could still withstand design live loads and a 100 mph wind from any direction.”[...]

could not cause collapse

John Skilling: 1964 (cited in: NIST NCSTAR 1-2, Appendix A)
http://www.nist.gov/el/disasterstudies/wtc/wtc_finalreports.cfm

The buildings have been investigated and found to be safe in an assumed collision with a large jet airliner (Boeing 707-DC8) traveling at 600 miles per hour. Analysis indicates that such collision **would result in only local damage and could not cause collapse.** (pp. 306 ff)

would not be endangered

Thomas W. Ennis: “Critics Impugned on Trade Center; Towers Called Safe”; New York Times February 15, 1964 <http://wayback.archive.org/web/20131227133223/http://www.nytimes.com/1964/02/15/nyregion/15WTC.html>

Richard Roth of the architectural firm of Emery Roth & Sons issued the statement defending the safety of the twin towers, which the firm, in association with Minoru Yamasaki, designed for the center.

Mr. Roth said that a structural analysis by the firm of Worthington, Skilling, Helle & Jackson had found that if a tower were hit by an airliner at 600 miles an hour, the damage to the tower would be only local and its occupants outside the immediate area of impact would not be endangered.

essentially a steel beam

Henry Hold and Company, LLC: “City in the Sky”; Times Books 2003

<http://911research.wtc7.net/wtc/analysis/design.html>

In 1965, real estate baron Lawrence Wien charged that the design of the Twins is structurally unsound. Richard Roth fired back with a 3-page telegram:

THE STRUCTURAL ANALYSIS CARRIED OUT BY THE FIRM OF WORTHINGTON, SKILLING, HELLE & JACKSON IS THE MOST COMPLETE AND DETAILED OF ANY EVER MADE FOR ANY BUILDING STRUCTURE. THE PRELIMINARY CALCULATIONS ALONE COVER 1,200 PAGES AND INVOLVE OVER 100 DETAILED DRAWINGS. ... BECAUSE OF ITS CONFIGURATION, WHICH IS ESSENTIALLY THAT OF A STEEL BEAM 209’ DEEP, THE TOWERS ARE ACTUALLY FAR LESS DARING STRUCTURALLY THAN A CONVENTIONAL BUILDING SUCH AS THE EMPIRE STATE BUILDING WHERE THE SPINE OR BRACED AREA OF THE BUILDING IS FAR SMALLER IN RELATION TO ITS HEIGHT. ... THE BUILDING AS DESIGNED IS SIXTEEN TIMES STIFFER THAN A CONVENTIONAL STRUCTURE. THE DESIGN CONCEPT IS SO SOUND THAT THE STRUCTURAL ENGINEER HAS BEEN ABLE TO BE ULTRA-CONSERVATIVE IN HIS DESIGN WITHOUT ADVERSELY AFFECTING THE ECONOMICS OF THE STRUCTURE. (pp. 134 ff)

some damage

1988 :: *Structural engineer Charlie Thornton explains that a fully loaded 747 — a three hundred ton element — crashing into a building that has been designed to carry thirteen thousand tons would probably not do anything to the major building. It could affect localized structural elements, knock out a column, and there could be some damage, but as far as a plane knocking a building over of that type, that would not happen.* [YT]

the building structure would still be there

Eric Nalder: “Twin Towers Engineered To Withstand Jet Collision”; The Seattle Times February 27 1993

<https://web.archive.org/web/20080614022913/https://community.seattletimes.nwsources.com/archive/?date=19930227&slug=1687698>

Skilling, based in Seattle, is among the world’s top structural engineers. He is responsible for much of Seattle’s downtown skyline and for several of the world’s tallest structures, including the Trade Center.

Concerned because of a case where an airplane hit the Empire State Building, Skilling's people did an analysis that showed the towers would withstand the impact of a Boeing 707.

"Our analysis indicated the biggest problem would be the fact that all the fuel (from the airplane) would dump into the building. There would be a horrendous fire. A lot of people would be killed," he said. "The building structure would still be there."

Skilling - a recognized expert in tall buildings - doesn't think a single 200-pound car bomb would topple or do major structural damage to a Trade Center tower. The supporting columns are closely spaced and even if several were disabled, the others would carry the load.

"However," he added, "I'm not saying that properly applied explosives - shaped explosives - of that magnitude could not do a tremendous amount of damage."

just a pencil puncturing a steel netting

January 25th, 2001 :: *The towers' construction manager, Frank deMartini, explains the building was designed to have a fully loaded 707 crash into it. He believes the building could probably sustain multiple impacts because the structure is like the mosquito netting on a screen door — an intense grid, and the jet plane is just a pencil puncturing that screen netting, really doing nothing.* [[YT](#)]

noone thought

September 11th, 2001, 9:05 a.m. :: In the lobby of the north WTC tower, just after the South Tower is hit, Fire Commissioner Thomas Von Essen speaks briefly to Fire Chief *Ray Downey*. According to Von Essen, Downey — who is a highly respected expert on building collapses — says to him, "You know, these buildings can collapse." Von Essen later recalls, "He just said it in passing, not that these buildings will collapse in 40 minutes and we have to get everybody out of here, or not that they'll collapse by tomorrow, or not that they necessarily will collapse at all. Just that they can collapse." But other firefighters do not appear to have shared this concern. According to the National Institute of Standards and Technology (NIST), the Fire Department command officers who are planning for operations inside the Twin Towers expect that there will "be localized collapse conditions on the damaged fire floors," but do "not expect that there [will] be any massive collapse conditions or complete building collapse." At the end of its three-year investigation of the WTC collapses, NIST will report, "No one interviewed indicated that they thought that the buildings would completely collapse." [National Institute of Standards and Technology, 9/2005, pp. 72 and 75-76] Deputy Fire Commissioner Lynn Tierney will meet up with Downey and others—including Von Essen—slightly later, on the south lawn of the WTC complex, where a new command center is set up. At that time, according to Tierney, Downey will only be concerned that the 360-foot antenna atop the North Tower will fall, and "No one ever thought the towers were going to come down." [Pittsburgh Tribune-Review, 9/11/2006] However, shortly before the first tower comes down, EMT Richard Zarrillo will be asked to relay a message to some senior firefighters that the mayor's Office of Emergency Management "says the buildings are going to collapse". And later in the day, Mayor Giuliani will recount that around the same time, he had been told "that the World Trade Center was going to collapse". The 9/11 Commission will state, "The best estimate of one senior [fire] chief, provided to the chief of the department sometime between 9:25 and 9:45, was that there might be a danger of collapse [of the South Tower] in a few hours, and therefore units probably should not ascend above floors in the sixties." The Commission does not state, however, whether this fire chief was referring to a total building collapse or just a localized collapse. [[historycommons](#)]

As It Happened

oh my god

September 11th, 2001, during and seconds after South Tower collapse :: *Reporters are at a loss of words, speak of a “huge explosion”, compare the collapse to a controlled demolition, speculate a large part fell off or that the towers were reduced to a pile of rubble.* [[YT](#)]

a witness

September 11th, 2001, minutes after North Tower collapse :: *Mark “Harley Guy” Walsh witnessed both collapses, which happened mainly due to structural failure; because the fires were just too intense.* [[YT](#)]

we believe

September 11th, 2:40pm :: Asked at a press conference about the cause of the explosions that brought the two buildings down, mayor *Rudy Giuliani* replies “We believe that it was caused by the after effects of the planes hitting the buildings.” [[YT](#), [transcript](#)] :: For the collection of oral histories by journalists who covered the attacks “Running Toward Danger - Stories Behind the Breaking News of 9/11” (July 16th, 2002), *Beth Fertig*, WNYC reporter, will remember: “I spent the rest of the afternoon at the mayor’s command center. The reporters were trying to figure out what happened. We were thinking that bombs had brought the buildings down. The mayor [Giuliani] talked to us and said he had no evidence of bombs.” (p. 203) [[Running Towards Danger](#)]

First reactions

lucky design choice

September 12th, 2001 :: *New Scientist* says a lucky design choice saved lives. Most buildings would have come down immediately, but the towers remained upright for nearly an hour. Eventually raging fires melted the supporting steel struts, but the time delay allowed hundreds of people to escape. It’s a good thing they didn’t fall over. [[New Scientist](#)]

shredding downward

September 12th, 2001 :: *Arizona Wildcat* reports that according to experts, although the impact of the jetliners was strong, it was the heat from the explosion that most likely caused the buildings to collapse and cites *Richard Ebeltoft*, a structural engineer and University of Arizona architecture lecturer, who speculates that flames fueled by thousands of gallons of aviation fuel melted the buildings steel supports. Once the top of the building lost balance, the floors appear to have begun shredding downward. [[Arizona Wildcat](#)]

designed to do that architecturally

September 12th, 2001 :: *CBS Local News NY* cite *PANY* sources explaining that the building actually collapsed the way it did – as if it was imploding unto itself – because it was designed to do that

architecturally in order to minimize collateral damage. Had it gone into a million different directions, it could have done even more damage. The towers were designed to fall that way if they were ever met with tragedy. [\[YT\]](#)

careful calculation or evil luck

September 12th, 2001 :: The *Los Angeles Times* article “2 Planes Hit Twin Towers at Exactly the Worst Spot” quotes experts *Nabih Youssef* (structural engineer heading the Tall Building Council in Los Angeles, expert on design and strength of skyscrapers), *Greg Fenves* (professor of civil engineering at UC Berkeley), *Ron Hamburger* (chief structural engineer for ABS Consulting in Oakland; past president of the Structural Engineers Assn. of California), *Scott Gustafson* (owner of Demtech Inc. of Blue Springs, Mo., one of the world’s leading demolition experts), *Hank Koffman*, *Ron Klemencic* (president of Skilling Ward Magnusson Barkshire (the Seattle firm that engineered the World Trade Center)), *James C. Anderson* (professor of civil engineering at USC), *Jon Magnusson* (chairman and chief executive of the Skilling firm) and *John Hooper* (structural engineer with the Skilling Ward Magnusson firm): the terrorists hit the buildings at their weakest spot to cause their disastrous collapse like hitting someone at the back of the knee, knowing what they were doing, as they showed some knowledge of physics in the attempt to make the hits as low as possible. Demolition of a building the size of the Twins would require hundreds of charges around the building, and it’s inconceivable anyone would be able to place that many charges, even with years of planning. *Hamburger* is “personally very surprised to see the entire building collapse”. The impact and heat generated burning jet fuel would suffice to destroy the buildings instead. One thing led to another, and it just kept snowballing – the terrorists were evil geniuses: one floor falls on top of the floor below it, and with one floor falling on top of another there’s no way to stop it. The south tower collapsed first, even though hit by the second plane, because the fireball was larger and because the plane hit the corner of the building, rather than the center, where there is more structural support. However, the tubular design is a famous, very well-designed structural system. Steel buildings in general are known for their strength – even less well-designed steel buildings survived the 1906 earthquake in San Francisco and the 1933 Long Beach quake. [\[LA Times\]](#)

so predictable

September 13th, 2001 :: *BBC* depict the towers’ core as a single steel reinforced concrete column, explain that the tower’s ultimate collapse was inevitable because once the steel frame on one floor had melted, it collapsed downwards, inflicting massive forces on the already-weakened floor below. Structural engineer *Chris Wise* says “it was the fire that killed the buildings. There’s nothing on earth that could survive those temperatures with that amount of fuel burning. The columns would have melted, the floors would have melted and eventually they would have collapsed one on top of each other” and “the top was acting like a huge piledriver” :: The buildings’ construction manager *Hyman Brown* says “Steel melts, and 24,000 gallons of aviation fluid melted the steel. Nothing is designed or will be designed to withstand that fire” and *John Knapton*, professor in structural engineering at Newcastle University, UK, knew the building would fall within two hours: “The eventual collapse of the twin towers was so predictable that the order should have been given to withdraw emergency services within an hour.” [\[BBC News\]](#)

doomed

September 13th, 2001 :: *Professor Z.P. Bažant* circulates first drafts of a simple analysis of the likely scenario. Assuming a collapse time of ~9s, with the stated aim to prove that the towers must have collapsed

and do so in the way seen, approximates that under most optimistic assumptions, the towers were doomed because the percentage of the kinetic energy dissipated plastically is of the order of 1%. The reason is the dynamic consequence of the prolonged heating of the steel columns to very high temperature. The heating caused creep buckling of the columns of the framed tube along the perimeter of the structure, which transmits the vertical load to the ground. Since the plastically dissipated energy $W[p]$ is, optimistically, of the order of 0.5 GN m, and the total release of gravitational potential energy is $W[g] = mg \cdot 2h = 4.2 \text{ GN m}$, even under the most optimistic assumptions by far, the plastic deformation can dissipate only a small part of the kinetic energy acquired by the upper part of building. :: The paper, co-authored by *Yong Zhou*, is published in the *Journal of Engineering Mechanics* January 1st, 2002. [[Source](#)]

limit state conditions

September 17th, 2001 :: *G. Charles Clifton*, HERA Structural Engineer, for *tenlinks.com* explains that the gravity and lateral load-resisting systems were designed to deliver the strength and stiffness required from a 110 story building with minimum dead load. This was achieved very well, with a steelwork weight of only 44.5 kg/m² floor area. The very light and open structure, superbly engineered to meet the design serviceability and ultimate limit state conditions on a building of this height and size, probably made the buildings more vulnerable to collapse from the aircraft impact than would have been the case for a more inefficient and heavier structural system. In his opinion the fires had a less important role to play in the collapse of both towers than the damage from the initial impact. It took both to cause the collapse, however the fire was in no way severe enough to have caused the collapse on its own. [[TenLinks.com](#)]

no clue

September 17th, 2001 :: According to the *enr.com* cover story "Massive Assault Doomed Towers - Terrorist Attacks Brings Down World Trade Center" by *Nadine M. Post* with *Sherie Winston*, *Jon D. Magnusson*, chairman-CEO of Skilling Ward Magnusson Barkshire Inc., Seattle, one of the successor firms of Skilling Ward Christiansen Robertson, structural engineer for the original World Trade Center, says: "From what I observed on TV, it appeared that the floor diaphragm, necessary to brace the exterior columns, had lost connection to the exterior wall." When the stability was lost, the exterior columns buckled outward, allowing the floors above to drop down onto floors below, overloading and failing each one as it went down. :: *Mark Loizeaux*, president of Controlled Demolition Inc., says: the 1,362-ft-tall south tower, which was hit at about the 60th floor, failed much as one would fell a tree. That is what was expected. But the 1,368-ft-tall north tower, similarly hit but at about the 90th floor, "telescoped". It failed vertically, rather than falling over. Regarding the cause of the telescoping, he says "I don't have a clue." [[ENR 1](#), [ENR 2](#)]

assured for years

September 20th, 2001 The *Los Angeles Times* article "A Weapon Unfathomed: Jet Fuel" by *John J. Goldman*, *David Zucchino* and *Matea Gold* remarks that the sight of those towers crumbling so quickly was never anticipated by generations of trade center officials and Fire Department commanders, who were assured for years by engineers that the towers could stand up to a crashing Boeing 707, quoting Fire Lt. *Andrew Graf* as saying "Our guys stayed in there so long because they didn't think the building would come down." :: Project engineer for the Twin Towers *Hyman Brown* says "we thought of the structural impact of a plane hitting, but not the burning fuel." [[LA Times](#)]

strong enough

October 2001 :: *Tufts Journal* quotes *Masoud Sanayeis*, professor of civil and environmental engineering, saying the buildings were strong enough to survive the impact of the airplanes but collapsed as a result of the ensuing fire. The buildings were designed using a tube-in-tube structural arrangement, creating a system that is very stiff and strong and able to resist lateral loads such as wind or an earthquake. The towers survived the initial impact. But the large amount of jet fuel delivered to the towers followed by explosions and subsequent fire weakened the floor systems and the columns. He suspects the fire created temperatures higher than what is normally experienced in office building fires caused by burning furniture or rugs or paper. The building still had to carry the massive loads of higher floors above the plane crash location. The structural columns could no longer carry the gravity loads of the floors above the fire. In addition, the heat could have weakened the floor systems above, as well as the floor-to-column connections. This combination caused the top-down collapse, producing a domino effect. The exterior columns that formed the outside tube of the World Trade Center buildings guided the self-contained collapse within these buildings. It looked like a designed and time-delayed implosion that took only a few seconds to bring each tower down. If there had been no fire, potentially the buildings could have survived. [[Tufts Journal](#)]

performed admirably

November 7th, 2001 :: *Scientific American* reports on a panel of Boston area-based civil and structural engineers who convened to discuss the fate of the superskyscrapers on the campus of the Massachusetts Institute of Technology. *Robert McNamara* says that the World Trade Center was probably one of the more resistant tall building structures. The buildings displayed a tremendous capacity to stand there despite the damage to a major portion of the gravity system. The lateral truss systems redistributed the load when other critical members were lost. It's a testament to the system that they lasted so long. According to him, the planes hit at just the right place. The truck bomb attack showed that the explosion at the bottom had little effect and that it's much easier to collapse a building from the top than the bottom. If they had hit the very top of the building, the fire damage wouldn't have had such a catastrophic effect. At the bottom, the columns are much heavier and stronger and so they would have taken a much larger load. :: *Eduardo Kausel*, M.I.T. professor of civil and environmental engineering, says the World Trade Center was never designed for the massive explosions nor the intense jet fuel fires that came next — a key design omission. Kausel says the towers performed admirably; they stood long enough for the majority of the people to be successfully evacuated. He approximates that about 30 percent of the collapse energy was expended rupturing the materials of the building, while the rest was converted into the kinetic energy of the falling mass, concluding that the the largest share of the kinetic energy was converted to heat, material rupture and deformation of the ground below. He goes on to explain that the intense heat softened or melted the structural elements—floor trusses and columns—so that they became like chewing gum, and that was enough to trigger the collapse. The floor trusses are likely to have been the first to sag and fail. As soon as the upper floors became unsupported, debris from the failed floor systems rained down onto the floors below, which eventually gave way, starting an unstoppable sequence. The dynamic forces are so large that the downward motion becomes unstoppable. He determines that the fall of the upper building portion down onto a single floor must have caused dynamic forces exceeding the buildings' design loads by at least an order of magnitude and performed computer simulations indicating the building material fell almost unrestricted at nearly the speed of free-falling objects. According to his hypothesis, the towers' resistive systems played no role. Otherwise the elapsed time of the fall would have been extended. Asked why the top did not tip over, he answers: "A tree is solid, whereas building is mostly air or empty space; only about 10 percent is solid material. Since there is no solid stump underneath to force it to the side, the building cannot tip over. It could only collapse

upon itself.” :: Unnamed, but well-informed correspondents are quoted with the suggestion that the aviation fuel fires burned sufficiently hot to melt and ignite the airliners’ aluminum airframe structures. Aluminum, a pyrophoric metal, could have added to the conflagrations. Hot molten aluminum could have seeped down into the floor systems, doing significant damage. [[Scientific American](#)]

avalanche theory

November 11th, 2001 :: *James Glanz* in “A NATION CHALLENGED: THE BUILDINGS; In Collapsing Towers, a Cascade of Failures” for the *NY Times* also cites *Jon Magnusson* as saying that the most likely series of events would involve the floor supports. If a member like the connections around the outside of the floors failed in a fire, that could mean the whole floor could go. :: The exterior columns could then have buckled under the tremendous weight above them, says *William F. Baker*, in charge of structural engineering at the architectural firm Skidmore, Owings & Merrill. :: *Eduardo Kausel* believes that the avalanche theory is correct. Once the columns on those few floors buckled, the rest of the columns had no chance of stopping the collapse: “they simply popped out of the way of the avalanche like matchsticks”. :: *Yogesh Jaluria*, professor of mechanical and aerospace engineering at Rutgers, is cited saying that the jet fuel immediately began burning at much higher temperatures than those of ordinary office fires, and that the temperature would have ‘jumped exponentially’ at the center of the impact zone. :: *Matthys Levy* doesn’t disagree that the floors must have collapsed at some point, but doesn’t think that’s enough to cause the building to go down. [[NY Times](#)]

within a nanosecond

November 19th, 2001 :: *John Seabrook* interviews *Jon Magnusson*, who says that ninety-nine per cent of all buildings would collapse immediately when hit by a 767 :: Among the dozens experts in the construction of tall buildings Seabrook has spoken to, only one said that he knew immediately, upon learning, from TV, of the planes’ hitting the buildings, that the towers were going to fall. This was *Mark Loizeaux*, the president of Controlled Demolition Incorporated that specializes in reducing tall buildings to manageable pieces of rubble. “Within a nanosecond,” he told. “I said, ‘It’s coming down. And the second tower will fall first, because it was hit lower down.’” [[New Yorker](#)]

a number of vulnerabilities

November 29th, 2001 :: *Ronald O. Hamburger* gives a *John A. Blume Distinguished Lecture*, explaining: “These buildings were incredibly strong, especially with respect to resisting dead loads and wind loads, but they also had a number of vulnerabilities. [...] We have reason to believe that, without the fire, the buildings could have stood indefinitely and been repaired.” Steel is born of fire. As it’s reheated, it expands and loses its rigidity. The extreme heat from the fires might have caused the steel floors to expand and bow, which may have caused the support columns to bend inward and buckle. Heat also may have caused the steel flooring to separate from the columns, or the columns themselves may have heated up and buckled outward. “Think of the impact of dropping a 25-story building straight down ... It was like a pile driver, which is why it collapsed as it did.” [[Stanford University Blume Earthquake Engineering Center](#)]

hasn’t fallen yet

December 6th, 2001 :: *Blair Kamin*, architecture critic, for *The Chicago Tribune* interviews senior vice president at Construction Technology Laboratories in north suburban Skokie, *W. Gene Corley*: “I said: ‘At

least it hasn't fallen yet. But if they don't get the fire out, it will.'" "The aircraft hits wounded the buildings, but they could have recovered if it weren't for the fire" :: The article also states that no one has authoritatively explained why One World Trade Center remained standing for 1 hour and 40 minutes while Two World Trade Center crashed to the ground just 56 minutes after impact. Another mystery is why smaller, nearby structures remained standing even though fire and steel from the twin towers rained down on them and even ripped huge chunks out of their facades. :: It also names [three possible collapse scenarios](#): exterior column, interior column or floor truss failure. [[Chicago Tribune](#)]

95 percent air

December 2001 :: *Thomas W. Eagar and Christopher Musso* offer one of the first peer-reviewed expert explanations with *Why Did the World Trade Center Collapse? Science, Engineering, and Speculation*, emphasizing: "Nearly every large building has redundant design that allows for loss of one primary structural member, such as a column. However, when multiple members fail, the shifting loads eventually overstress the adjacent members and the collapse occurs like a row of dominoes falling down. [...] As the joists on one or two of the most heavily burned floors gave way and the outer box columns began to bow out-ward, the floors above them also fell. The floor below (with its 1,300 t design capacity) could not support the roughly 45,000 t of ten floors (or more) above crashing down on these angle clips. This started the domino effect that caused the buildings to collapse within ten seconds [...] It has been suggested that it was fortunate that the WTC did not tip over onto other buildings surrounding the area. [...] the building is not solid; it is 95 percent air and, hence, can implode onto itself. [...] a 500,000 t structure has too much inertia to fall in any direction other than nearly straight down. The World Trade Center was not defectively designed. [...] As scientists and engineers, we must not succumb to speculative thinking when a tragedy such as this occurs. Quantitative reasoning can help sort fact from fiction, and can help us learn from this unfortunate disaster." :: [[JOM](#)]

The Following Years

would have collapsed without fire

January 9th, 2002 :: *Richard Gewain*, a senior engineer at Maryland-based Hughes Associates and a member of the federally designated panel of engineers probing the collapses, says the major insult was the plane knocking out the columns, and destroying the connections to the floors. Conceivably, the building would have collapsed without the fire. The impact was so tremendous that each building's fireproofing was rendered irrelevant. [[Source](#), [Mirror](#)]

as a result, the building collapsed

March 6th, 2002 :: Funded by the *Federal Emergency Management Agency*, Dr. *Abolhassan Astaneh-Asl*, professor of civil and environmental engineering at the University of California, Berkeley, under the Directorate of Engineering of the National Science Foundation (NSF), is one of five expert witnesses invited to testify before the Committee on Science of the U.S. House of Representatives at the hearing "Learning from 9/11 – Understanding the Collapse of the World Trade Center". Dr. Astaneh-Asl's presentation includes a movie of a simulated crash of a passenger aircraft onto a generic steel structure created with MSC.Dytran and MSC.Marc Nonlinear Software. While the impact of the crash substantially weakened the structure, "the remaining steel beams and columns melted because of the extreme temperatures. Also buckling in the trusses

was possible, especially after the damaged crossbeams were unable to provide lateral support. As a result, the load could no longer be distributed and the building collapsed.” [[Source](#)]

one after another

April 30th, 2002 :: *NOVA (PBS)*, for “Why the Towers Fell”, create an animation showing how the columns were a loose bundle of straws, gaining all their stiffness solely from the floor slabs, and then of a pancake collapse that leaves the core standing up. *Matthys Levy* (author of “Why buildings fall down”) explains: “As the steel began to soften and melt, the interior core columns began to give. Then you had this sequential failure that took place where it all pancaked — one after the other.” [[YT](#), [transcript \(PBS\)](#)]

proper parameters

June 2002 :: *Lu Xinzhen* and *Jiang Jianjing* present a LS-DYNA FEA “Simulation for the Collapse of WTC after Aeroplane Impact”. They say the simulation results are very close to the real situation, which means that such type of special damage process can be recurred on the computer with proper parameters and numerical model. The results show that the direct reason for the collapse is the softening of steel under fire and the chain reaction damage of floors under the impact load of upper floors. If the fire resistance and the ductility of the structure are improved, the collapse may be avoided. As the fracture of steel is considered in this analysis, the failure plastic deformation is discussed as parameters, whose value is set as 0.5%, 1% and 5% respectively. If the fracture plastic strain of steel structure is 0.5%, the chain collapse will take place entirely. However, if the fracture strain is improved to 1%, the impact energy of upper floors will be absorbed by the lower structures and the chain collapse will be stopped at about 100m under the airplane impact zone. When the fracture strain is improved to 5%, only part of the structure near the airplane impacting zone will be damaged, and no chain collapse will take place. Hence, if the structure has enough ductility to absorb the energy of upper floors’ collapse, the chain damage will be controlled. Even though the influence of heap load is considered, the towers still have much larger chance to escape from the entire collapse. [[Source](#)]

FEMA: rapidly converted energy

September 1st, 2002 :: *FEMA* publishes the “World Trade Center Building Performance Study” and finds that the structure was able to successfully redistribute the building weight after the plane impacts to the remaining elements and to maintain a stable condition, and that the structure may have been able to remain standing in this weakened condition for an indefinite period. However, fires heated floor framing and slabs, causing expansion, which developed additional large stresses in some members. As the temperature of column steel increased, the yield strength and modulus of elasticity degraded and the critical buckling strength of the columns decreased and initiated buckling, a significant effect in the failure of the interior core columns. Once collapse initiated, much of the potential energy was rapidly converted into kinetic energy. As the large mass of the collapsing floors above accelerated and impacted on the floors below, it caused an immediate progressive series of floor failures, punching each in turn onto the floor below, accelerating as the sequence progressed in a pancake-type collapse of successive floors. [[FEMA](#)]

no biggie

September 11th, 2002 :: For *Naples Daily News*, *Ralf Kircher* interviews *Bernard Panto*, who worked in various positions as an engineer on the World Trade Center construction project. He had heard that a plane had hit the World Trade Center, but had thought to himself that it was nothing to worry about. “No biggie,” he said to himself. “Can’t knock that building down.” [[Naples Daily News](#)]

no structural flaw or weak link

October 22nd, 2002 :: *James Glanz* and *Eric Lipton* report in *Expert Report Disputes U.S. On Collapse*: “Experts commissioned in a \$4 billion insurance case involving the World Trade Center have come to conclusions that fundamentally contradict a federal investigation into what caused the twin towers to collapse. The new analysis, according to several experts who have examined the confidential findings, holds that the unusual engineering design of the towers did not contribute to the collapses, and that the damage caused by the planes and the resulting fires made the failures of the buildings inevitable. The analysis, paid for by Larry A. Silverstein, who held the lease on the trade center, finds that there was no structural flaw or weak link that could have led to the collapse. The earlier inquiry concluded that unconventional lightweight floor supports in the twin towers probably softened and then failed, leading to a progressive collapse of the entire buildings. [...] Produced by some of the most prominent engineering firms in the country, the analysis is the first to suggest a detailed, step-by-step, alternative explanation for the destruction of the towers. It concludes that the damage to the exterior and interior support columns caused by the hijacked planes, and the devastating fires that erupted inside, rendered vertical steel support columns unable to hold up the weight of the buildings above the floors of impact.” [[New York Times](#)]

bolts popped and fell apart

October 28th, 2002 :: *Jaime Holguin* for *CBS* reports that the single-bolt connections in the framework of the World Trade Center popped and fell apart, causing the floors to collapse on top of each other. According to an analysis conducted by a team of researchers from the *Massachusetts Institute of Technology*, the bolts did not properly secure the towers’ steel floor trusses. The bolts were pulled toward the center of the buildings while the floor trusses sagged. [[CBS News](#)]

might have remained standing

February 5th, 2003 :: According to an analysis by *James G. Quintiere* of the University of Maryland, College Park, thinks the thickness of the surviving fire insulation, rather than the destruction of insulation during the impacts, explains why the towers collapsed when they did. If the insulation had matched that in the north tower, the south tower would have stayed standing longer. If both towers had had insulation over 50 millimetres thick, they might not have collapsed at all. His analysis calls into question the safety of other buildings constructed to the same standards as the twin towers. However, the Port Authority of New York, the owner of the twin towers, rejects his theory. :: *Frank Lombardi*, the Port Authority’s chief engineer, insists that the thickness of the insulation is irrelevant. He says the impacts dislodged much of the fireproofing on the trusses. This view is supported by *Gene Corley*, who led the FEMA’s investigation last year. “I do not believe the insulation was substandard,” he says. However, he concedes that if extra insulation had been applied and had remained in place after the planes hit, the buildings would have remained standing longer. :: FEMA investigators should have looked more closely at this issue, *Quintiere* says. “We have a

distorted, flawed and incomplete presentation to the public of why the WTC twin towers collapsed due to fire.” [[New Scientist](#)]

too fragile to sustain such motions

April 10, 2003 :: Researchers at MIT compile eight articles for the book *The Towers Lost and Beyond*. The second, *Inferno at the World Trade Center, NY* by *Eduardo Kausel*, explains: “From an engineering point of view, there were three causes to the massive structural damage that led to ultimate failure. These are the impact of the aircraft, the subsequent fireball, and most importantly, the raging fire caused by the vast amounts of jet fuel carried by the planes. Burning fuel must have also cascaded down floor openings to the levels below. [...] Local collapse of the floors caused the heat-weakened columns to lose their lateral support, which under the intense weight of the floors above the level of the fire caused them to buckle, break and roll out like matchsticks. At that point, the upper floors began to fall wholesale onto the structure below, and as they gained momentum, their crushing descent became unstoppable. Indeed, with two fairly simple dynamic models I developed in the hours following the collapse, I determined that the fall of the upper building portion down the height of a single floor must have caused dynamic forces exceeding the design loads by at least an order of magnitude (i.e. more than 10 times the weight of the upper floors). Thus, there was no way in the world that the columns below could have taken this large overload, and as these gave way, an avalanche down the building ensued causing the 110 story towers to collapse in about 12 seconds in what was practically a free fall. [...] In essence then, the towers did not collapse like trees because the structures, despite their strength, were too fragile to sustain such motions. [...] From a practical viewpoint, the chance that any individual building out of hundreds of thousands in the United States might suffer an attack is so small that it would not make economic sense to attempt making them jet-crash proof” [[Eduardo Kausel: The Towers Lost and Beyond](#)]

vulnerable point

November 24th, 2003 :: *The Sunday Times*, in an article called “Kamikaze attackers may have known twin sisters’ weak spot”, cite a top British structural engineer, *Gordon Masterson*, who says the airplanes crashed into the Twins at their most vulnerable spot, which suggests the suicide attackers may have known where to strike. Structural damage from this impact, and the searing heat of blazing aviation fuel, then combined to bring down the top 20 floors. The accumulating weight caused both buildings to collapse, floor by floor, faster and faster. If they had struck right at the top of the tower that would have had a greater effect on the foundations, but the fire would have been more contained. However, they struck in both cases at an extremely vulnerable point. :: *Marc Mimram* says very tall buildings are prey to a phenomenon called moment tensor inversion - a shock from a seismic wave or impact that ripples through the building but has a whiplash, leveraged effect on its higher floors. [[Sunday Times](#)]

collapse could be avoided

December 1st, 2003 :: *Abdolhassan Astaneh-Asl* in “World Trade Center Collapse, Field Investigations and Analyses” opines that the highly redundant exterior tube of the World Trade Center with many closely spaced columns was able to tolerate the loss of many columns and support the gravity while almost all occupants who could use a stairway escaped to safety. The collapse of the towers was most likely due to the intense fire initiated by the jet fuel of the planes and continued due to burning of the building contents. It is also the opinion of the author that had there been better fireproofing installed to delay the steel structure,

specially the light weight truss joists and exterior columns from reaching high temperature until the content of the buildings burned out, probably the collapse could be avoided and the victims above the impact area rescued. [[Source](#)]

9/11 Report: hollow steel shaft

July 22nd, 2004 :: The *National Commission on Terrorist Attacks Upon the United States*, an independent, bipartisan commission created by congressional legislation and the signature of President George W. Bush, issue their final report which states that the interior core of the buildings was a hollow steel shaft. [[Source](#)]

only way they could fail

July 23rd, 2004 :: *Liz Else* interviews *Mark Loiseaux* for *The New Scientist* article “Baltimore Blasters”, who says when he saw that it was an airliner that hit, he remembered the long clear span configuration from the central core to the outer skin of the World Trade Center from a school report he did, and *CDI* had just taken down two 40-storey structures in New York. “I still had some cellphone numbers so when the second plane hit I said: ‘Start calling all the cellphones, tell them that the building is going to come down.’ [...] And I sat there watching, I picked up the phone and I called a couple of people on the National Research Council Committee involved in assessing the impact of explosives. They said: ‘What do you think this is, that they’re going to fail, they’re both going to fail?’ The expression around was they’re going to pancake down, almost vertically. And they did. **It was the only way they could fail. It was inevitable.** And it was horrific.” Asked whether they could have been built in such a way that they would have withstood the impact, he replies: “Bad question - they did withstand the impact. The correct question is could they have been built to withstand the consequences, the fire? [...] I’ll defer to the reports coming out, but I will say - is society willing to pay for it? It’s far cheaper to take the battle to terrorists than let them bring it to us.” [[New Scientist](#)]

comes straight down

October 21st, 2004 :: According to *Boulder Weekly*, construction manager for the World Trade Center and University of Colorado civil engineering professor *Hyman Brown* explains that the airplane fuel and the fire-suppression system caused the building to collapse. The fire-suppression system blocks off five-floor blocks so the fire can’t go up or down. The steel in that five-floor area melts. All the tonnage above the five-floor area comes straight down when the steel melts. That broke all the connections, and that caused the building to collapse. [[Boulder Weekly](#)]

just had to lose some of their structural strength

February 9th, 2005 :: *PM* examines the evidence and consults the experts for *9/11: Debunking The Myths* and are “able to debunk each of these assertions with hard evidence and a healthy dose of common sense. We learned that a few theories are based on something as innocent as a reporting error on that chaotic day. Others are the byproducts of cynical imaginations that aim to inject suspicion and animosity into public debate [...] The collapse of both World Trade Center (WTC) towers—and the smaller WTC 7 a few hours later—initially surprised even some experts. But subsequent studies have shown that the WTC’s structural integrity was destroyed by intense fire as well as the severe damage inflicted by the planes. However, that explanation hasn’t swayed conspiracy theorists, who contend that all three buildings were wired with

explosives in advance and razed in a series of controlled demolitions. [...] Experts agree that for the towers to collapse, their steel frames didn't need to melt, they just had to lose some of their structural strength—and that required exposure to much less heat" :: [[Popular Mechanics](#)]

NIST Draft comments

June 2005 :: After a six-week period for public review of their 10000-page report, *NIST* publishes the “Public Comments Received by NIST on DRAFT Reports”. *John Lyle*, Arup, objects that results from 5 FEM indicate that the natural frequency of the structure is not significantly altered by a large hole in the outer tube and missing parts of the core (p. 197); that the global model ignored that the impact scenario was shorter than the natural period of the towers; misses clarification whether P-Δ effects were taken into account as result from swaying (p. 183) :: *Barbara Lane*, Arup, does not agree with NIST's conclusion that only impact induced fire proofing damage caused the collapse. Loss of fire proofing may have reduced the time to collapse, but the analysis presented does not prove that it would have prevented collapse, particularly when so much emphasis is placed, in this regard, on the tests carried out. She believes that the repeated statements about fire protection are a little misleading. (pp. 308, 322) :: *Dave Johnston*, BOMA International, asks NIST exactly how many building collapses have there been.....ever? (p. 354) :: Professor *James Quintiere* does not believe that NIST has presented a convincing argument for their collapse hypotheses for WTC 1 and 2. NIST fails to document the rationale for the speedy elimination of the steel in this incident. NIST has not presented clear and sufficient evidence that the aircraft impacts caused the elimination of insulation, especially from the core columns. The NIST report reads like a scientific enterprise using computer simulations that have never been used (or validated) in this way before. The October surprise in the NIST investigation was the assertion that all of the core column insulation was knocked off by the airplane impacts. NIST has relied on state-of-the-art computer models that are at the forefront of their technologies. However, these models have not been proven comprehensively for less complex incidents than the WTC. Can an engine possibly hit a core column without hitting anything on the floor occupancy and structure? That does not seem possible, so how can an engine damage a core colum? The global collapse mechanism of the buildings must be made as clear as possible. A vague answer expressed by the current NIST working hypothesis is not sufficient. If the core of the answers are really revealed and understood, NIST should be able to explain them in simple fundamental physics, and not shroud them in computer graphics. (pp. 374 ff) :: *Jon Magnusson*, PE, SE, Hon. AIA, says NIST has provided absolutely no data showing there has been even a single death as a result of progressive collapse anywhere in the U.S. The consequences of the WTC attack did NOT constitute a case of progressive collapse. The WTC design MET AND BETTERED all currently proposed progressive collapse standards and code provisions (p. 554) [[NIST](#)]

NIST: does not actually include

September 2005 :: With the specific objective (p. xxxix) to determine why and how WTC 1 and WTC 2 collapsed following the initial impacts of the aircraft, *NIST* in the “Federal Building and Fire Safety Investigation of the World Trade Center Disaster: Final Report of the National Construction Safety Team on the Collapses of the World Trade Center Tower” examine more than 170 areas on 16 perimeter column panels, but only three columns have evidence that the steel reached temperatures above 250°C. Only two core column specimens have sufficient paint remaining to make such an analysis, and their temperatures did not reach 250 °C. Using metallographic analysis, NIST determine that there was no evidence that any of the samples had reached temperatures above 600 °C (pp. 176-177). In state-of-the-art computer simulations, none of the columns with intact insulation reached temperatures over 300 °C. Only a few isolated truss members with intact insulation were heated to temperatures over 400 °C in the WTC 1 simulations and to

temperatures over 500 °C in the WTC 2 simulations. In WTC 1, if the fires had been allowed to continue past the time of building collapse, complete burnout would likely have occurred within a short time since the fires had already traversed around the entire floor, and most of the combustibles would already have been consumed. In WTC 2, if the fire simulation were extended for 2 hours past the time of building collapse with all windows broken, the temperatures in the truss steel on the west side of the building (where the insulation was undamaged) would likely have increased for about 40 min before falling off rapidly as the combustibles were consumed. Temperatures of 700 °C to 760 °C were reached over approximately 15 percent of the west floor area for less than 10 min. Approximately 60 percent of the floor steel had temperatures between 600 °C and 700 °C for about 15 min. Approximately 70 percent of the floor steel had temperatures that exceeded 500 °C for about 45 min. At these temperatures, the floors would be expected to sag and then recover a portion of the sag as the steel began to cool. The temperatures of the insulated exterior and core columns would not have increased to the point where they would have experienced significant loss of strength or stiffness. (p. 184) NIST finds the towers withstood the impacts and would have remained standing were it not for the dislodged insulation and the subsequent multi-floor fires. (p. 174). The focus of the investigation however is on the sequence of events from the instant of aircraft impact to the initiation of collapse for each tower. For brevity in the report, this sequence is referred to as the ‘probable collapse sequence,’ although it (includes little analysis of | does not actually include) the structural behaviour of the tower after the conditions for collapse initiation were reached and collapse became inevitable. (footnotes on pp. xxxvii & 82) [[NIST](#)]

mechanics of progressive collapse: $W_g > W_p$

June 26th, 2006 :: Professor Z. P. Bažant and Mathieu Verdure, at the U.S. National Congress of Theoretical and Applied Mechanics, Boulder, Colorado, present “Mechanics of Progressive Collapse: Learning from World Trade Center and Building Demolitions”. The destruction of the Towers was not only the largest mass murder in U.S. history, but also a big surprise for the structural engineering profession. No experienced structural engineer watching the attack expected the WTC towers to collapse. No skyscraper has ever before collapsed due to fire. The fact that the WTC towers did, beckons deep examination. Further calculations beyond the 2002 “Simple Analysis” are found superfluous, but a theory describing the progressive collapse dynamics beyond the initial trigger, with the WTC as a paradigm, could still be very useful for other purposes, especially for learning from demolitions. They go on to show that progressive collapse will be triggered if the total (internal) energy loss during the crushing of one story (equal to the energy dissipated by the complete crushing and compaction of one story, minus the loss of gravity potential during the crushing of that story) exceeds the kinetic energy impacted to that story. Regardless of the load capacity of the columns, there is no way to deny the inevitability of progressive collapse driven by gravity alone. Also, some critics are addressed, who have been under the mistaken impression that collapse cannot occur if (because of safety factors used in design) the weight mg of the upper part is less than the load capacity F_0 of the floor, which led them to postulate various strange ideas (such as “fracture wave” and planted explosives). However, since the energy loss up to the point at which the load-deflection diagram intersects the line $F=mg(z)$ is smaller than the kinetic energy of the impacting mass $m(z)$ (Fig. 3), it is clear that this impression is erroneous. What matters is energy, not strength, nor stiffness. :: **March 2007** the paper will be published in the *Journal of Engineering Mechanics*. [[Source](#)]

super-heated jet fuel

August 21st, 2006 :: CBC interviews Lee Hamilton, 9/11 Commission co-chair. He says “the engineers and the architects have studied this thing in extraordinary detail, and they can tell you precisely what caused the

collapse of those buildings. What caused the collapse of the buildings [...] was that the super-heated jet fuel melted the steel super-structure of these buildings and caused their collapse. There's a powerful lot of evidence to sustain that point of view, including the pictures of the airplanes flying into the building." [\[CBC\]](#)

very high strength steel not allowed to save cost

September 25th, 2006 :: *Karen Auguston Field*, Editor-in-Chief for *Design News*, cites *Abolhassan Astaneh-Asl* explaining at *MSC. Software's* 2006 "Virtual Product Development Conference" that the simulation model shows the plane slicing right through the outer walls of the as-built building like it was a thin soda can. "Because of their unique design and the use of the so called 'steel bearing wall' tube structural system, which as far as we know has never been used before or after its application in the WTC towers, the buildings essentially showed no resistance to the impact of a medium-sized plane flying into them at about 450 miles per hour. [...] the truss joists collapsed first, leaving the exterior columns of probably two floors in the impact area with no bracing but still under gravity load from the floors above. As the columns heated up and reached temperatures of nearly 1,000F, their strength was reduced to less than half the design strength and they started to buckle. When the columns buckled, the top portion of the building, losing its supports, was pulled down by gravity and dropping on the floors below, pancaking the floors one after another and leading to progressive collapse in an almost perfect vertical direction of the pull of gravity force." He also notes that designers chose to fabricate many of the building columns out of very high strength steel [90 ksi steel as opposed to the more typical 36-65 ksi steel], which is not allowed by the structural design codes then and is still not allowed in current codes. "But the World Trade Center did not need to obtain a permit from City Hall. Because of special status as Port Authority of New York and New Jersey, they could make such choices outside the prevailing codes." This choice allowed builders to use less steel in the columns [two to three times thinner than typical columns] to save cost. [\[Design News\]](#)

gravity alone must have caused just that

March 31st, 2008 :: To check whether the recent allegations of controlled demolition, which rest on the free fall hypothesis, have any scientific merit and to prove that they do not, *Z. P. Bažant*, *Jia-Liang Le*, *Frank R. Greening* and *David B. Benson* present "What Did and Did Not Cause Collapse of WTC Twin Towers in New York". It is shown that the observed size range (0.01 mm—0.1 mm) of the dust particles of pulverized concrete is consistent with the theory of comminution caused by impact, and that less than 10% of the total gravitational energy, converted to kinetic energy, sufficed to produce this dust — whereas more than 150 tons of TNT per tower would have to be installed to produce the same pulverization. Also, initial speculation that very high temperatures were necessary to explain collapse are revised based on NIST's findings, since steel temperature $\geq 150^{\circ}\text{C}$ sufficed to trigger the viscoplastic buckling of columns. To get convinced of the inevitability of gravity driven progressive collapse, further analysis is, for a structural engineer, superfluous, but nevertheless needed to dispel false myths. It is no surprise at all that the towers collapsed essentially on their footprint. Gravity alone must have caused just that. [\[Source 1\]](#), [\[Source 2\]](#)

NIST FAQ: multiple factors

September 19, 2011 :: The *NIST FAQ* state that the WTC towers collapsed because: (1) the impact of the planes severed and damaged support columns, dislodged fireproofing insulation coating the steel floor trusses and steel columns, and widely dispersed jet fuel over multiple floors; and (2) the subsequent unusually large number of jet-fuel ignited multi-floor fires (which reached temperatures as high as 1,000

degrees Celsius) significantly weakened the floors and columns with dislodged fireproofing to the point where floors sagged and pulled inward on the perimeter columns. This led to the inward bowing of the perimeter columns and failure of the south face of WTC 1 and the east face of WTC 2, initiating the collapse of each of the towers. Both photographic and video evidence — as well as accounts from the New York City Police Department aviation unit during a half-hour period prior to collapse — support this sequence for each tower. NIST's findings do not support the “pancake theory” of collapse, which is premised on a progressive failure of the floor systems in the WTC towers. Instead, the NIST investigation showed conclusively that the failure of the inwardly bowed perimeter columns initiated collapse and that the occurrence of this inward bowing required the sagging floors to remain connected to the columns and pull the columns inwards. Thus, the floors did not fail progressively to cause a pancaking phenomenon. [\[NIST\]](#) The *NIST FAQ* for WTC 7 clarify that progressive collapse did NOT occur in the WTC towers. First, the collapse of each tower was not triggered by local damage or a single initiating event. Second, the structures were able to redistribute loads from the impact and fire-damaged structural components and subsystems to undamaged components and to keep the building standing until a sudden, global collapse occurred. Had a hat truss that connected the core columns to the exterior frame not been installed to support a TV antenna atop each WTC tower after the structure had been fully designed, it is likely that the core of the WTC towers would have collapsed sooner, triggering a global collapse. Such a collapse would have some features similar to that of a progressive collapse. The Twins were brought on by multiple factors, including structural damage caused by the aircraft impact, extensive dislodgement of the sprayed fire-resistive materials or fireproofing in the impacted region, and a weakening of the steel structures created by the fires. [\[NIST\]](#)

Ten Years later ...

all factors

September 21st, 2011 :: According to a theory advanced by materials scientist *Christian J. Simensen* of SINTEF Materials and Chemistry, Norway, a mixture of water from sprinkler systems and molten aluminum from melted aircraft hulls created explosions that led to the collapses of the Twin Towers: both scientific experiments and 250 reported disasters suffered by the aluminium industry have shown that the combination of molten aluminium and water releases enormous explosions. [\[Science Daily\]](#) Professor *Bažant*, according to “New Twin Tower Collapse Model Could Squash 9/11 Conspiracies”, thinks that the official explanation suffices, as he had explained it in six papers in leading journals. In his opinion, all factors related to the collapse have been accounted for. [\[Live Science\]](#)

knew right away

May 20, 2013 :: For “Bringing down the House” (*U.S. News*), *Michael Satchell* interviews Mark Loizeaux of *CDI*. As experts in buildings’ vulnerabilities, they knew right away what few Americans realized. “I told Doug immediately that the tower was coming down, and when the second tower was hit, that it would follow,” remembers Mark. The Loizeaux brothers watched first responders streaming into the doomed towers and tried frantically, and unsuccessfully, to phone in warnings. [\[U.S. News\]](#)

different degrees of ductility, fracturing and end support flexibility

September 4th, 2016 :: In response to experiments by *Korol* and *Sivakumaran*, *Jia-Liang Le* and *Z.P. Bažant* concede that, if the column ends were rigidly supported and if the ductility of steel was unlimited, then the

simple plastic three-hinge mechanism with constant bending moments, would have dissipated ~3.5-times as much energy than considered in previous studies, but posit that calibration by matching of the video record of initial collapse implies that this energy must have been reduced to about 2/3 of the energy predicted by the three-hinge model. This estimated 2/3 reduction must have been caused by the fracturing of steel and by the flexibility of spandrel beams which reduced the rotations of the plastic hinges at column ends. With this update of input data, all the observed features of the WTC collapse remain to be closely matched by the gravity-driven mechanics of progressive collapse. [[Source](#)]

collapse must have been spontaneous

February 13th, 2017 :: Responding to an article published earlier by *Europhysics News*, *Jia-Liang Le* and *Z.P. Bazant* reiterate in “Mechanics-based mathematical studies proving spontaneity of post-impact WTC towers collapse” that the towers were designed not to fail due to airplane impact damage – and that they didn’t. The progressive collapse must have been spontaneous, gravity-driven, and that after impact, no external weakening of the structure was needed to explain the collapse. The main cause of the total collapse of the towers damaged by impact was a fire of enormous proportions. The kinetic energy of the top part exceeded by an order of magnitude the energy required for complete buckling of all the columns of the cold story. In the lower floors, one story got squashed within mere 0.07 s. The air ejected from the story must have reached, Mach 1. The sonic booms heard, the rapidly expanding dust clouds and the wide ejection of debris are no surprise. From the viewpoint of physics, and structural mechanics in particular, it is perfectly clear that no WTC demolition took place. The collapse was triggered by an atypical fire ignited simultaneously in a large volume. It was driven by gravity, and was spontaneous. In hindsight, it was, under the given circumstances, inevitable. Experience from mining and tunneling shows that such 0.01mm small concrete particles could be produced by explosives only if the alleged authors installed about 150 tons of TNT into small holes drilled into all concrete floor slabs of each tower. Critics do not explain how such a massive operation, requiring many workers, could have been carried out in secrecy, no one leaking it later to the public. [[Europhysics News](#)]

Twenty Years later ...

many different factors working together in a complex fashion

August 18th, 2021 :: *S. Shyam Sunder* recaps in a *NIST Blog* post titled *20 Years Later: NIST’s World Trade Center Investigation and Its Legacy*: “The WTC buildings collapsed from many different factors working together in a complex fashion. [...] We have reviewed other WTC studies that have come out since we completed our investigation, including those that consider alternative hypotheses for the WTC building collapses. Based on our exhaustive analysis and the evidence available to us, NIST continues to stand behind the findings and recommendations of our investigation.” [[Taking Measure](#)]

would still be standing without the fires

September 8th, 2021 :: *Terry Grant* interviews *Barzin Mobasher*, Arizona State University civil engineering professor, for *ASU News*, opines: “Were it not for the fire, engineers could secure the building, strip out the damaged components and repair and retrofit. [...] The potential energy stored in the structure due to weight of the building itself at the floor heights was sufficient to create a self-driving and accelerating domino collapse effect much like a landslide that gains speed through the process.” [[ASU News](#)]

like an axe head driven through wood

September 8th, 2021 :: *Donald Dusenberry* in *What Structural Engineers Learned from 9/11* explains: “The four sides of each building formed a tube with window slots. This tube was designed to carry the weight of the building and its contents directly downward to the foundations. The walls acted as vertical beams, wide and tall as the buildings, that bent to the side to absorb wind loads. [...] Failure finally occurred when the intense fires inside the towers weakened floor systems that braced the exterior walls. When the walls buckled, the upper portions of the buildings fell as blocks and were driven through the lower floors as they fell, much like an axe head driven through wood to split it.” [[Scientific American](#)]

widely accepted by experts

September 8th, 2021 :: *David Oswald, Erica Kuligowski and Kate Nguyen* clarify in *9/11 conspiracy theories debunked: 20 years later, engineering experts explain how the twin towers collapsed*: “The fact that the towers withstood initial impacts also spawned one of the most common conspiracy theories surrounding 9/11: that a bomb or explosives must have been detonated somewhere within the buildings. These theories have developed from video footage showing the towers rapidly collapsing downwards some time after impact, similar to a controlled demolition. But it is possible for them to have collapsed this way without explosives. It was fire that caused this. And the fire is believed to have come from the burning of remaining aircraft fuel. [...] The floors’ failure pulled the columns back inwards, eventually leading to them buckling, and the floors collapsing. The collapsing floors then fell on more floors below, leading to a progressive collapse. This explanation, documented in the official reports, is widely accepted by experts as the cause of the twin towers’ collapse. It is understood the South Tower collapsed sooner because it suffered more damage from the initial aircraft impact, which also dislodged more fireproofing material. [...] The events of 9/11 may have been challenging to foresee, but the fires that led to the towers’ collapse could have been better prepared for.” [[The Conversation](#)]

much like a hot spaghetti noodle

September 11th, 2023 :: *Dr. Douglas Schmucker* presents a review of the critical collapse sequence in the second-year course “CvEEN 2140: Strength of Materials”. With the aid of the official Building Performance Summary, he depicts the collapse sequence as “most plausibly resulting from a combination of impact and debris damage to both structural members and fire retardant on those members, partial collapse of floors due to either heat from the building content fires or impact damage, and then with failure of the floors buckling of the columns. Buckling of exterior columns, which can be seen directly in video and photographs, rapidly progressed to ‘pancaking’ of floor after floor as the energy of elevated floors moving downwards could not be absorbed.” The primary heat-related effects were from simultaneous fires on multiple floors, floors that had experienced a total loss of or compromised fire retardant and suppression systems. “The effect on the floor system was to cause it to drape much like a hot spaghetti noodle will lose its stiffness.” [[University of Utah](#)]

steel frames didn’t need to melt

September 11th, 2025 :: *PM* reiterates, in a *Special Report: Debunking the 9/11 Conspiracy Theories of the World Trade Center*, the statements made 20 years earlier: “The WTC’s structural integrity was destroyed by

intense fire as well as the severe damage inflicted by the planes. [...] Experts agree that for the towers to collapse, their steel frames didn't need to melt, they just had to lose some of their structural strength—and that required exposure to much less heat.” :: The piece addresses initiation (heat, fireproofing loss) without discussing post-initiation progression. :: [[*Popular Mechanics*](#)]

Exhibit B

A Diagnostic Material Model:

The Rate-Dependent Collapse Threshold

A recurring objection to this analysis is that it “proposes no alternative explanation.” This reflects a category error: the analysis is methodological, not mechanistic. However, the objection invites a useful diagnostic question:

What material properties would be required for the observed behavior to follow generically from the stated mechanisms?

Answering this clarifies the burden of proof.

The Anti-Newtonian Solid

Standard materials exhibit predictable responses: Newtonian fluids (stress proportional to shear rate, $\sigma = \eta\dot{\gamma}$); dilatant fluids like cornstarch slurry (viscosity increases with shear rate — hit fast, acts solid); structural steel (strain-rate *hardening* — higher yield strength under rapid loading).

Now consider the inverse: a material supporting large static loads with high capacity, but offering vanishing resistance under rapid dynamic impact. Call this an **Anti-Newtonian Solid**: $\sigma = \eta\dot{\gamma}^{-k}$, $k > 1$. At low strain rates, $\sigma \rightarrow \infty$ (high resistance); at high strain rates, $\sigma \rightarrow 0$ (progressive “mush”).

Real steel does not behave this way — it exhibits strain-rate hardening, not softening, and its fracture toughness scales with cross-section, not against it. Yet for progressive collapse to proceed as described — rapidly, symmetrically, without arrest, across dozens of stories — the structure must behave *as if* it possessed Anti-Newtonian properties: robust enough under service conditions that the towers stood for decades, yet offering minimal resistance once dynamic descent begins, with some threshold velocity implicitly separating the two regimes. Had ordinary steel instead followed $\sigma = \sigma_0(\dot{\gamma}/\dot{\gamma}_{ref})^n$, $n > 0$ — real rate-hardening — rapid impacts would *increase* resistance and favor arrest. Achieving the inverse would require some combination of synchronized thermal weakening, connection failure thresholds, geometric nonlinearity, or coordinated redistribution suppression — none of which are generic properties of steel structures; each is a conditional outcome requiring a specific structural state to be independently demonstrated.

This is not impossible. It is simply not a generic property of steel structures.

Viscoplastic Buckling Extension

For columnar structures, couple this to effective modulus:

$$E_{\text{eff}} = E_0 (1 + (\dot{\gamma} / \dot{\gamma}_0)^{-k})^{-1}$$

Then Euler buckling load becomes rate-dependent:

$$P_{cr} = \pi^2 E_{eff} I / L^2$$

Under slow gravity: $E_{eff} \approx E_0 \rightarrow$ normal capacity

Under impact ($v \approx \sqrt{2gh} \sim 15 \text{ m/s}$): $\dot{\gamma} \gg \dot{\gamma}_0 \rightarrow E_{eff} \rightarrow 0 \rightarrow$ catastrophic buckling

This formalizes the intuitive claim: collapse propagation implicitly assumes that dynamic loading drives resistance toward zero.

Energy Implications

Under the Anti-Newtonian model, energy dissipation per story:

$$W_d = \int \sigma d\epsilon \approx \eta \dot{\gamma}^k \epsilon_{max}$$

As impact velocity increases ($\dot{\gamma} \uparrow$), dissipation *decreases*—consistent with observed rapid descent.

But real steel dissipates *more* energy under dynamic loading, not less. The tension is resolved only if:

- thermal weakening reduces σ_0 dramatically, or
- connection failures bypass plastic dissipation, or
- comminution absorbs energy via different pathways

Each is a contingent claim. None are generic.

The Satirical Core

The Anti-Newtonian Solid is a **reductio**: it shows what material properties would make the claimed mechanism trivially inevitable.

Real structures do not behave this way.

Therefore, claims that observed behavior follows “generically” from gravity and fire must either:

1. Show that fire+damage creates effective Anti-Newtonian conditions broadly across parameter space,
or
2. Retract from “inevitable” to “plausible under specific conditions”

The first is a modeling burden. The second is honest epistemic positioning.

Changelog v1.1 [2026-09-13]

Monitum

REWRITE: opening metaphor
REWRITE: redundant disclaimers removed, text tightened for better flow

Part I: Epistemic Limits

ADDITION: Section 5.5 The Evidential Burden of Inevitability Claims (2026-01-29).
ADDITION: Section 5.5.1 The Duhem-Quine Insulation of Core Hypotheses.
ADDITION: Section 5.5.2 Epistemic Closure by Administrative Decision.
ADDITION: New rows Summary Table: Epistemic Tensions
REMOVED: "What This Document Argues"

Part II: Dialectic Inventory

ADDITION: Section 4.3.1 The Coefficient of Restitution Paradox (formerly 4.3.5 in earlier drafts).
ADDITION: Section 4.3.2 The Acceleration Ceiling Under Pure Inelastic Accretion
ADDITION: Section 4.3.3 d'Alembert's Principle
ADDITION: Section 4.3.4 What NIST's Own Record Says About the Core.
ADDITION: Section 4.4 Ensemble Sensitivity Analysis of the Discrete Floor Model
ADDITION: Section 4.6.2 Spatiotemporal Coherence
ADDITION: Section 4.8.5 The Size Effect Law
ADDITION: Sections 4.9 & 4.10 (Exegesis of localization assumptions, Severity Parameter Γ , ξ_{out} , R).
ADDITION: Section 6.5.1 The Load-Sharing Switch (F_{c_sys} vs F_{c_iso}).
ADDITION: Section 6.5.2 The Snap-Through Cascade and the Localization-Kinematics Tension.
ADDITION: Section 6.7.1 A Worked Example: The Measured and the Unmeasured Mobilization Factor.
ADDITION: Sections 6.11.1 through 6.11.6 Case studies, "Compute Defense".
REPLACED: Variable name collisions fixed & standardized ($\gamma \rightarrow \Gamma$, $W_g \rightarrow E_g$, $W_p \rightarrow E_d$, $\kappa_{out} \rightarrow \xi_{out}$, $\beta \rightarrow R$)
REMOVED / MOVED: Original Section 3.3.5 Anti-Newtonian Solid: Exhibit B.
MERGED / REWRITE: Original Sections 3.1, 3.2, 3.6 consolidated & rewritten into new 3.1 & 4.6-4.10 framework
REWRITE: Refined, expanded Section 4.7 (Scale Arguments) & 5.3 (Quantitative Energy Asymmetry)

Part III: Chronology of Closure

ADDITION: Section 2.5.3 The Europhysics Incident
ADDITION: Section 5.5 The Unfalsifiable Comparator
ADDITION: Section 5.6 A Non-Sinister Account of the Same Pattern
REWRITE: Chapter 3 tightened, reframed as 3.2 The Single Defensible Point and A Note on This Document's Own Exposure
REWRITE: General corrections to balance tone

Timeline & References

ADDITION: New Timeline entries added (Kausel, Dusenberry, Oswald, Hamburger, Popular Mechanics 2025).
ADDITION: New References added (Němec, Kosciuk/Robbin, Cherepanov, Hoffman, Schneider, Sezen, Wu, Isobe, Wong-Cayley, Mungan, Hamm-Géminard, etc.)
ADDITION: "Concordance with cited sources" table

Exhibits & Appendix

REPLACED: historical Timeline was extracted and formatted as a standalone Exhibit A
REPLACED: "Anti-Newtonian Solid" thought experiment moved from main text (old 3.3.5) to Exhibit B

General / Formatting

FIXED: Syntax and formatting errors.
TWEAKS: Sub- and superscripts, math blocks, and citation syntax optimized for .md $\rightarrow$.odt $\rightarrow$.pdf workflow.
REWRITE: Redundant disclaimers regarding "not proposing alternative theories" pruned
REWORK: Summary Table & Overviews Pts I & II

FRONT EPICYCLOID: {"velocities":["9","-11","25"],"radii":["5.7","-14.8","4.5"],"phases":["0","0","0"]}